



MOED REFERENCE GUIDE

Comprehensive User Manual

Plan engagements. Keep relationship history.
Coordinate follow-up. Manage communication.
Protect and recover shared records.

ONE CONNECTED WORKFLOW

Inquiry > Engagement > Presentations

People > Chron and email > Follow-up

Financial closeout > Relationship history

Application source 2.0.8 | September 10, 2026

Reviewer, editor, and administrator reference

[Open contents >](#)

Clickable chapters, topic finder, bookmarks, and related-topic navigation

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How to use this manual

A complete reference with illustrated walkthroughs

Navigate in four ways

Use the clickable contents for chapters; use the bookmarks panel for sections and screenshots; follow Related chapters links for connected work; or use the alphabetical topic finder. Every page has Contents, Topic finder, Previous, Next, and Chapter start links.

Scope and edition

This manual describes the MOED browser application at source version 2.0.8, updated on September 10, 2026. It covers reviewer, editor, and administrator workflows and the configured Mattermost integration. Deployment-specific settings may change names, reminder windows, available integrations, or branding.

The 13 chapters follow the sidebar User Manual. Each begins with the reference guidance and continues with illustrated walkthroughs. Section and figure headings also appear as PDF bookmarks.

Screenshots and example records

Screenshots were captured from the current source in an isolated local preview. Cedar Grove Community, Jordan Parker, Alex Morgan, Example Conference, Avery Host, Casey Speaker, and their events and inquiries are fictional training examples. A few existing preview records and the preview account name may appear in directory or account views. No production messages were sent and no invitations were submitted.

The previews have no connected Mattermost service. Mail is disabled or uses a local development log; no email was queued during screenshot capture. Those capabilities are documented from the application guide and source; screenshots label the configuration limitation. Empty financial/statistics views represent a new training record, not a claim that the feature is unavailable.

Screenshots show selected viewports and form sections. Scroll in the application to continue longer forms. Zoom into the PDF for fine control labels. Most screenshots use the light theme; dark theme and mobile navigation are illustrated separately.

Meaning of access labels

Reviewer: read records, export, inspect statistics, and manage the personal account.

Editor: additionally create, edit, archive/restore, manage work, and correspond.

Administrator: additionally govern users, audit, backups, and permanent deletion.

Related chapters: [Roles and access \(p. 8\)](#) / [Daily dashboard \(p. 10\)](#) / [Profile and security \(p. 75\)](#)

CHAPTER 01

Getting Oriented

MOED keeps event planning, people, communication history, and next actions connected instead of scattering them across separate systems.

Related chapters: [Roles and access \(p. 8\)](#) / [Daily dashboard \(p. 10\)](#) / [Profile and security \(p. 75\)](#)

The Record Model at a Glance

An Inquiry holds a possible booking while it is being qualified. Once booked, it creates an Engagement linked to an Organization and its Contacts. These records can carry Chron Log history and Follow-Up Work.

Use the Sidebar

The main areas are always in the left sidebar. Under Work, open **Email Templates** to maintain reusable engagement messages. On a narrow screen, use the menu button in the top bar and tap outside the panel or press **Esc** to close it.

Switch the Theme

Select **Dark Theme** or **Light Theme** near the bottom of the sidebar. MOED remembers the choice in this browser.

Open Your Account

Select your name and picture to edit your profile. Calendar, account security, this manual, and sign-out controls are grouped in the lower sidebar.

A Practical First Session

- 01 Complete your profile.** Add your name, verified email, phone, and picture in [My Profile](#).
- 02 Protect the account.** Enroll an authenticator and store the one-time recovery codes from [Account Security](#).
- 03 Check the daily view.** Use the [Dashboard](#) to scan upcoming engagements, assigned work, and missing details.
- 04 Find the relationship.** Open the organization and relevant contacts before an event to review relationship details,

financial history, Chron, and open work.

- 05 Leave a clear next action.** When a conversation creates a commitment, record the communication in Chron and create an owned, dated task.

Browse Longer Lists

Use **Records per page**, where available, to choose 20, 50, or 100 records. Each list remembers your preferred size in this browser. Use the page numbers, **Previous**, or **Next** to move through results. Search and sorting apply to the full result list, so you do not need to search each page separately.

Common Action Icons

View - Open details

Edit - Change a record

Start - Begin a task

Complete - Finish a task

Archive - Hide reversibly

Restore - Return to active

Delete - Remove permanently

Archive first. Archiving is reversible and keeps history. Permanent deletion is limited to administrators, requires freshly confirmed administrator access, and cannot be undone.

Save, cancel, and recover a draft

Required fields are marked in the forms. Correct the listed errors and save again if validation fails. Cancel returns without saving the current form.

When you create a related organization or contact from an inquiry/contact form, MOED carries the draft back to the original form. Confirm the selected relationship after returning.

If another person changes a record while you are editing, MOED may report a conflict. Preserve your unsaved text, reload the latest record, and reapply only the changes that still belong. A reload is also the first step after an expired request.

Keyboard users can use the Skip to main content link, Tab/Shift+Tab, and the visible focus indication. Reduce-motion preferences suppress attention animations while retaining due-state color cues.

CHAPTER 01 / ILLUSTRATED WALKTHROUGH

The online manual and navigation

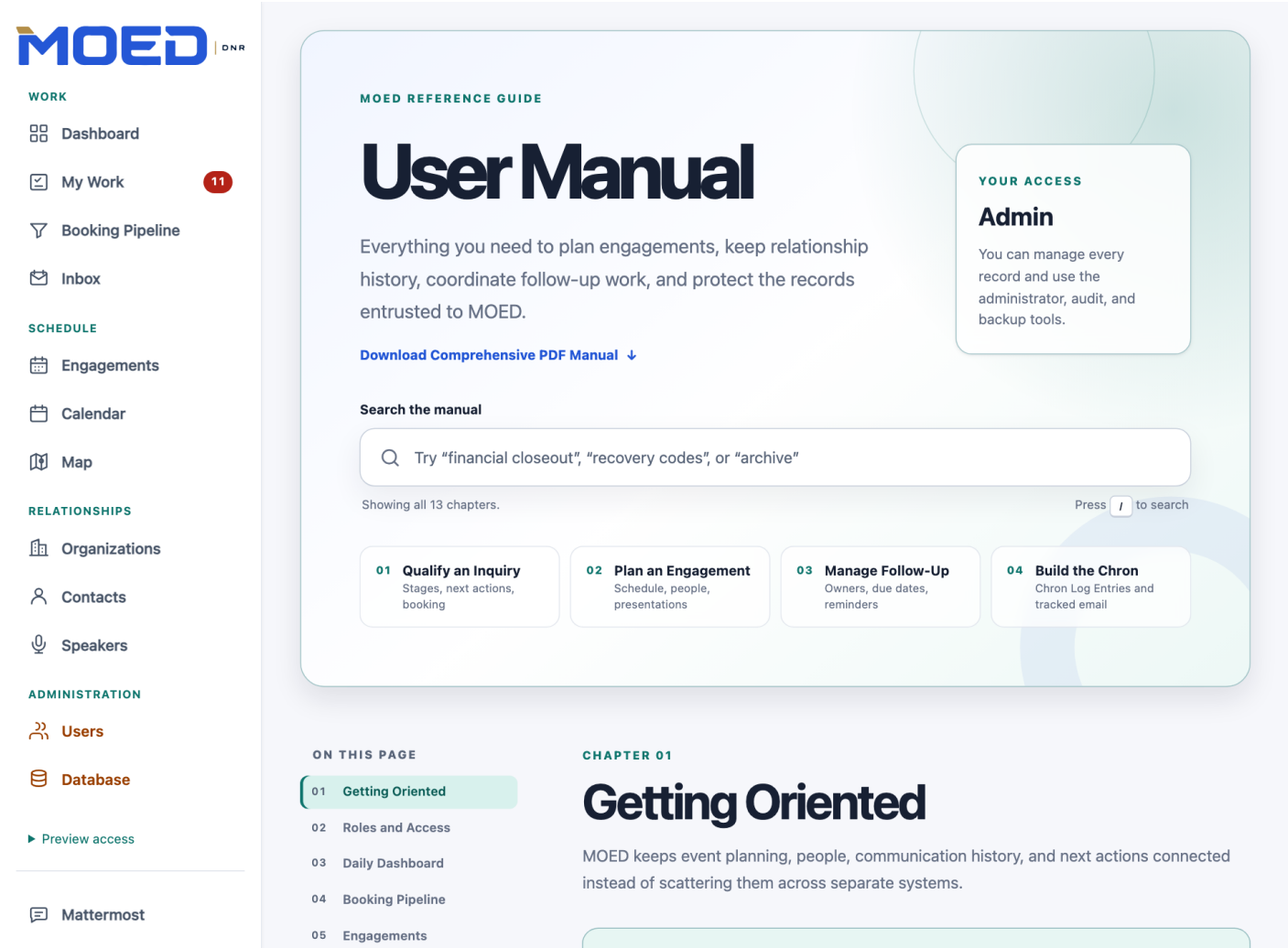


Figure 01. The online manual and navigation. Actual preview screen; fictional records.

- 01 Open User Manual near the bottom of the sidebar. Its 13 chapters match this PDF.
- 02 Use the chapter rail or popular-topic cards to jump to an area.

- 03 Press / in the online manual to focus search. In this PDF use your reader's Find command.
- Related chapters: [Roles and access \(p. 8\)](#) / [Daily dashboard \(p. 10\)](#) / [Profile and security \(p. 75\)](#)

CHAPTER 01 / ILLUSTRATED WALKTHROUGH

Dark theme

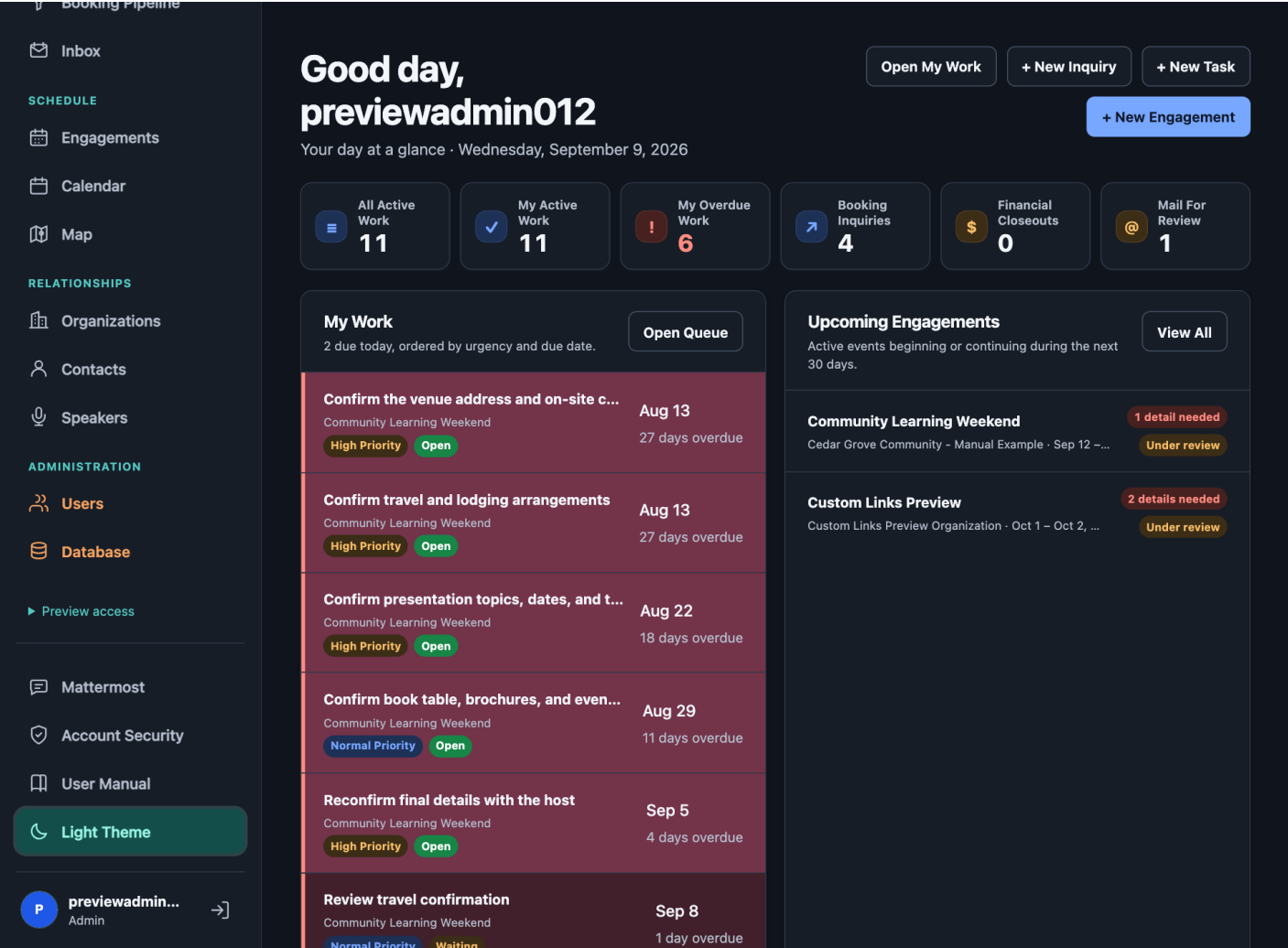


Figure 02. Dark theme. Actual preview screen; fictional records.

01 Select Dark Theme in the sidebar to use the dark palette.

02 Select Light Theme to return. The preference is remembered by this browser.

Related chapters: [Roles and access \(p. 8\)](#) / [Daily dashboard \(p. 10\)](#) / [Profile and security \(p. 75\)](#)

CHAPTER 01 / ILLUSTRATED WALKTHROUGH

Navigation on a phone

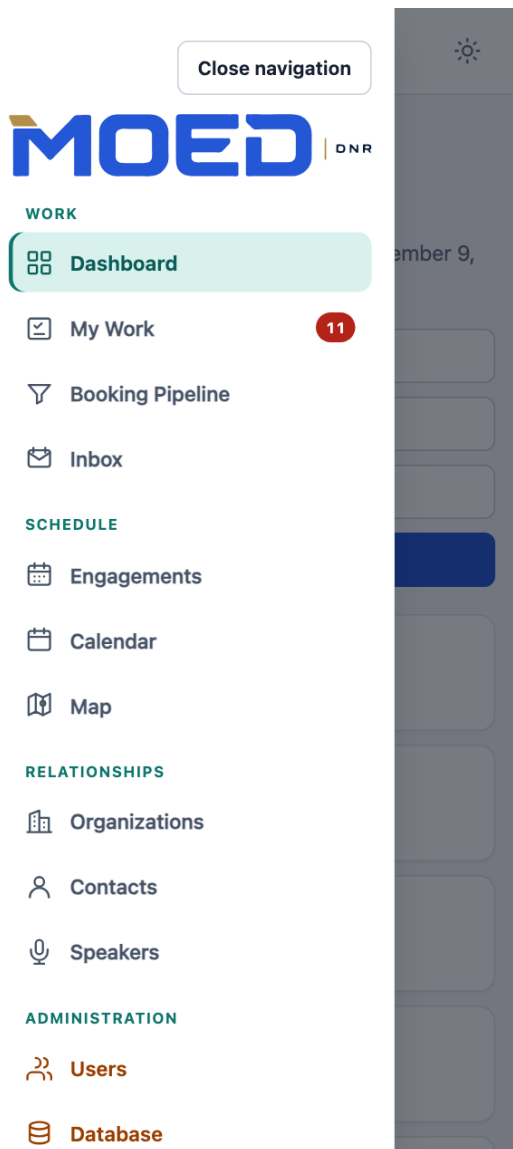


Figure 03. Navigation on a phone. Actual preview screen; fictional records.

01 Tap the menu button in the top bar to open navigation.

02 Choose an area, tap outside the panel, or press Esc to close it.

03 Long lists and forms continue vertically; use their own filters and actions.

Related chapters: [Roles and access \(p. 8\)](#) / [Daily dashboard \(p. 10\)](#) / [Profile and security \(p. 75\)](#)

CHAPTER 02

Roles and Access

Everyone can read the shared operational picture. Editing and destructive actions are deliberately narrower.

Related chapters: [Profile and security \(p. 75\)](#) / [Administration \(p. 87\)](#)

Capability	Reviewer	Editor	Administrator
View records, Chron, work, map, and exports	Yes	Yes	Yes
Manage own profile, calendar links, and security	Yes	Yes	Yes
Create and edit inquiries, engagements, organizations, contacts, and speakers	No	Yes	Yes
Manage tasks, Chron entries, closeouts, and engagement email	No	Yes	Yes
Add, edit, archive, and restore email templates	No	Yes	Yes
Archive and restore shared records	No	Yes	Yes
Manage users, audit history, backups, and permanent deletion	No	No	Yes

Read and Export

Reviewers can inspect shared records, follow links between them, use search and filters, copy presentation briefs, view presentation and QR-code PDFs, review link statistics, and subscribe to the calendar.

Coordinate and Maintain

Editors add reviewer capabilities plus record creation, editing, archive/restore, task management, Chron maintenance, financial closeout, engagement email, and inbound email review.

Govern and Recover

Administrators add user lifecycle controls, audit inspection, encrypted backups, permanent deletion, and access to deployment signals. Sensitive actions require fresh password and 2FA confirmation.

CHAPTER 02 / ILLUSTRATED WALKTHROUGH

Preview role permissions

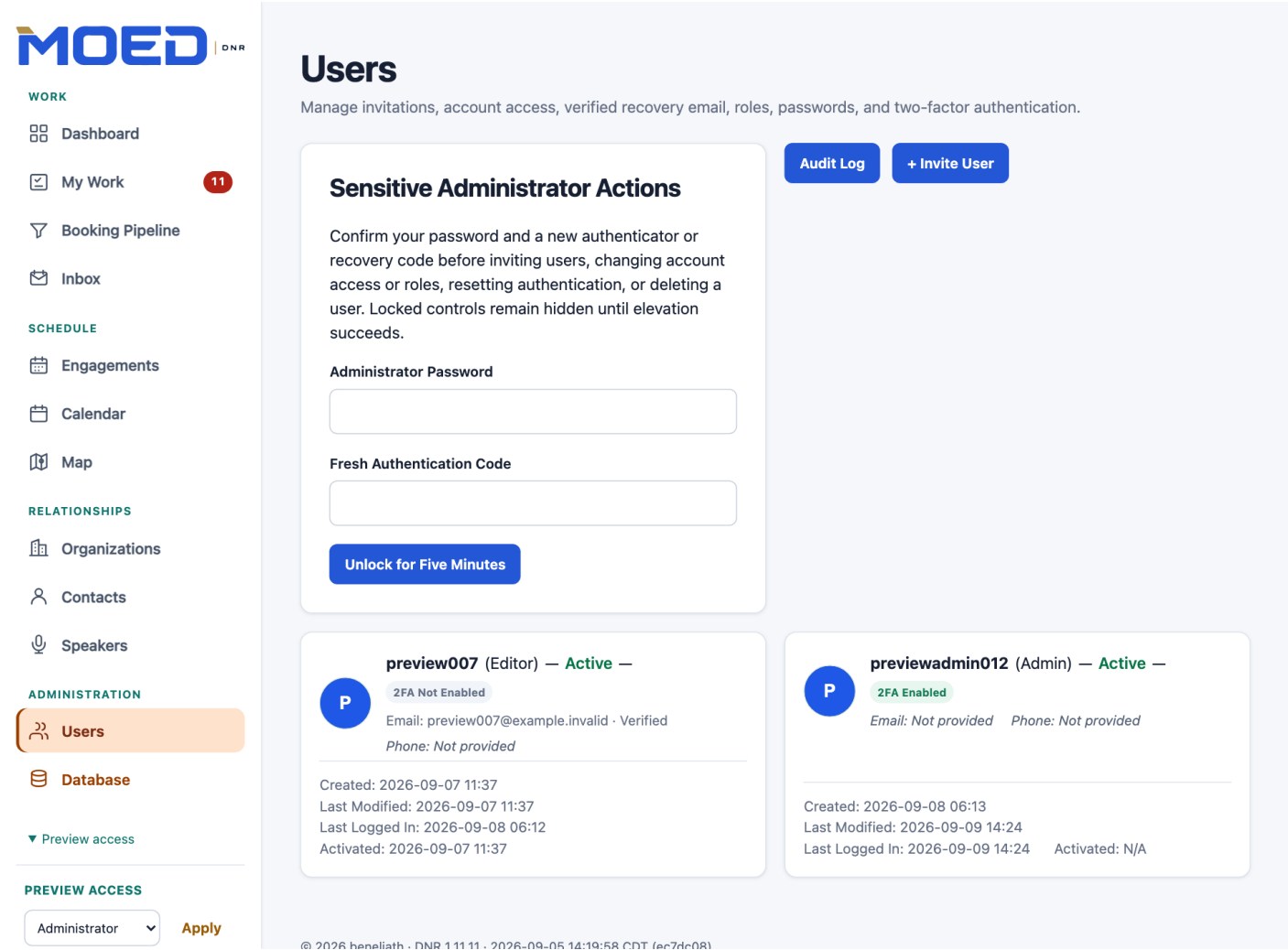


Figure 04. Preview role permissions. Actual preview screen; fictional records.

- 01** Administrators can expand Preview access in the sidebar and select Editor or Reviewer.

02 Apply changes the menus and effective permissions. Return to Administrator ends the preview.
- 03** Preview actions affect application data. Use a separate sample environment when practicing.

Related chapters: [Profile and security \(p. 75\)](#) / [Administration \(p. 87\)](#)

CHAPTER 03

Daily Dashboard

The Dashboard answers three questions: what is coming, what is yours, and what is incomplete.

Related chapters: [Booking pipeline \(p. 12\)](#) / [Engagements \(p. 19\)](#) / [Work queue \(p. 54\)](#)

Daily Summary

The compact top cards distinguish all active work from your own active and overdue work. Shared counts link to the booking pipeline, financial closeouts, and - when available to your role - inbound mail awaiting review.

Primary Panels

My Work leads with assigned tasks by urgency and due date, alongside **Upcoming Engagements**. Open the **Booking pipeline** disclosure below Needs Attention for inquiry next actions and pipeline health.

My Work and Inquiry Next Actions show a short due date with a second line such as **1 day overdue**, **Due today**, or **Upcoming**. Items more than 3 days overdue pulse red in a repeating cascade to draw your attention. For inquiries, this is based on the next-action due date.

Event Readiness

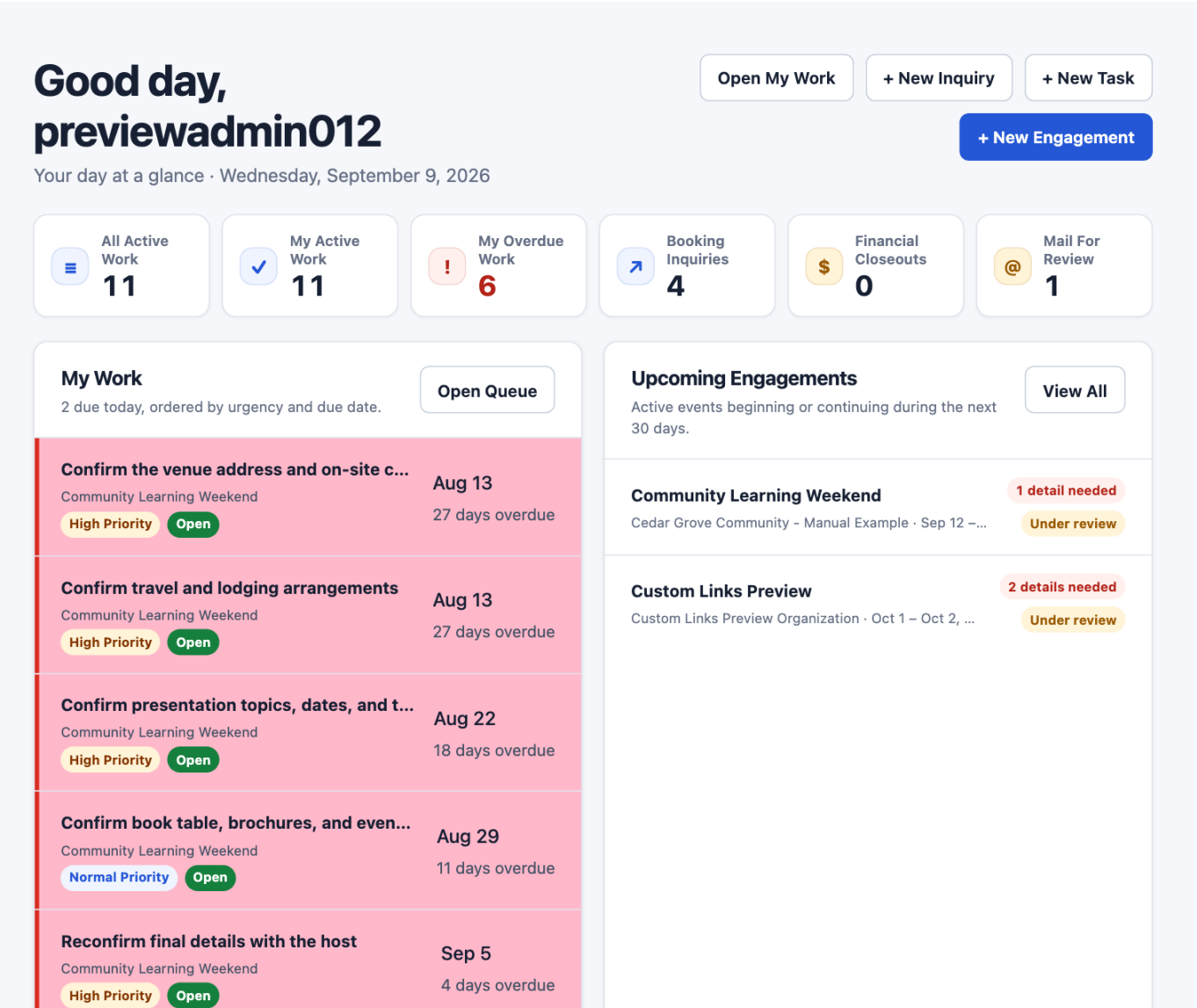
Readiness flags an unconfirmed event, missing venue address, missing presentation, or missing event contact assignment. Open the event to complete the missing details.

Financial Closeouts

Ended events remain here until actual giving/income, lodging, and travel receipts are finalized. The age indicator shows how many days the report has been waiting.

CHAPTER 03 / ILLUSTRATED WALKTHROUGH

Start the day on the Dashboard



CHAPTER 04

Inquiry and Booking Pipeline

Use an Inquiry for a possible event that is not booked yet. It keeps early communication and follow-up visible without creating a partly known Engagement.

Related chapters: [Engagements \(p. 19\)](#) / [Work queue \(p. 54\)](#) / [Chron and email \(p. 61\)](#)

Inquiry First, Engagement After Booking

The Inquiry is the source record for qualification. Converting it creates the Engagement once an organization, title, and preferred date range are known. The original Inquiry then becomes read-only and remains linked for reference.

Work the Pipeline

- 01 Capture the request.** Select **New Inquiry** in the Booking Pipeline. Add the organization and contact when known; they can remain blank during early triage.
- 02 Assign the next action.** Choose an owner and priority, then state the concrete next step and its due date. The pipeline cards and Dashboard emphasize this action.
- 03 Move the stage.** Use **New > Contacted > Qualified > Awaiting Details > Proposal Sent** as the normal path. Move backward when the facts warrant it. Declining requires a reason and can be reopened later.
- 04 Keep the history together.** Open the Inquiry and use the **Activity**, **Correspondence**, and **Tasks** tabs to keep decisions, email, and follow-up work together. Open a message from Correspondence to review delivery or retry a failed delivery while the Inquiry is active.
- 05 Review the booking.** Select **Convert to Engagement** when the request is ready. Check the imported engagement details, availability warnings, and which open tasks should move. Clearing every task checkbox moves none. If a next action remains, explicitly carry it forward as a task or resolve it with a reason; the original action and your decision remain in history. Warnings require acknowledgement but remain a deliberate, auditable decision.

- 06 Continue in the Engagement.** Conversion creates the Engagement, assigns the Inquiry's primary contact as Primary Host when available, moves selected work, and adds the standard event checklist. Replies to earlier Inquiry correspondence continue into the converted Engagement.

Filter the Board

Search request text, organizations, contacts, and locations. Use the owner and next-action filters to find your inquiries, unassigned work, overdue actions, or requests that still need a next step.

Use the Starting Templates

Initial response, request details, date options, and proposal follow-up templates are editable before sending. Sent messages appear in Correspondence and Chron with delivery status.

Keep the Source Record

Do not copy pre-booking Chron Log Entries by hand. The booked Inquiry retains its source email, stage history, Chron, correspondence, and any work not moved to the Engagement. Resolve or reassign active Inquiries before archiving their Contact or Organization.

The default board shows active inquiries and those booked this month. Use the **Stage** filter for **Booked**, **Declined**, or **All Stages**. In All Stages, Declined starts on a separate row. The stages wrap to fit your screen; scroll down to see the remaining rows.

CHAPTER 04 / ILLUSTRATED WALKTHROUGH

Read and filter the booking board

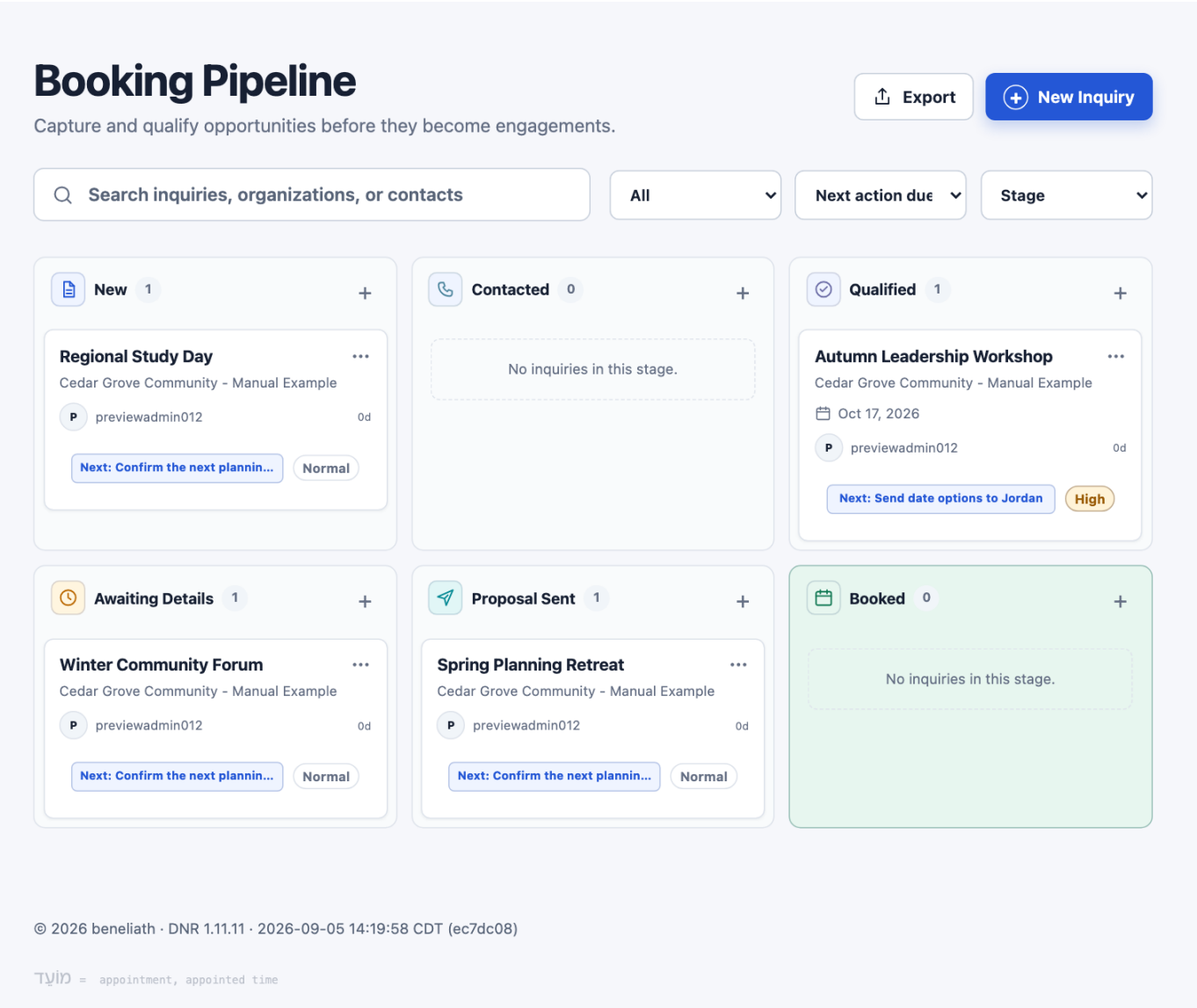


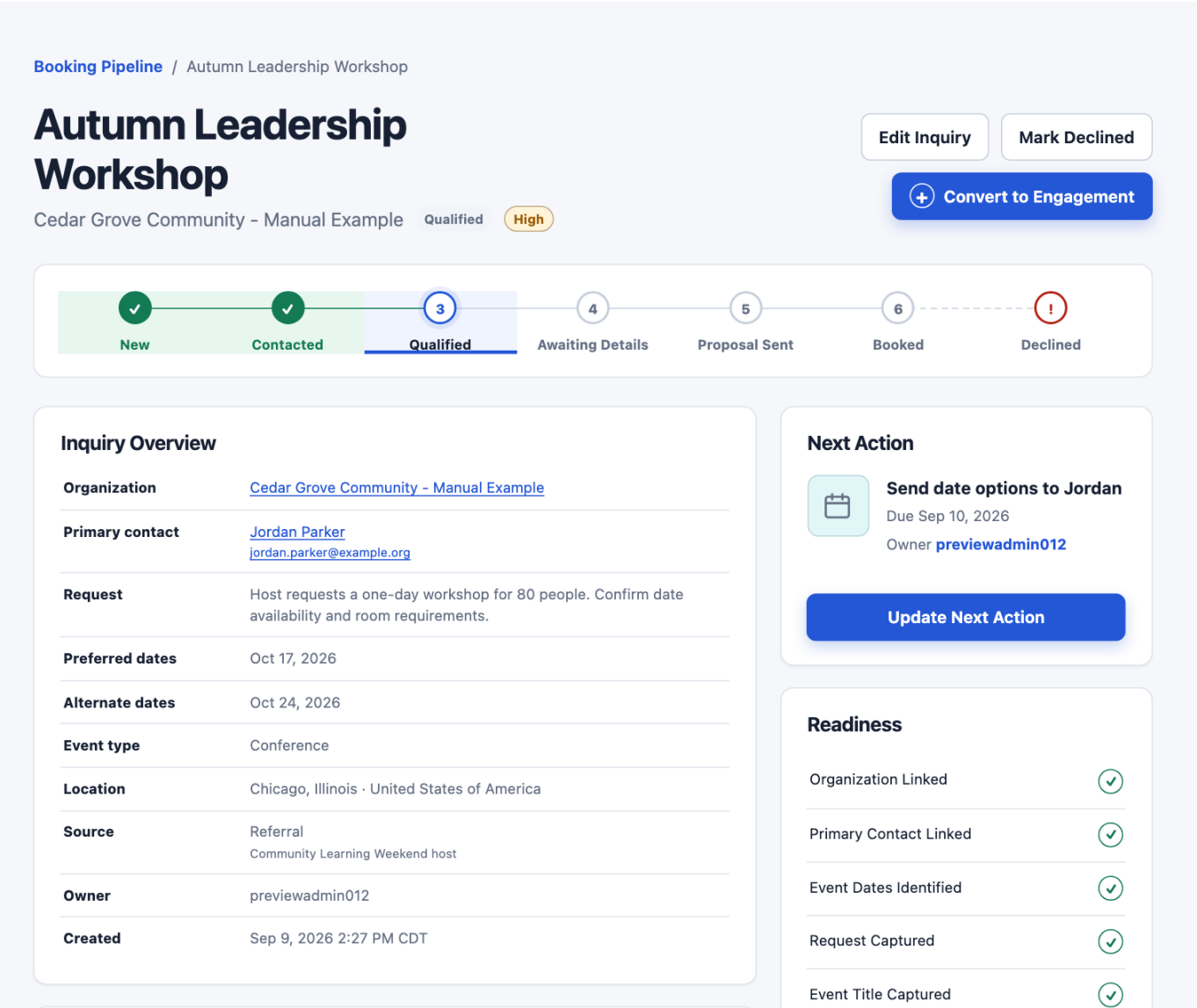
Figure 06. Read and filter the booking board. Actual preview screen; fictional records.

- 01 Search requests and narrow by owner, stage, or next action.
- 02 Open a card to review the request and its next step.

- 03 The default view includes active inquiries and bookings from this month. Other stages remain available in the filter.
- Related chapters: [Engagements \(p. 19\)](#) / [Work queue \(p. 54\)](#) / [Chron and email \(p. 61\)](#)

CHAPTER 04 / ILLUSTRATED WALKTHROUGH

Review an inquiry



CHAPTER 04 / ILLUSTRATED WALKTHROUGH

Capture the request and date options

Booking Pipeline / Inquiry / Edit

INQUIRY #1

Edit Inquiry

Keep the request and its next action current.

* Required fields

01

Request

Capture enough context to qualify the opportunity without inventing event details.

Inquiry Title

Autumn Leadership Workshop

Organization

Search organizations

Cedar Grove Community - Manual Example

Create organization

Primary Contact

Search compatible contacts

Parker, Jordan · Cedar Grove Community - Manual Example

Create contact

Choose an organization to narrow contacts. Standalone contacts can be selected when no organization is chosen. Creating a related record keeps this inquiry draft.

What Is Being Requested?

Host requests a one-day workshop for 80 people. Confirm date availability and room requirements.

Figure 08. Capture the request and date options. Actual preview screen; fictional records.

- 01

Enter the request title and summary. Choose the organization and primary contact when known.
- 02

Record preferred dates, alternate dates, event type, and location.
- 03

Choose an owner, priority, source, next action, and due date, then save.
- Related chapters:

Engagements (p. 19) / Work queue (p. 54) / Chron and email (p. 61)

CHAPTER 04 / ILLUSTRATED WALKTHROUGH

Give the inquiry a next action

Address Line 2

City

Chicago

State

Illinois

Zipcode

Country

 United States of America

03

Ownership & Next Action

Every active inquiry should make its next move visible.

Owner

previewadmin012 · Admin

Priority

High

Source

Referral

Source Detail

Community Learning Weekend host

Next Action

Send date options to Jordan

Due

09/10/2026

Cancel

Save Changes

© 2026 beneliath · DNR 1.11.11 · 2026-09-05 14:19:58 CDT (ec7dc08)

Tylin = appointment, appointed time

(n^2 - 1) / n, ..., (n^2 - 1) / n

Figure 09. Give the inquiry a next action. Actual preview screen; fictional records.

- 01 Keep the next action concrete, such as sending date options or requesting room details.
- 02 Set a due date so the pipeline and Dashboard can highlight it.

- 03** Use Source Detail for a referral, mailbox, form, or campaign reference.

Related chapters: [Engagements \(p. 19\)](#) / [Work queue \(p. 54\)](#) / [Chron and email \(p. 61\)](#)

CHAPTER 04 / ILLUSTRATED WALKTHROUGH

Convert a qualified inquiry

Booking Pipeline / Autumn Leadership Workshop / Convert

Convert Inquiry to Engagement

Review the imported details and resolve scheduling warnings before creating the engagement.

1Details

2Availability

3Create

Engagement Details

Organization

Cedar Grove Community - Manual Example

Event Title

Autumn Leadership Workshop

Start

Oct 17, 2026

End

Oct 17, 2026

Event Type

Conference

Caller / Owner

previewadmin012

JP

Jordan Parker

Primary host

Reconcile the inquiry next action

Send date options to Jordan · Due Sep 10, 2026

☐ Carry forward as an engagement task

☐ Resolve at booking

Availability Check

✓

No direct date overlap detected

No active engagement or qualified inquiry overlaps this date range.

Figure 10. Convert a qualified inquiry. Actual preview screen; fictional records.

01 Review the proposed engagement and any schedule conflicts.

02 Choose the open inquiry tasks to move; clearing every box moves none.

03 Carry an unresolved next action forward as a task, or resolve it with a reason.

04 Acknowledge any availability warnings before confirming the conversion.

Related chapters: [Engagements \(p. 19\)](#) / [Work queue \(p. 54\)](#) / [Chron and email \(p. 61\)](#)

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CHAPTER 04 / ILLUSTRATED WALKTHROUGH

Write inquiry correspondence

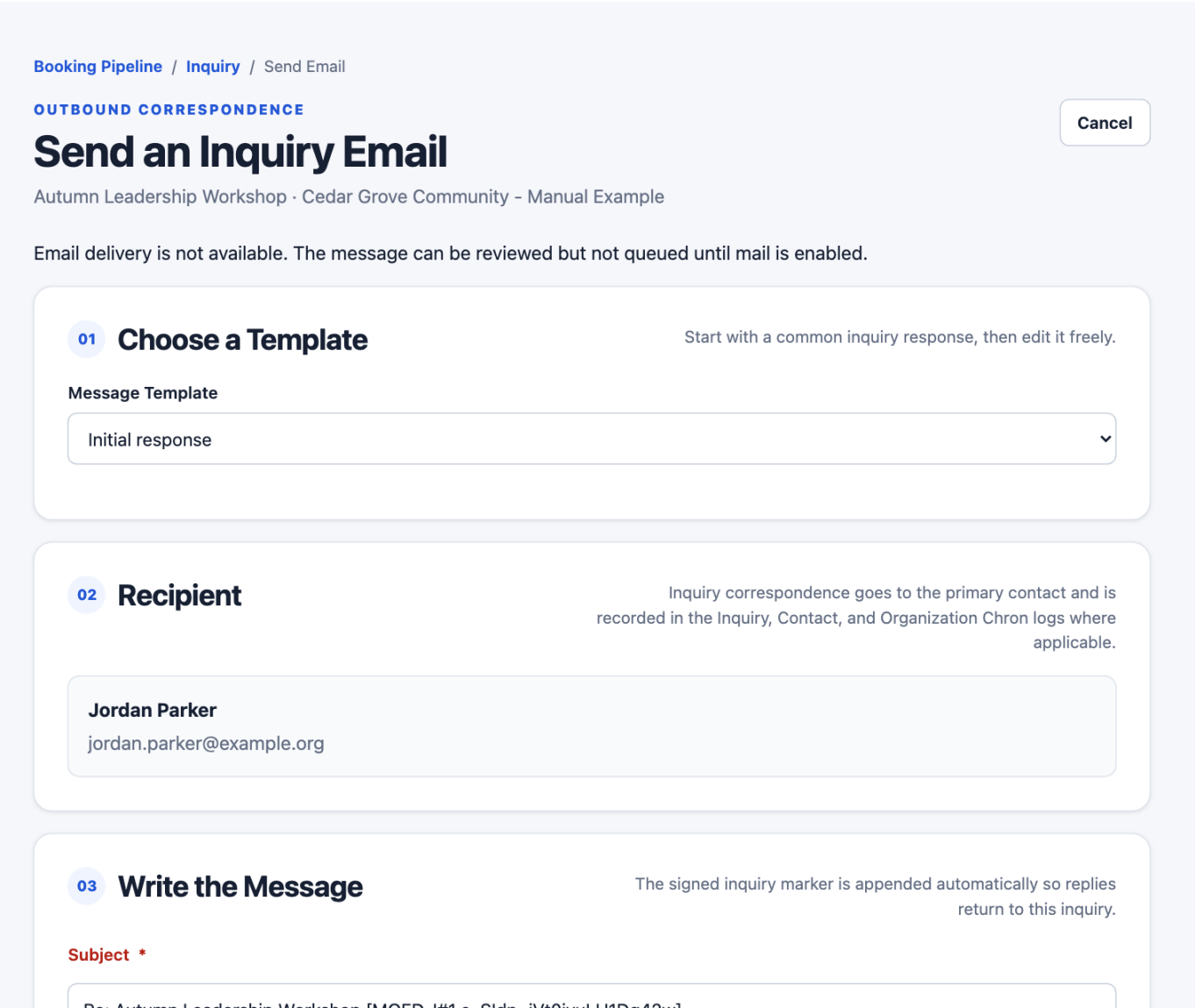


Figure 11. Write inquiry correspondence. Actual preview screen; fictional records.

- 01** Choose Initial response, Request details, Date options, or Proposal follow-up.

02 Verify the primary contact, then edit the subject and plain-text message.
- 03** Queue the message when ready; its signed inquiry marker keeps replies connected.

Screenshot context: The screenshot preview has mail delivery disabled. The form is shown without sending a message.

Related chapters: [Engagements \(p. 19\)](#) / [Work queue \(p. 54\)](#) / [Chron and email \(p. 61\)](#)

CHAPTER 05

Engagements

An engagement is the operational record for one event: the schedule, assigned people, presentations, logistics, planning state, communication history, work, and final receipts.

Related chapters: [People and organizations](#) (p. 42) / [Work queue](#) (p. 54) / [Chron and email](#) (p. 61) / [Map and calendar](#) (p. 68)

Find and Read an Engagement

Search Broadly

The Engagements search checks event titles and descriptions, organizations, contacts, caller names, active Chron text, and Chron authors. Unquoted words are alternatives; words inside one quoted group must all match. Terms shorter than three characters are ignored.

Example: conference "travel confirmed"

Filter Precisely

Switch between current and archived records, filter by lifecycle, then sort by organization, date, confirmation, or lifecycle. The summary cards show current lifecycle totals.

The table shows event dates, lifecycle, confirmation, and whether financial closeout is open, closed, or not applicable. Select the title or View icon for the full record.

Create or Edit an Engagement

01 Choose the organization. The event must belong to one active organization. If it is new, create the organization first.

02 Name and schedule the event. Enter a required event title, an optional description, a valid start and end date, and an event type. Use Other when the preset types do not fit.

03 Assign event contacts. Select multiple existing contacts and give each any applicable event roles: Primary host, On-site contact, Billing, Travel, or Materials. Search the directory to add another person, or use Add new contact repeatedly. New contacts and organization affiliations are saved with the event.

04 Add presentations. Choose a saved speaker from the dropdown; add missing speakers in [Relationships > Speakers](#). Presentation details can be filled in later.

Dates must fall within the event range, and duration is entered in minutes. Record actual attendance afterward, including zero when applicable. At least one complete presentation is required before the engagement can be Confirmed.

05 Capture logistics. Record book-table and brochure permissions, travel coverage, planned compensation, travel/lodging estimates, lodging type, and the physical event location.

06 Set planning states and ownership. Choose lifecycle and confirmation, then select the Caller when that person should own the initial standard checklist. If no Caller is selected, the checklist belongs to the user creating the engagement. Add an initial Chron entry when there is context worth preserving.

Lifecycle

Active	The event is moving forward.
Postponed	The original is paused and may link to its replacement.
Canceled	A reason is required; open event tasks are canceled automatically.
Completed	The event has finished. Finalizing its financial report also sets this state.

Confirmation

Work in Progress	Details are still being assembled.
Under Review	The plan is ready for checking or decision.
Confirmed	The commitment is agreed and at least one complete presentation exists.

Lifecycle and confirmation are independent. A postponed event can retain its former confirmation state, while a completed event may reflect the confirmation state it reached during planning.

Presentation Files, QR Codes, and History

PDF Speaker Notes

Attach one PDF up to 100 MB to each presentation with **Choose PDF**, then save. Select **View PDF Speaker Notes** on the event detail page to open it in a new tab. Use **Replace PDF** or **Remove current PDF** on the edit page and save to update it.

Use this file for notes you intend to share: anyone with its QR code can open the PDF without signing in.

Speaker QR Codes

Saving creates separate codes for each presentation's available speaker resources: website, bio, donations, connection, blog, books, custom links, and notes. Select a displayed code to copy it. Download PNG or SVG images when you prefer to save a file for your slides or handouts.

The Speaker Notes code appears after its PDF is saved. Replacing the PDF keeps the same code. Removing the file hides the code; adding notes again reuses the code and its statistics.

Select **View QR Codes PDF** to view, save, or print a single landscape sheet of that presentation's labeled codes, schedule, and venue. Missing schedule details show **To be confirmed**. Disabled codes are labeled; notes without a PDF and previous speakers' codes are left out. Viewing the sheet does not add visits.

Each saved presentation has a nearby **Save Changes** button. It saves the engagement form, including other changes on the page. Presentations can be archived and restored separately from the event. When restoring, update any date that falls outside the event's current date range. Administrators can permanently delete presentations after confirming their access.

When postponing or canceling an event, link it to a replacement from the same organization. MOED displays both "rescheduled as" and "rescheduled from" references and prevents circular links.

Manage Email Templates

Open [Email Templates](#) under **Work** in the sidebar, or select **Manage Email Templates** in the engagement email composer. This library supplies engagement email in the browser and Mattermost. Inquiry emails use their own starting templates.

01 Add or edit a template. Select **+ New Email Template** , or open an active template by its name or Edit action.

Enter the template name, subject, and plain-text message. The original Booking confirmation, Travel and lodging request, Final-detail reconfirmation, Presentation schedule, and Post-event thank-you templates can all be edited.

02 Personalize the starting text. Expand **Personalize with event fields** . Click where the field belongs in the subject or message, then choose a field button. Event name, organization, dates, location, and speaker names are available; Presentation schedule belongs in the message only. For example, [Welcome to {{event_name}}](#) uses that engagement's event name when composing. The routing marker is added automatically.

03 Choose suggestions and order. Select the event contact roles to suggest when this template is chosen. Leave them clear to choose contacts when composing. Lower **Display order** numbers appear first. Speakers are always a separate sender choice; inserting Speaker names into the message does not add recipients.

04 Save and review. Select **Create Template** or **Save Changes** . Choose the saved template in an engagement email to review the filled subject, message, and recipients before queuing. When you entered the library from an engagement, **Return to Email** takes you back to its composer.

Archive and Restore

Search the library by name or subject. **Archive** removes a template from new emails in both composers. Open **Archived** to view it or select **Restore** to make it available again. Restore an archived template before editing it.

Delete and Access

Editors and administrators can add, edit, archive, and restore templates. Reviewers can view them. Only administrators can permanently **Delete** an archived template, after confirming their access. Deletion cannot be undone; sent messages, their template names, and Chron history remain available.

Changes apply to future messages. A draft already open in a composer keeps its reviewed text. If its template is archived or deleted before sending, review the preserved draft as **Custom message** or choose another template. If another session changed a template while you were editing, keep a copy of your draft, reload, and review the latest version before saving. **Custom message** is always available, even when every saved template is archived.

Review QR Code Activity

- In the **Presentations** tab, select **Statistics** beneath a code to see its visits. Open a speaker's record and choose **Statistics** for activity across that speaker's presentations.
- Choose a date range to focus on a recent event or a longer period. In the speaker view, use **Link Type** to focus on notes, donations, or **Custom Links**, then select **Apply Filters**.
- Use the charts to see when and where links were opened. Hover or tap for details, or expand the data tables for exact counts. These are visits to links, not an attendance count. Speaker Notes records the link visit before opening the PDF, including cached delivery; opening a copied direct PDF address does not add a visit.

To change where an existing code leads, editors and administrators can open that code's **Statistics**, edit **Destination**, and select **Save Link**. The printed code continues to work. Clear **Link enabled** and save to stop access through that code. For Speaker Notes, replace the PDF on the presentation instead of editing a destination.

Editing a speaker's profile does not change destinations already shared with audiences. Previous speakers' links and notes remain separate when you change a presentation's speaker.

Send and Track Event Email

Editors and administrators can send email from an active engagement. Use **Send Email** near the page heading or **Send Message** in the Correspondence section. Every message remains easy to find there afterward.

- 01 Choose a starting template.** Choose an active template from the shared library, or **Custom message** to write from scratch. Event fields are filled from this engagement. Template text is only a starting point; edit the subject and body before sending. Use [Manage Email Templates](#) to learn how to maintain the library.
- 02 Select recipients.** Use the Primary host, On-site contact, Billing, Travel, Materials, or Speaker shortcuts, or select individual recipient cards. Speakers assigned to active presentations are optional and unchecked by default. Choose the Speaker shortcut or check a speaker card to include them. You may send to speakers alone. Changing templates keeps your speaker selection. Only assigned contacts and speakers with a valid email address can receive the message. Every unique address receives a separate email, so recipients never

see one another. A speaker and contact sharing one address receive only one copy.

- 03 Review the routing marker.** The exact engagement marker is included in every template subject and is added automatically if removed. When inbound mail is enabled, replies return through the shared mailbox and the marker helps place them in the right engagement Chron.
- 04 Optionally append the event brief.** The share-safe brief includes the public event schedule, venue, description, and presentations. It deliberately excludes Chron, internal notes, compensation, giving, and financial-closeout information. Preview it before queuing.
- 05 Send and check the result.** Select **Queue Email**, then use the Correspondence list to see whether each recipient is waiting, sent, or failed. If a recipient remains failed, open the message and select **Retry Failed Deliveries**.

One Place to Check

Open the outbound message to review what was sent, who it was sent to, and the result for each address.

Automatic Chron History

The message is added to the Engagement, Organization, and selected Contact Chron logs. Select **View Outbound Message** from a Chron entry to return to its delivery record.

View, Share, and Close the Event

The event detail page shows its unique email routing marker. The built-in outbound composer inserts it automatically. When using another mail application, select the copy icon immediately beside the marker and keep it in the email subject or plain-text body when that message should route to the engagement's Chron log.

In the **Presentations** tab, open **Export Presentation** at the bottom of a presentation card. Briefs include that presentation's speaker, schedule, duration, and attendance, with its event and venue context.

Copy Text

Copies a plain-text presentation brief for messages or notes.

Copy MD

Copies a Markdown-formatted presentation brief for systems that support structured text.

View PDF

Opens a branded brief for the selected presentation in your browser for viewing, saving, or printing.

View QR Codes PDF

Opens a themed landscape sheet containing only that presentation's labeled QR codes.

Editors and administrators can use **Financial Closeout** to save a receipt draft while work is still pending. Leave unknown amounts blank; enter zero only when nothing was received. Drafts are excluded from finalized financial history. The original planning estimates remain unchanged. Every event task due on or before the last active presentation must first be marked Completed; canceled tasks still block until truthfully resolved, while later and undated tasks do not hold closeout. Confirm that the amounts are final to complete the engagement; later corrections retain the original close date and record the update.

Archive and Delete Carefully

Archive removes the engagement from current views but keeps its history available for restoration. Permanent deletion removes the event and all event-specific information, including presentations, Chron, tasks, contact role assignments, and its financial report. Prefer archive unless removal is explicitly required.

Presentation reference and file history

A complete presentation needs its topic, date, time, selected speaker, duration, and expected attendance. Its date must be inside the engagement range before the event can be confirmed.

PDF Speaker Notes displays upload metadata, including who uploaded the current file and when, where available. Old files without stored uploader metadata can show an unavailable uploader.

Speaker Notes QR links open the PDF directly without signing in. Replacement keeps the code; removal makes the notes unavailable. A visit through the notes short link counts before the PDF opens, including cached delivery. Opening a copied direct PDF URL does not count. Internal QR sheets do not add visits.

Archive and restore presentations separately from the engagement. A restored presentation must fit the current dates. Changing its speaker changes the current resource

set; previous speaker codes remain part of historical statistics.

Readiness, closeout, and lifecycle effects

Use the readiness panel to find missing presentations, contact details, or planning information. Resolve the source record and revisit the panel.

Canceling an engagement requires a reason and cancels its open event tasks. A task canceled for convenience still fails a required Completed prerequisite for financial closeout.

Changing event dates updates generated active task dates unless a date was manually overridden. Postponed/canceled originals can link to a replacement from the same organization; circular reschedule links are rejected.

Reset presentation statistics

Administrators can choose Reset Presentation Statistics from a presentation card. Fresh administrator confirmation is required before Reset Statistics to Zero.

The reset permanently removes every recorded link visit for that presentation across all dates, including disabled and previous-speaker links. It preserves QR codes, destinations, uploaded notes, and presentation details. Other presentations are unaffected, and future visits start counting from zero.

Restore archived presentations

Open the engagement editor and choose Restore Archived Presentations. Select the desired presentations, correct their dates or times when needed, and choose Restore Selected.

Restored presentation dates must fit the event's current range. The active engagement and selected speaker must be valid. Administrators can delete an archived presentation permanently from the restore view after fresh confirmation.

CHAPTER 05 / ILLUSTRATED WALKTHROUGH

Find an engagement

Engagements

Manage event lifecycle, confirmation, schedules, and speaking commitments.

Active2

Postponed0

Canceled0

Completed0

title, organization, contact, chron log text, "and"/or user

Current

Archived (0)

Lifecycle: AllActivePostponedCanceledCompleted

Sort: Organization ↑Date ↑Confirmation ↑Lifecycle ↑

EVENT TITLE	EVENT DATE(S)	LIFECYCLE	CONFIRMATION	CLOSEOUT	ACTIONS
Community Learning Weekend	Sep 12–13, 2026	Active	under review	Open	   
Custom Links Preview	Oct 1–2, 2026	Active	under review	Open	   

© 2026 beneliath · DNR 1.11.11 · 2026-09-05 14:19:58 CDT (ec7dc08)

᠓ᠶᠢᠨ = appointment, appointed time

$$\begin{pmatrix} 11 & -1 & 1 & -1 & 11 \\ 6 & 6 & & & \end{pmatrix} \cdot \begin{pmatrix} 1 & -1 & 1 & -1 & 1 \\ -1 & 1 & -1 & 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 & -1 & 1 & -1 & 1 \\ -1 & 1 & -1 & 1 & -1 \end{pmatrix}$$

Figure 12. Find an engagement. Actual preview screen; fictional records.

- 01 Search the complete list, then use lifecycle and current/archive filters.

02 Sort by organization, date, confirmation, or lifecycle.
- 03 Open the event title or View action. Financial closeout is a separate status from planning confirmation.

Related chapters: [People and organizations](#) (p. 42) / [Work queue](#) (p. 54) / [Chron and email](#) (p. 61) / [Map and calendar](#) (p. 68)

CHAPTER 05 / ILLUSTRATED WALKTHROUGH

Use the engagement workspace

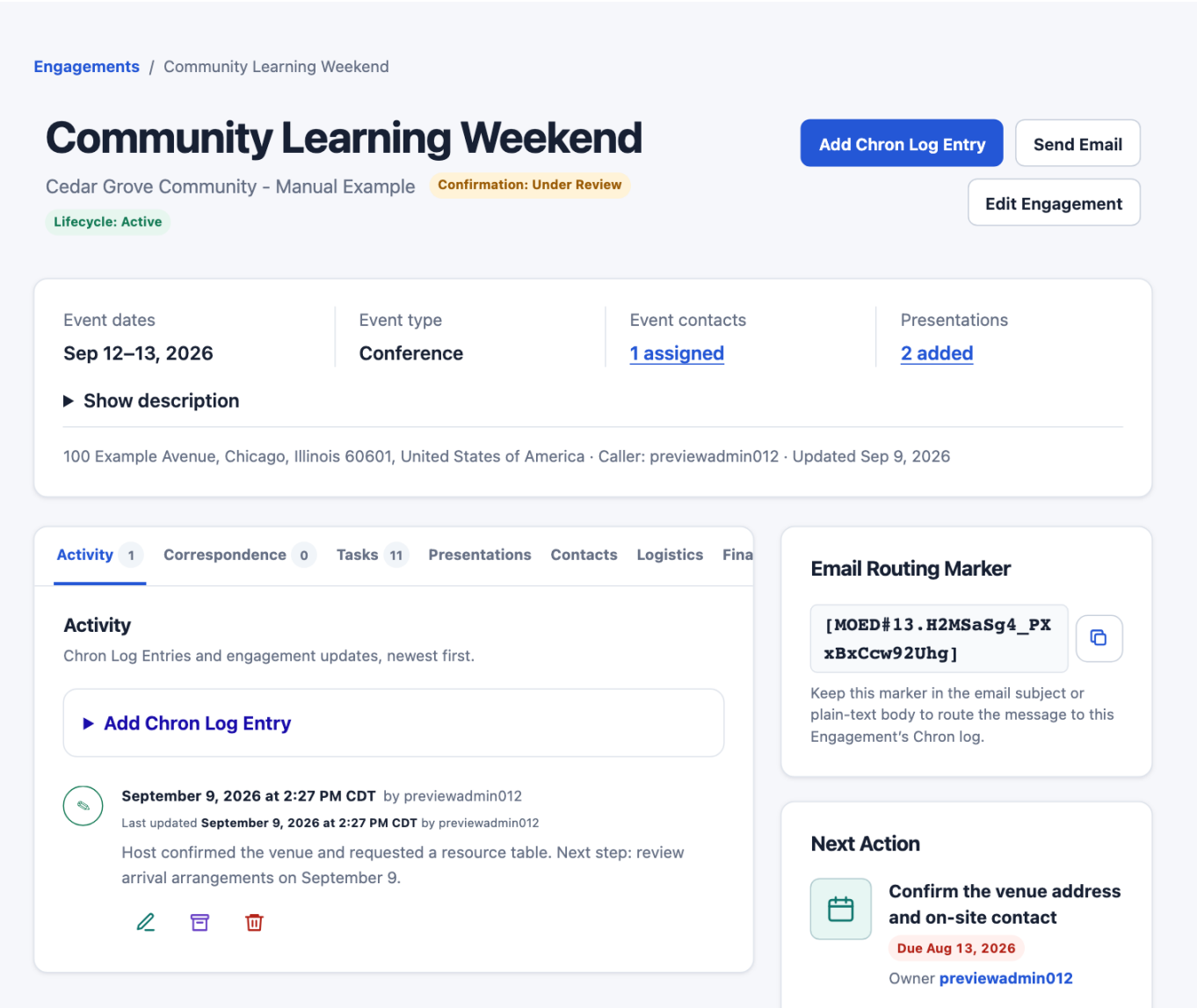


Figure 13. Use the engagement workspace. Actual preview screen; fictional records.

- 01 Read the event summary and right-side readiness and next-action information.

02 Switch among Activity, Correspondence, Tasks, Presentations, Contacts, Logistics, and Financials.
- 03 Use the nearby routing-marker copy control when continuing a conversation in another email application.

Related chapters: [People and organizations](#) (p. 42) / [Work queue](#) (p. 54) / [Chron and email](#) (p. 61) / [Map and calendar](#) (p. 68)

CHAPTER 05 / ILLUSTRATED WALKTHROUGH

Edit event details and schedule

Engagements / Edit Engagement

Edit Engagement

Update event details, schedule, presentations, and logistics.

* Required fields

Event Details & Schedule

Organization

Cedar Grove Community - Manual Example

Start *

09/12/2026

End *

09/13/2026

Event Title

Community Learning Weekend

Event Type

conference

Event Description

Two practical sessions with discussion, a resource table, and time for host follow-up. Fictional manual example.

Figure 14. Edit event details and schedule. Actual preview screen; fictional records.

01

Choose an active organization and enter the event title and description.

02

Set valid start and end dates. Choose Other and describe the type when presets do not fit.

03

Assign a Caller for the initial checklist; later Caller changes do not transfer existing tasks.

Related chapters:

People and organizations (p. 42) / Work queue (p. 54) / Chron and email (p. 61) / Map and calendar (p. 68)

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CHAPTER 05 / ILLUSTRATED WALKTHROUGH

Assign event-specific contact roles

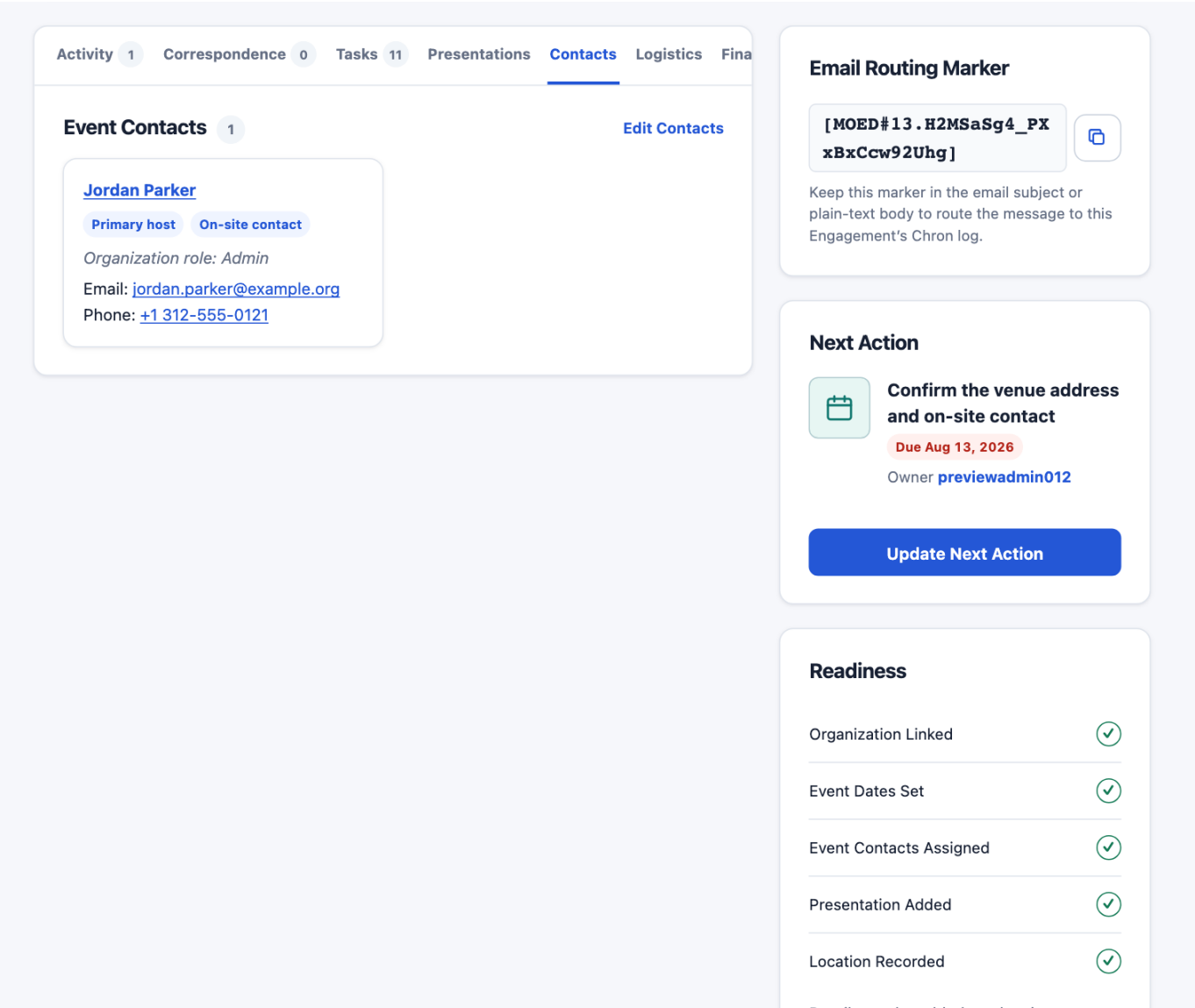


Figure 15. Assign event-specific contact roles. Actual preview screen; fictional records.

- 01 Open Contacts to see the people assigned to this event and their roles.
- 02 In Edit Engagement, add existing contacts or create new people. Assign Primary host, On-site contact, Billing, Travel, and Materials as applicable.

- 03 These roles are separate from a contact's organization affiliations.

Related chapters: [People and organizations](#) (p. 42) / [Work queue](#) (p. 54) / [Chron and email](#) (p. 61) / [Map and calendar](#) (p. 68)

CHAPTER 05 / ILLUSTRATED WALKTHROUGH

Presentation details and audience resources

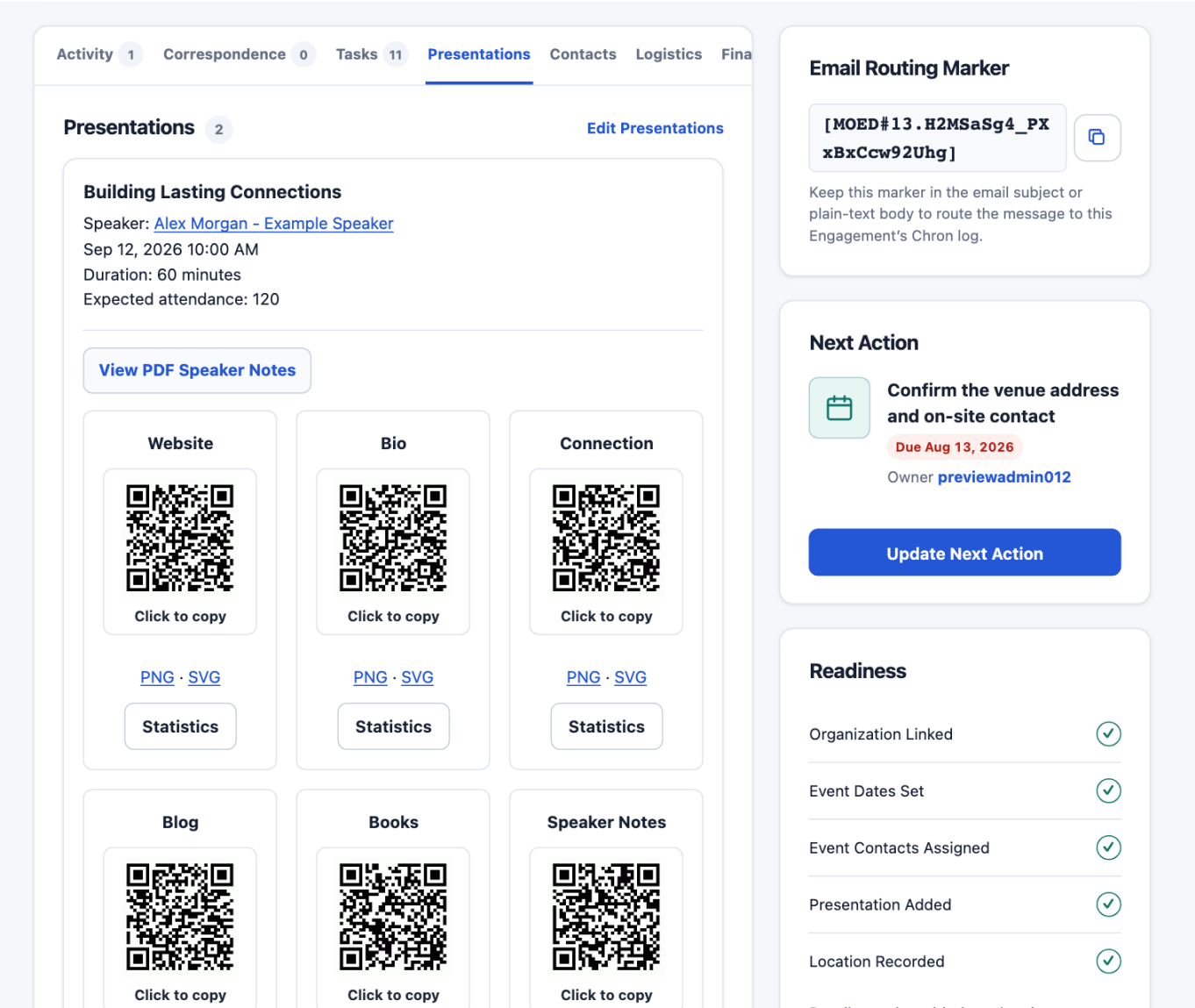


Figure 16. Presentation details and audience resources. Actual preview screen; fictional records.

- 01** Open Presentations and review each speaker, title, schedule, duration, and attendance.

02 Use Copy Text, Copy MD, or View PDF for a presentation brief.
- 03** View saved speaker notes, copy a QR image, download PNG/SVG, or open the QR Codes PDF.

Related chapters: [People and organizations](#) (p. 42) / [Work queue](#) (p. 54) / [Chron and email](#) (p. 61) / [Map and calendar](#) (p. 68)

CHAPTER 05 / ILLUSTRATED WALKTHROUGH

Edit a presentation and speaker notes

Topic/Title

Building Lasting Connections

Date

09/12/2026

Time

10:00

☒ AM ☐ PM

Speaker

Alex Morgan - Example Speaker — alex.morgan@exam

Duration (minutes)

60

Expected Attendance

120

Actual Attendance

Presentation Files & QR Codes

PDF Speaker Notes may be up to 100 MB. Keep each save under 120 MB total. QR codes are generated when you save.

PDF Speaker Notes

Anyone with the Speaker Notes QR code can open this PDF without signing in.

[View community-learning-speaker-notes.pdf](#) Uploaded Sep 9, 2026 2:27 PM CDT By previewadmin012

0.0 MB ☐ Remove current PDF

Replace PDF

No replacement selected

View QR Codes PDF

Website

Bio

Connection

Figure 17. Edit a presentation and speaker notes. Actual preview screen; fictional records.

- 01

Choose a saved speaker and enter topic, date, time, duration, and expected attendance.
- 02

Choose PDF or Replace PDF to attach public speaker notes up to 100 MB. Save to upload it.
- 03

Record actual attendance after the event. Zero is valid when nobody attended.
- 04

The nearby Save Changes button saves the whole engagement form.
- Related chapters:

[People and organizations \(p. 42\)](#) / [Work queue \(p. 54\)](#) / [Chron and email \(p. 61\)](#) / [Map and calendar \(p. 68\)](#)
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CHAPTER 05 / ILLUSTRATED WALKTHROUGH

Review event logistics

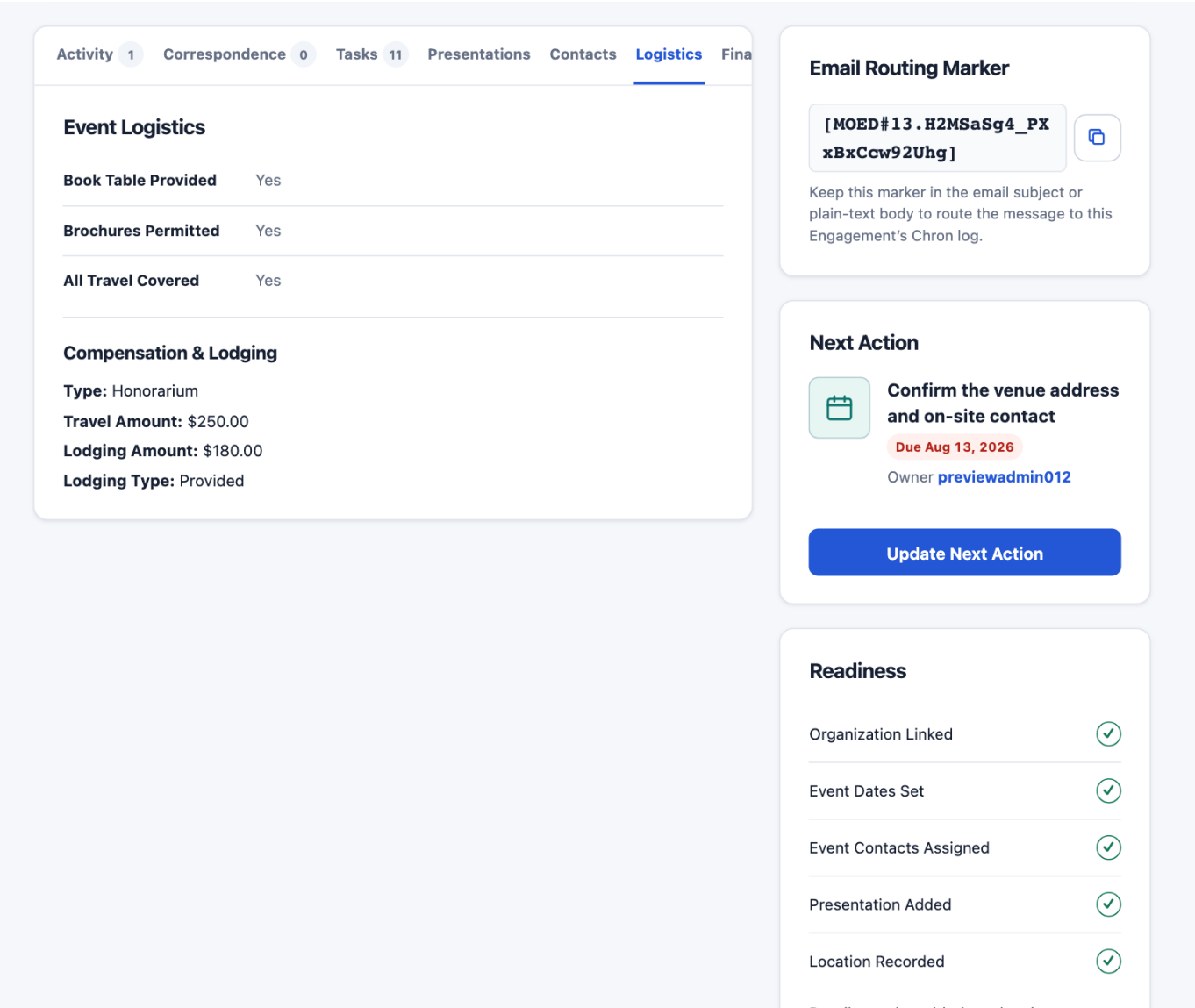


Figure 18. Review event logistics. Actual preview screen; fictional records.

- 01** Open Logistics to review the event address, materials, travel, compensation, and lodging.

02 Planning estimates describe arrangements; actual received amounts belong in Financial Closeout.
- 03** Keep the physical event location accurate for the map and presentation exports.

Related chapters: [People and organizations](#) (p. 42) / [Work queue](#) (p. 54) / [Chron and email](#) (p. 61) / [Map and calendar](#) (p. 68)

CHAPTER 05 / ILLUSTRATED WALKTHROUGH

Lifecycle and rescheduling

Lifecycle state

Active

Confirmation status

Under Review

Caller

previewadmin012

Chron Log

Entries are shown newest first. Archived entries are removed from this page.

New Chron Log Entry

Add scheduling notes, important information, or reminders.

Add Chron Log Entry

September 9, 2026 at 2:27 PM CDT by previewadmin012

Last updated September 9, 2026 at 2:27 PM CDT by previewadmin012

Host confirmed the venue and requested a resource table. Next step: review arrival arrangements on September 9.

Figure 19. Lifecycle and rescheduling. Actual preview screen; fictional records.

- 01 Use Active, Postponed, Canceled, or Completed to describe the operational state.

02 Keep Work in Progress, Under Review, or Confirmed as the separate planning confidence.
- 03 Provide a cancellation reason. When rescheduling, link the replacement event from the same organization.

Related chapters: [People and organizations](#) (p. 42) / [Work queue](#) (p. 54) / [Chron and email](#) (p. 61) / [Map and calendar](#) (p. 68)

CHAPTER 05 / ILLUSTRATED WALKTHROUGH

Review presentation QR activity

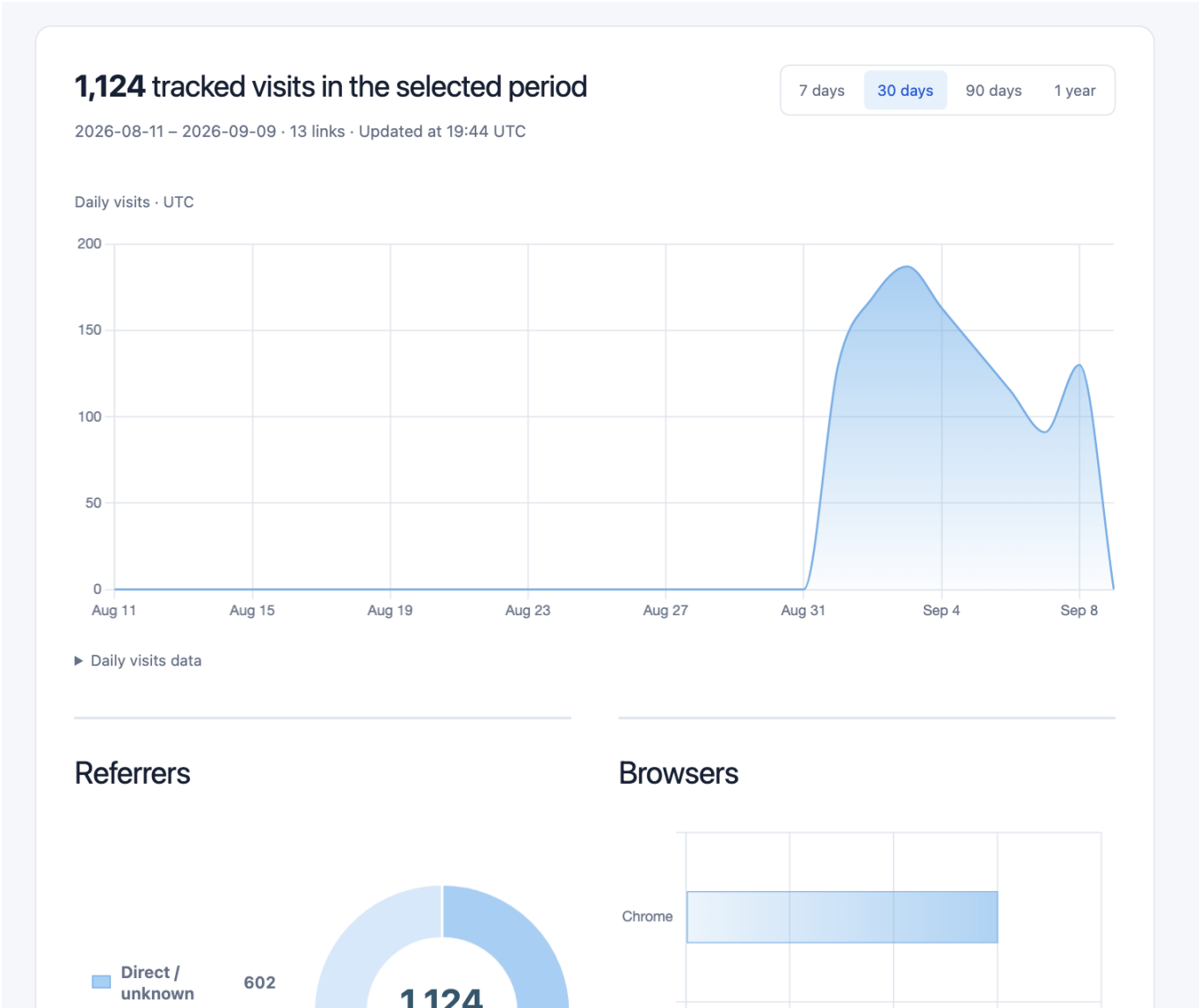


Figure 20. Review presentation QR activity. Actual preview screen; fictional records.

- 01** Select a date range and review the visits and distribution charts.

02 A presentation code measures its own activity; the speaker view combines that speaker's presentations.
- 03** Open a single link to change its destination or enable/disable it. Administrators reset all statistics for a presentation from that presentation's card.

Screenshot context: The chart uses fictional, seeded visit counts to illustrate the report.

Related chapters: [People and organizations \(p. 42\)](#) / [Work queue \(p. 54\)](#) / [Chron and email \(p. 61\)](#) / [Map and calendar \(p. 68\)](#)

CHAPTER 05 / ILLUSTRATED WALKTHROUGH

Review financial closeout

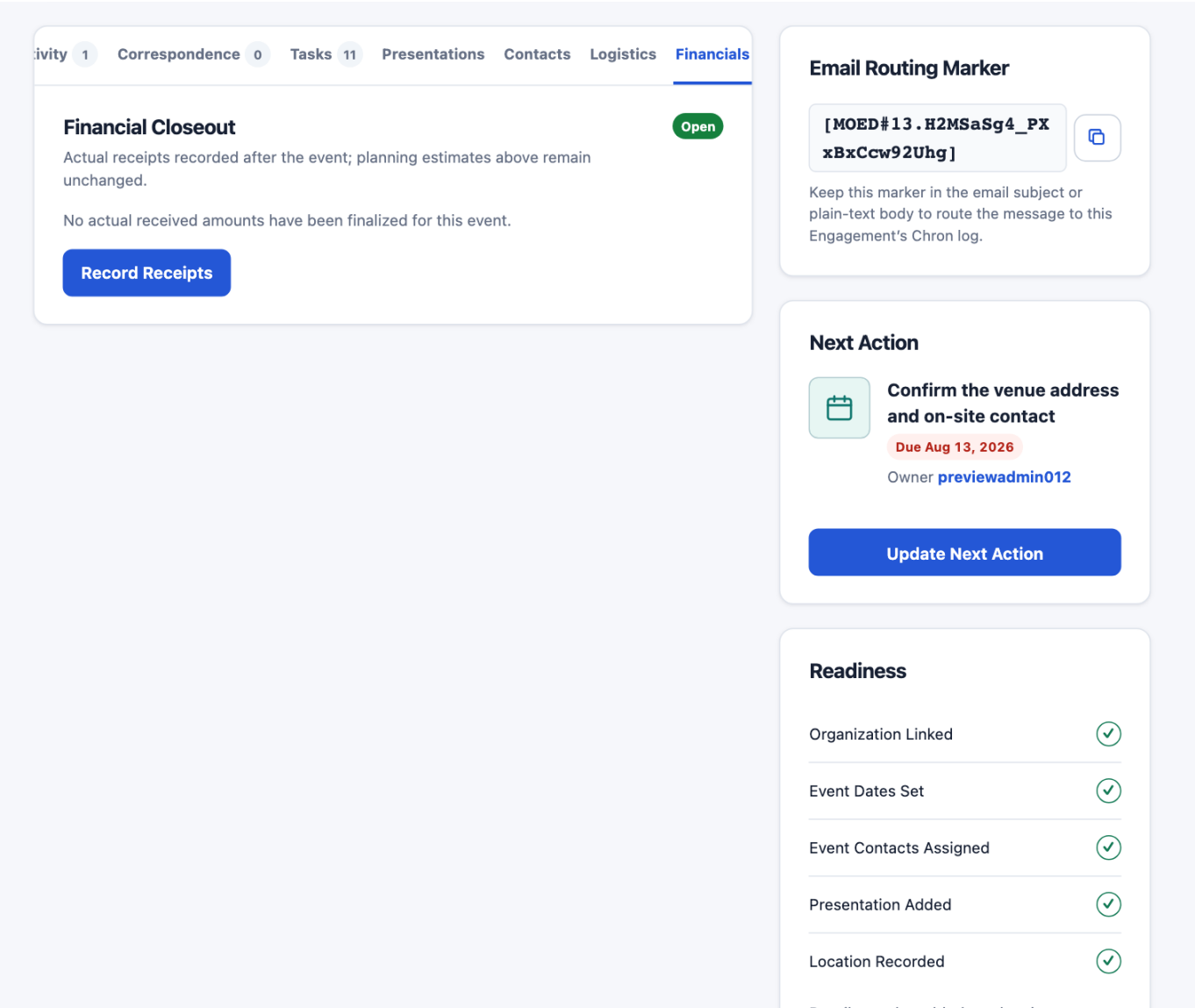


Figure 21. Review financial closeout. Actual preview screen; fictional records.

01 Open Financials to see whether a report is open, a draft, or final.

02 Use Financial Closeout to enter giving/income, lodging, and travel actually received.

03 A draft does not enter finalized organization financial history.

Related chapters: [People and organizations](#) (p. 42) / [Work queue](#) (p. 54) / [Chron and email](#) (p. 61) / [Map and calendar](#) (p. 68)

CHAPTER 05 / ILLUSTRATED WALKTHROUGH

Draft and finalize received amounts

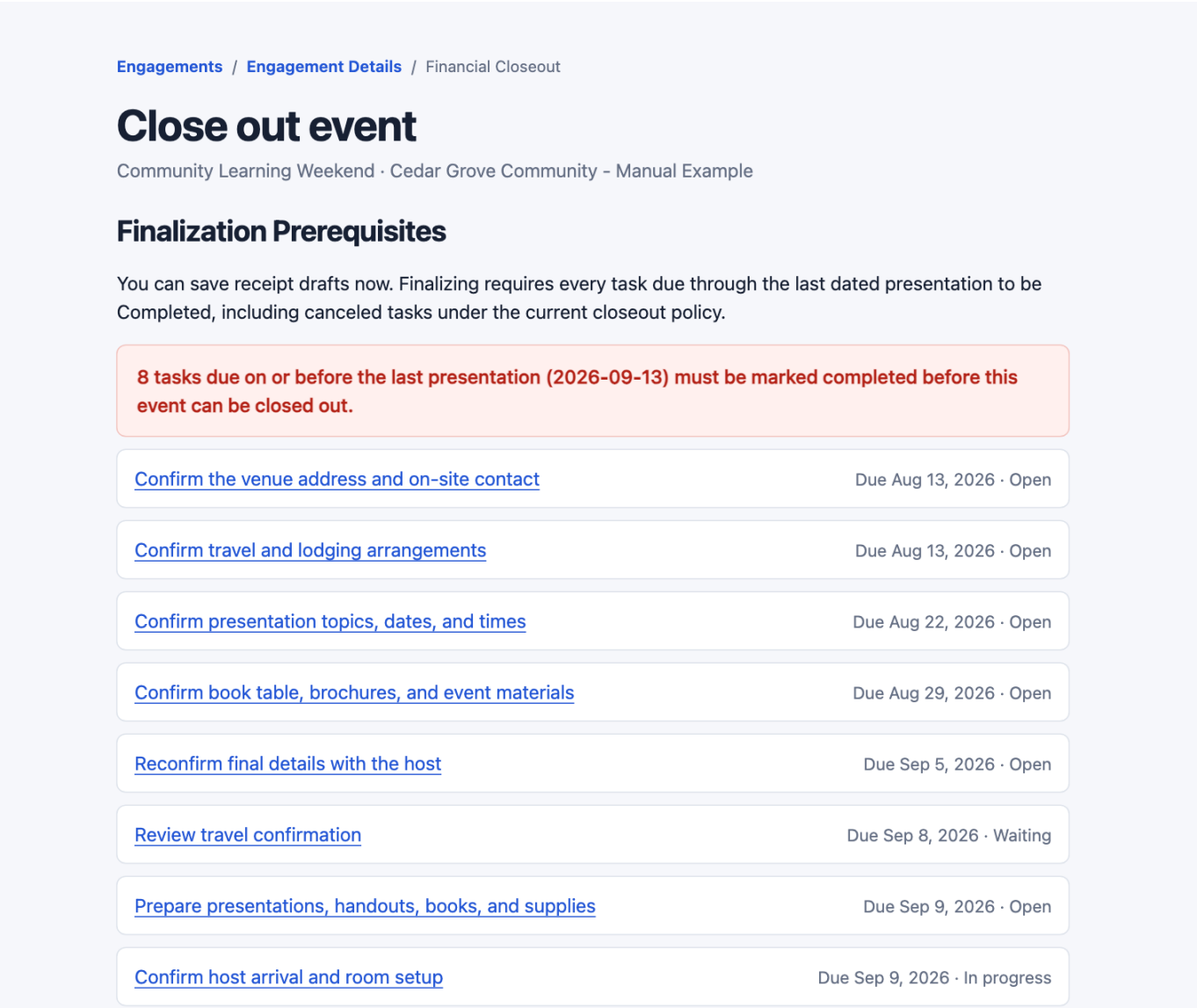


Figure 22. Draft and finalize received amounts. Actual preview screen; fictional records.

- 01

Leave unknown draft amounts blank; enter zero only when nothing was received.

02

Resolve every task due on or before the last active presentation by marking it Completed. Canceled tasks still hold closeout.
- 03

Check the final-amount confirmation, then Finalize and close event when prerequisites are satisfied.

04

For a final report, use correction to update amounts while retaining the original close date.
- Related chapters: [People and organizations](#) (p. 42) / [Work queue](#) (p. 54) / [Chron and email](#) (p. 61) / [Map and calendar](#) (p. 68)

CHAPTER 05 / ILLUSTRATED WALKTHROUGH

Manage the email template library

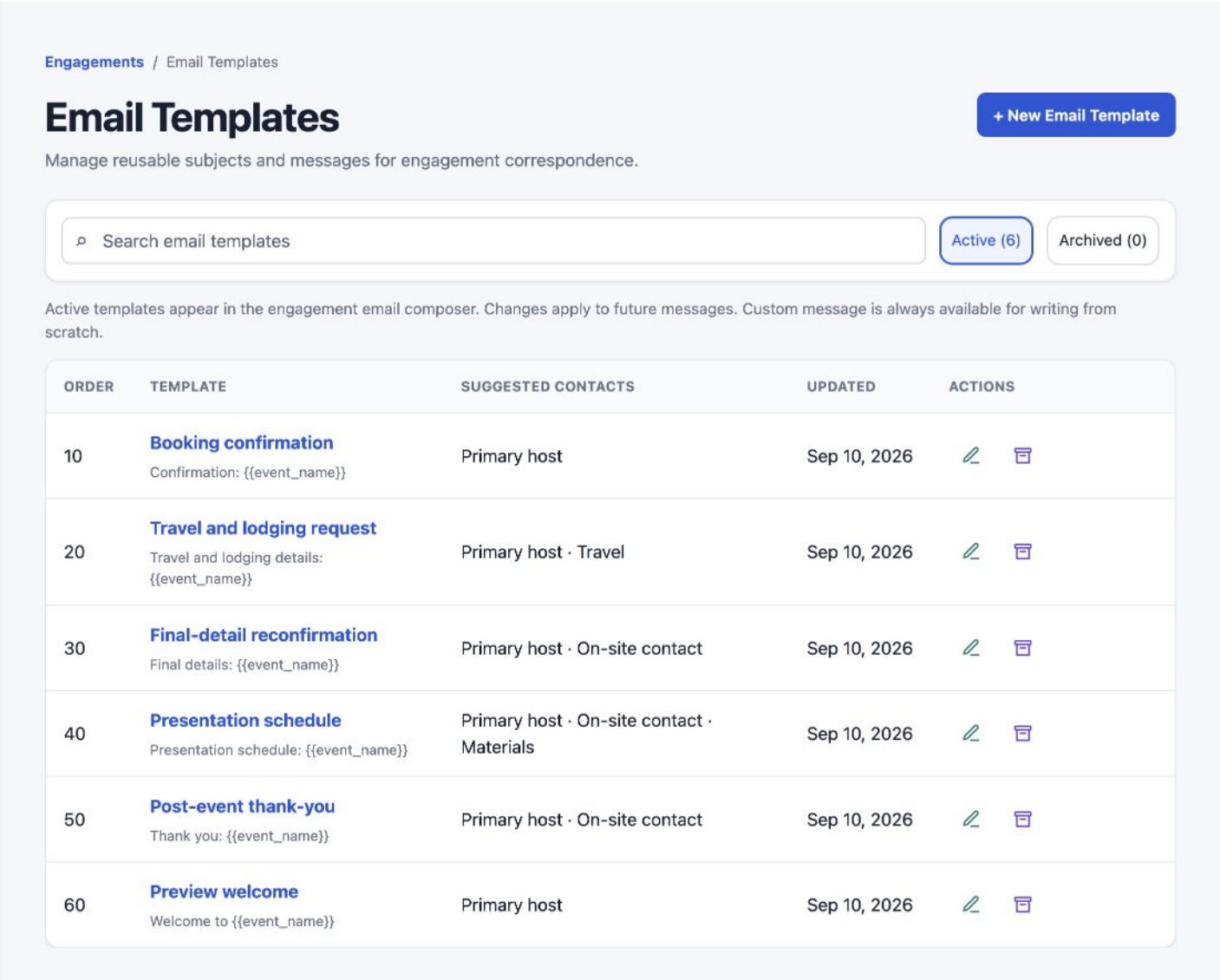


Figure 23. Manage the email template library. Actual preview screen; fictional records.

01 Open Email Templates under Work in the sidebar, or Manage Email Templates in an engagement email.

02 Search by name or subject. Select + New Email Template, a template name, or its Edit action.

03 Archive removes a template from new messages. Open Archived and choose Restore to use it again.

04 Only administrators can permanently delete an archived template after confirming access. Sent messages and history remain unchanged.

Screenshot context: Fictional records from the isolated template preview; no email was queued during capture.

Related chapters: [People and organizations \(p. 42\)](#) / [Work queue \(p. 54\)](#) / [Chron and email \(p. 61\)](#) / [Map and calendar \(p. 68\)](#)

CHAPTER 05 / ILLUSTRATED WALKTHROUGH

Add or edit reusable email text

Email Templates / Edit Email Template

Edit Email Template

Set the starting subject, message, and suggested contacts for engagement emails.

* Required fields

Template Details

Template name

Preview welcome

Subject

Welcome to {{event_name}}

The engagement routing marker is added automatically when composing the email.

Plain-text message

Hello,

Thank you for hosting {{event_name}} on {{event_dates}}.

Speakers: {{speaker_names}}

► Personalize with event fields

Figure 24. Add or edit reusable email text. Actual preview screen; fictional records.

- 01** Enter a clear template name, a subject, and the starting plain-text message.
- 02** Use event fields such as {{event_name}} in the text. The engagement routing marker is added automatically when composing.

- 03** Existing templates can be edited too. Save Changes updates the template for future selections, leaving sent messages intact.
- 04** If another session changed the template, preserve your edits, reload, and review the current version before saving.

Screenshot context: Fictional records from the isolated template preview; no email was queued during capture.

Related chapters: [People and organizations \(p. 42\)](#) / [Work queue \(p. 54\)](#) / [Chron and email \(p. 61\)](#) / [Map and calendar \(p. 68\)](#)

CHAPTER 05 / ILLUSTRATED WALKTHROUGH

Personalize fields and suggest recipients

▼ Personalize with event fields

These fields are filled from the engagement when you choose the template. Click in the subject or message, then choose a field to insert it.

Event name {{event_name}}

Organization name {{organization_name}}

Event date range {{event_dates}}

Event start date {{event_start_date}}

Event end date {{event_end_date}}

Event location {{event_location}}

Speaker names {{speaker_names}}

Presentation schedule (message only) {{presentation_schedule}}

Recipient Suggestions

Suggest event contacts when this template is selected. The sender can change recipients and include speakers before sending.

☒ Primary host ☐ On-site contact ☐ Billing ☐ Travel ☐ Materials

Display order

60

Lower numbers appear first in the template library and email composer.

Cancel

Save Changes

Figure 25. Personalize fields and suggest recipients. Actual preview screen; fictional records.

- 01

Expand Personalize with event fields. Place the cursor in the subject or message, then select a field button.
- 02

Presentation schedule is available in the message only. Speaker names inserts text; it does not select recipients.
- 03

Choose suggested event contact roles. Leave all clear to choose contacts when composing. Lower Display order numbers appear first.
- 04

Save the template. Open an engagement email, choose it, and review the filled text and recipients before queuing.
- Screenshot context:

Fictional records from the isolated template preview; no email was queued during capture.
- Related chapters:

[People and organizations \(p. 42\)](#) / [Work queue \(p. 54\)](#) / [Chron and email \(p. 61\)](#) / [Map and calendar \(p. 68\)](#)

CHAPTER 05 / ILLUSTRATED WALKTHROUGH

Copy or export a presentation brief

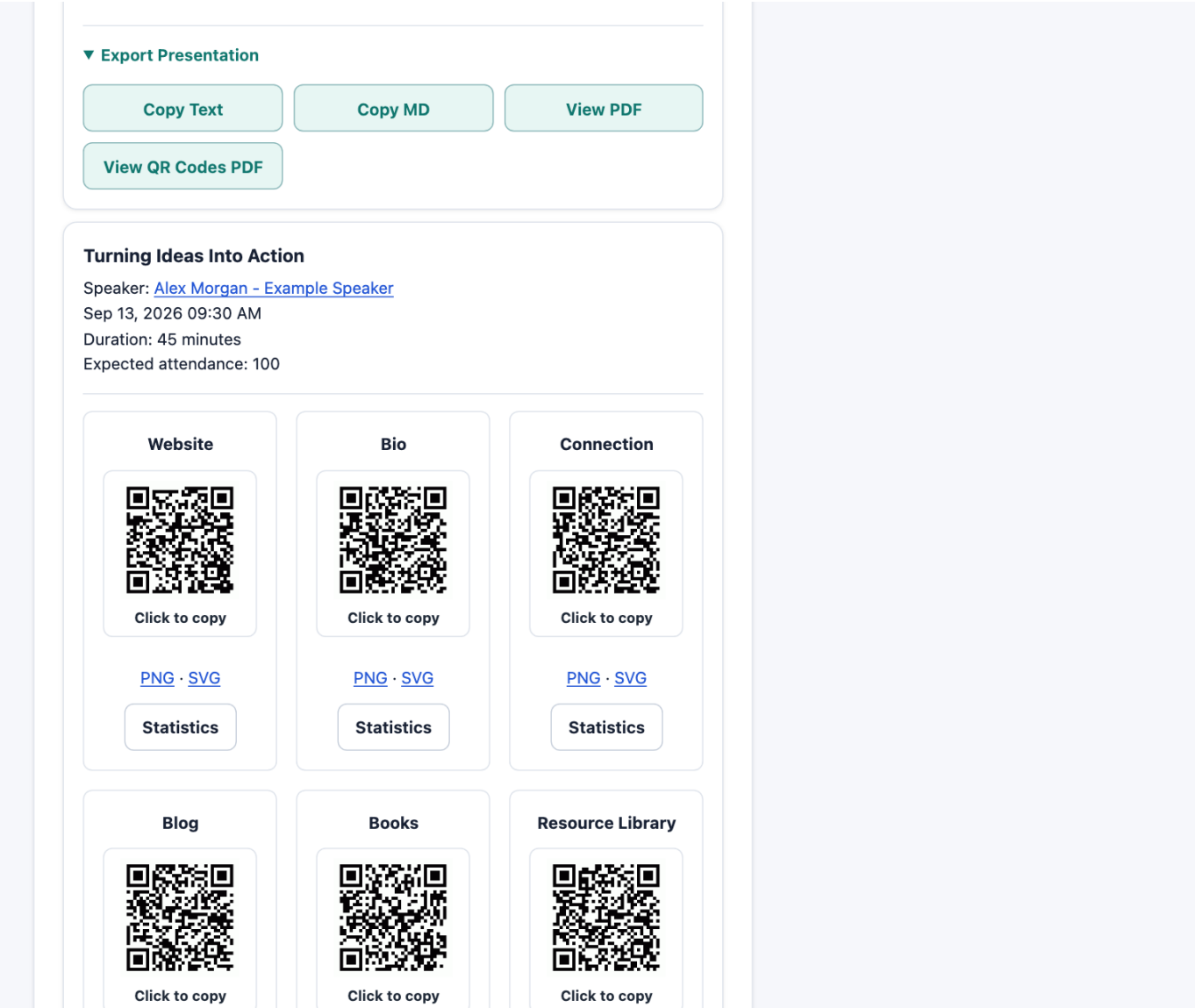


Figure 26. Copy or export a presentation brief. Actual preview screen; fictional records.

01 In Presentations, expand Export Presentation at the bottom of the card.

02 Copy Text produces plain text; Copy MD produces Markdown.

03 View PDF opens the branded presentation brief. View QR Codes PDF opens the landscape resource sheet for viewing, saving, or printing.

Related chapters: [People and organizations](#) (p. 42) / [Work queue](#) (p. 54) / [Chron and email](#) (p. 61) / [Map and calendar](#) (p. 68)

CHAPTER 05 / ILLUSTRATED WALKTHROUGH

Change an existing QR destination

[Community Learning Platform](#)

Building Lasting Connections · Alex Morgan - Example Speaker



Short URL

<http://127.0.0.1:9087/surls/7a1e4e4e9af23319>[Download PNG](#) · [Download SVG](#)

Destination

<https://example.org>☒ Link enabled

Save Link

Page 1 of 1

Figure 27. Change an existing QR destination. Actual preview screen; fictional records.

01 Review the specific code and its activity.**02** Edit Destination and save to redirect an already-published resource code without changing its image.**03** Clear Link enabled and save to stop that code from opening its destination.**04** For speaker notes, replace the PDF on the presentation; its destination is managed automatically.**Related chapters:** [People and organizations](#) (p. 42) / [Work queue](#) (p. 54) / [Chron and email](#) (p. 61) / [Map and calendar](#) (p. 68)

CHAPTER 05 / ILLUSTRATED WALKTHROUGH

Reset one presentation's statistics

Reset Presentation Statistics

Building Lasting Connections

Community Learning Weekend · Alex Morgan – Example Speaker

This permanently clears all recorded QR and link visits for this presentation, across every date, including disabled links and links for previous speakers. This cannot be undone.

QR codes, destinations, uploaded notes, and presentation details stay the same. New visits start counting from zero. Other presentations are unaffected.

Reset Statistics to Zero

Cancel

Figure 28. Reset one presentation's statistics. Actual preview screen; fictional records.

01 Administrators open Reset Presentation Statistics on the presentation card.

02 Confirm the exact presentation and obtain fresh administrator access.

03 Reset Statistics to Zero permanently clears all dates and all its links, including disabled links and previous speakers.

04 Codes, destinations, notes, and other presentations stay intact; new visits start at zero.

Screenshot context: The reset review screen is illustrated; no statistics were deleted.

Related chapters: [People and organizations \(p. 42\)](#) / [Work queue \(p. 54\)](#) / [Chron and email \(p. 61\)](#) / [Map and calendar \(p. 68\)](#)

CHAPTER 05 / ILLUSTRATED WALKTHROUGH

Enter actual receipts

Back to Event Tasks

Actual receipts

Leave an amount blank when it is not known yet. Enter 0 only when you have confirmed no amount was received. Every category must be entered before finalizing.

Giving / income received *

\$ Not entered

Lodging received *

\$ Not entered

Travel received *

\$ Not entered

Total entered: \$0.00

0 of 3 categories entered

Closeout notes

Optional context, payment references, or correction reason

☐ I confirm these are the final received amounts and this event is ready to be financially closed

Save draft

Finalize and close event

Cancel

Figure 29. Enter actual receipts. Actual preview screen; fictional records.

01 Enter known Giving/income, Lodging, and Travel received amounts.

02 The draft total shows the entered values and completion state.

03 Save draft while prerequisites remain unresolved. Enter all three amounts and confirm they are final before finalizing.

Related chapters: [People and organizations](#) (p. 42) / [Work queue](#) (p. 54) / [Chron and email](#) (p. 61) / [Map and calendar](#) (p. 68)

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CHAPTER 05 / ILLUSTRATED WALKTHROUGH

Read a finalized financial report

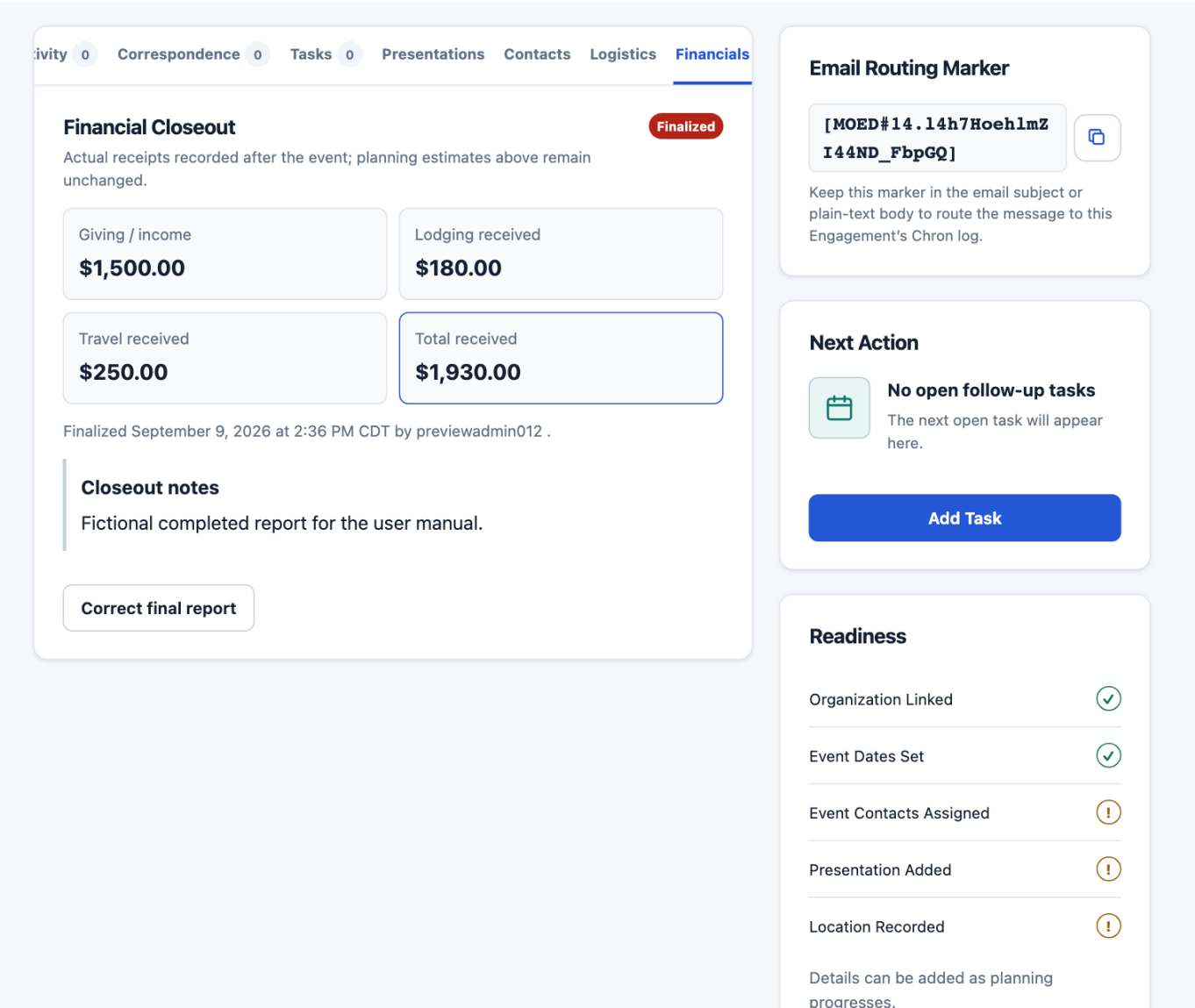


Figure 30. Read a finalized financial report. Actual preview screen; fictional records.

01 Open Financials on the completed event to see its final received amounts.

02 The report contributes to the organization's finalized financial history.

03 Use the correction control when an amount needs to change; the original closer and close date are retained.

Screenshot context: This completed report contains fictional amounts.

Related chapters: [People and organizations \(p. 42\)](#) / [Work queue \(p. 54\)](#) / [Chron and email \(p. 61\)](#) / [Map and calendar \(p. 68\)](#)

CHAPTER 06

Organizations, Contacts, and Speakers

Organizations hold relationship history, contacts identify the people involved, and speaker profiles keep presentation details and audience resources ready to reuse.

Related chapters: [Booking pipeline \(p. 12\)](#) / [Engagements \(p. 19\)](#) / [Chron and email \(p. 61\)](#)

Relationship Record

Store the organization name, notes, affiliation, distinctives, website, phone, fax, and physical and mailing addresses. You can use one address for both or maintain them separately.

Only the organization name is required; add address and relationship details when known. A new organization can create its first contact at the same time, and can add more contacts before saving.

Person Record

Store first and last name, optional primary organization, additional organization affiliations with their own roles or titles, email, phone, incidental notes, and an optional JPEG, PNG, or WebP photo up to 5 MB. A contact can be saved without an organization. Use Create Organization in the contact form to add a missing organization without losing the contact details or selected photo.

The organization role is Pastor, Admin, or Other. When you choose Other, enter the custom description in **Describe Other Role** beside the Role field. Additional organizations each have their own role or title, such as Pastor at a church and Chairman at a research center. Event-specific roles are assigned separately on each engagement.

Enter Organization Addresses

- Choose the country before the region. The Country picker includes countries worldwide and defaults to **United States of America** for a new address.
- For a United States address, the region field is labeled **State** and provides a dropdown of state names.
- For a Canadian address, the region field is labeled **Province** and provides a dropdown of province and territory names.

- For every other country, the field is labeled **State/Province** and accepts typed regional information.
- The same behavior applies to both physical and mailing addresses. Select **Same as Physical Address** when the mailing address does not differ.

Lists, Details, and Financial History

- Search organizations by their details or active contacts; sort by name and switch between Active and Archived.
- The organization list shows location, active contacts, last giving, and lifetime giving from finalized reports.
- An organization detail page summarizes lifetime giving, last and average event giving, lodging, travel, and recent finalized reports.
- Search contacts by person or organization, sort by last name or organization, and choose 20, 50, or 100 records per page.
- A contact detail page links to all its organizations and shows that person's Chron and open follow-up work.

Keep Speaker Profiles Ready to Use

Open **Relationships > Speakers** to find a speaker by name, email, or phone. Select **Name** to switch alphabetical order, then open the speaker's name or View icon for their details. Administrators and editors can use **New Speaker** or **Edit Speaker**; reviewers can view profiles and statistics.

- Save the speaker's name, email address, and phone number once, then choose that speaker when adding presentations.
- Add a biography and an optional photo to make the profile useful and recognizable. Photos can be JPEG, PNG, or WebP, up to 5 MB.
- Add the speaker's website and other audience links. The written **Bio** is for the profile; **Bio URL** is the web page opened by the biography QR code.
- For videos, social pages, or other resources, use **Speakers > Edit Speaker > Custom Links**. Give each link a clear name and full web address; select **Add Custom Link** for more, then **Save speaker**.

Saving a new custom link adds a separate QR code to that speaker's existing presentations, and future presentations receive their own codes. Renaming or removing a profile link keeps already-published links and their statistics intact. To change or disable a shared code, select **Statistics** beneath that code on its presentation.

Speakers remain available permanently and cannot be archived or deleted. Keep their existing profiles current instead of creating duplicates.

Organization Archive Rule

An organization cannot be archived while it still has active contacts or engagements. Archive those child records first, or move them to another active organization. Restoring a contact or engagement also requires its organization to be active.

Birthdays and organization affiliations

A contact can also store an optional birthday. Birthday-enabled calendar subscriptions create yearly all-day entries; they do not expose the birth year or internal contact notes.

Additional affiliations are independent relationships, each with its own role or title. A contact can remain associated with another organization when one organization is removed. Event-specific assignments belong to each engagement.

Browse organizations

Figure 31. Browse organizations. Actual preview screen; fictional records.

- 03** Use Active/Archived, sorting, and Records per page to manage a longer directory.

Related chapters: [Booking pipeline \(p. 12\)](#) / [Engagements \(p. 19\)](#) / [Chron and email \(p. 61\)](#)

CHAPTER 06 / ILLUSTRATED WALKTHROUGH

Review relationship history

Organizations / Organization Details

Cedar Grove Community - Manual Example

Relationships, activity, and upcoming engagements.

Edit Organization

Add Chron Log Entry

Contacts

1 people

Jordan Parker

Next engagement

Community Learning Weekend

Sep 12–13, 2026

Follow-up

Open tasks

Record a conversation

▼ Profile, addresses, and notes

Affiliation

Community education

Distinctives

Weekend learning and service

Website

https://example.org

Phone

+1 312-555-0120

Fax

Not specified

Physical Address

100 Example Avenue, Chicago, Illinois, 60601 United States of America

Notes

Fictional training organization. Host prefers a planning call two weeks before each event.

Figure 32. Review relationship history. Actual preview screen; fictional records.

01

Use Activity for organization-level Chron.

02

Switch to Contacts, Engagements, Tasks, or Financials for related information.

03

Use Edit Organization to maintain details and add or move contacts.

Related chapters:

Booking pipeline (p. 12) / Engagements (p. 19) / Chron and email (p. 61)

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CHAPTER 06 / ILLUSTRATED WALKTHROUGH

Read organization financial history

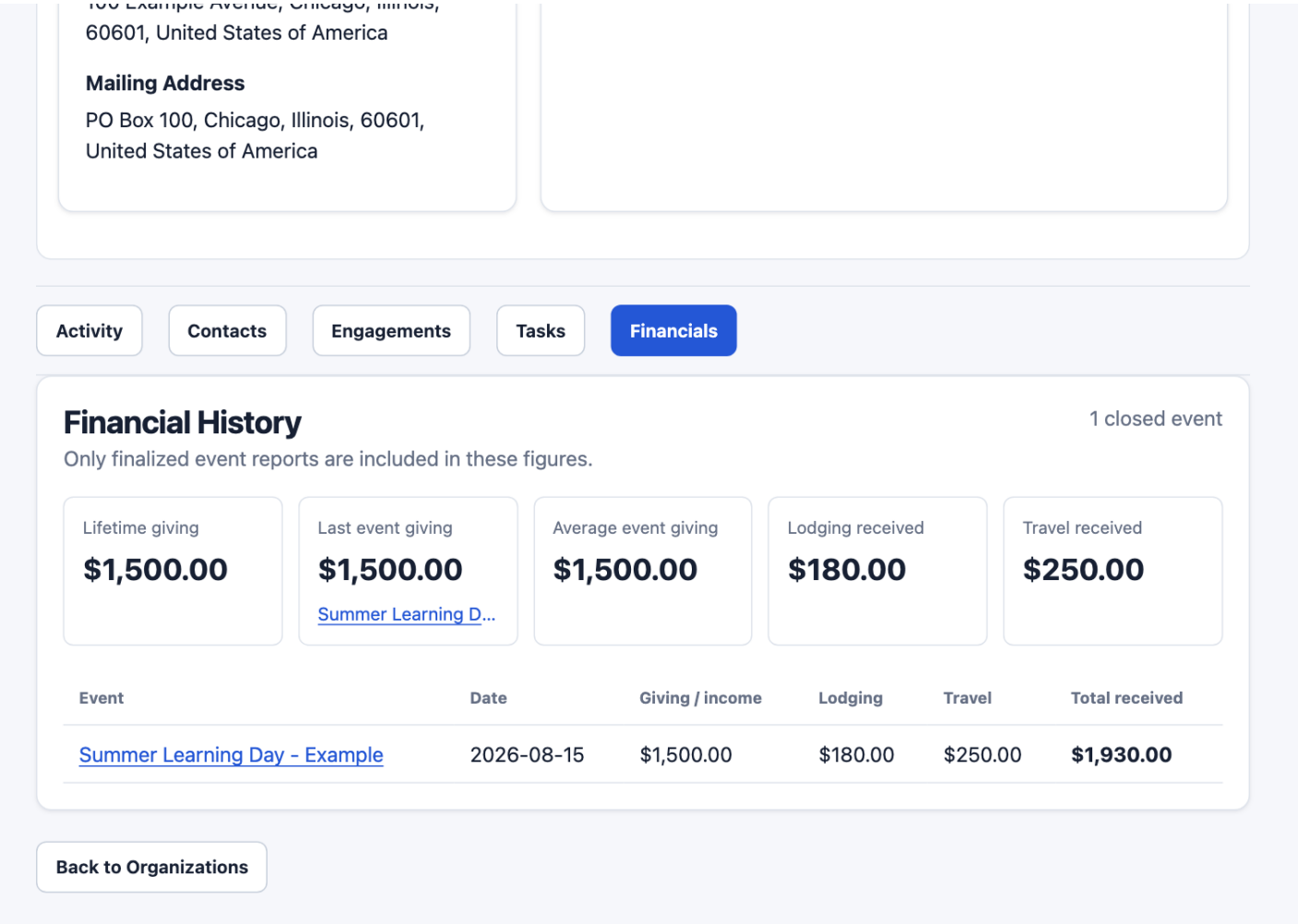


Figure 33. Read organization financial history. Actual preview screen; fictional records.

- 01

Open Financials for finalized giving, lodging, travel, and event reports.
- 02

Draft closeout amounts do not affect these totals.
- 03

Open an event report for the underlying figures and correction history.
- Screenshot context:

Amounts are fictional examples.
- Related chapters:

[Booking pipeline \(p. 12\)](#) / [Engagements \(p. 19\)](#) / [Chron and email \(p. 61\)](#)

CHAPTER 06 / ILLUSTRATED WALKTHROUGH

Organization identity and addresses

Organizations / Edit Organization

Edit Organization

Update organization information and addresses.

Organization Name *

Cedar Grove Community - Manual Example

Notes

Fictional training organization. Host prefers a planning call two weeks before each event.

Affiliation

Community education

Distinctives

Weekend learning and service

Website URL

https://example.org

Phone

 +1 ▾

(312) 555-0120

Env

Figure 34. Organization identity and addresses. Actual preview screen; fictional records.

- 01

Only the organization name is required. Add affiliation, distinctives, contact channels, and notes when known.
- 02

Choose country before state or province. US and Canadian regions use dropdowns; other countries accept typed regions.
- 03

Use Same as Physical Address when mailing details match.
- Related chapters:

[Booking pipeline \(p. 12\)](#) / [Engagements \(p. 19\)](#) / [Chron and email \(p. 61\)](#)

Browse contacts

Contact

+ New Contact

Find the people connected to every organization and engagement.

Parker

Clear

Active

Archived

Sort: Last Name ↑

Primary organization ↑

Showing contacts matching "Parker".

CONTACT	ORGANIZATIONS	PHONE NUMBER	EMAIL ADDRESS	ACTIONS
JP Parker, Jordan	Cedar Grove Community - Manual Example Primary Admin	+1 312-555-0121	jordan.parker@example.org	

© 2026 beneliath · DNR 1.11.11 · 2026-09-05 14:19:58 CDT (ec7dc08)

Tylo = appointment, appointed time

(
6_6) ()
(Y_))
(il), (il), ((l,-))

Genesis 49: 9, 10 ... Revelation 5:5
Do you see Him?

Figure 35. Browse contacts. Actual preview screen; fictional records.

- 01 Search by person or organization.
 - 02 Sort by last name or organization and adjust Records per page.
 - 03 Open a contact to review their own history and all affiliations.

Related chapters: [Booking pipeline \(p. 12\)](#) / [Engagements \(p. 19\)](#) / [Chron and email \(p. 61\)](#)

CHAPTER 06 / ILLUSTRATED WALKTHROUGH

Read a contact record

Contacts / Contact Details

Parker, Jordan

Admin at Cedar Grove Community - Manual Example

Edit Contact

Add Chron Log Entry

JP

Organizations and roles

[Cedar Grove Community - Manual Example](#)

Primary Admin

Email

[jordan.parker@example.org](#)

Phone

+1 312-555-0121

Birthday

Not specified

Notes

Fictional primary host. Coordinates venue access and speaker arrival.

Activity

Communication history for this contact, newest first.

Add Chron Log Entry

September 9, 2026 at 2:27 PM CDT by previewadmin012

Last updated September 9, 2026 at 2:27 PM CDT by previewadmin012

Figure 36. Read a contact record. Actual preview screen; fictional records.

- 01

Check name, role, email, phone, photo, and organization links.
- 02

Keep person-specific conversations in this contact's Chron.
- 03

Create follow-up work here to retain the relationship context.
- Related chapters:

[Booking pipeline \(p. 12\)](#) / [Engagements \(p. 19\)](#) / [Chron and email \(p. 61\)](#)

CHAPTER 06 / ILLUSTRATED WALKTHROUGH

Maintain affiliations and a photo

Contacts / Edit Contact

Edit Contact

Update contact information and roles at each organization.

* Required fields

Primary organization

Cedar Grove Community - Manual Example

First Name *

Jordan

Last Name *

Parker

Primary role *

Admin

Additional organizations

A person can serve several organizations. Add each organization and the role they hold there. Make primary switches their primary organization and keeps the previous organization and role in this list.

Additional organization

Select an organization

Role or title at this organization

For example, Chairman

Make primary

Remove

Add another organization

Email *

Figure 37. Maintain affiliations and a photo. Actual preview screen; fictional records.

01

Enter names and contact channels; an organization is optional.

02

Set the primary organization role and add additional organization affiliations with their own roles.

03

Use Other with Describe Other Role for a custom title.

04

Upload a JPEG, PNG, or WebP photo up to 5 MB and save. Create Organization preserves the contact draft.

Related chapters:

Booking pipeline (p. 12) / Engagements (p. 19) / Chron and email (p. 61)

Find a saved speaker

Figure 38. Find a saved speaker. Actual preview screen; fictional records.

- Related chapters:** [Booking pipeline \(p. 12\)](#) / [Engagements \(p. 19\)](#) / [Chron and email \(p. 61\)](#)

CHAPTER 06 / ILLUSTRATED WALKTHROUGH

Speaker profile and resources

Speakers / Speaker Details

Alex Morgan - Example Speaker

Speaker

Edit Speaker

Statistics



Email

alex.morgan@example.org

Phone

[+1 312-555-0122](tel:+13125550122)

Bio

Alex is a fictional speaker used to demonstrate planning, presentation resources, and follow-up.

Links

Website URL

<https://example.org>

Donation URL

Not specified

Blog URL

<https://example.org/articles>

Resource Library

<https://example.org/resources>

Bio URL

<https://example.org/bio>

Connection URL

<https://example.org/connect>

Books URL

<https://example.org/books>

Back to Speakers

Figure 39. Speaker profile and resources. Actual preview screen; fictional records.

01

Review the reusable speaker identity, biography, and audience links.

02

Open Statistics to review activity across the speaker's presentations.

03

Edit the profile to maintain shared details. The written biography and Bio URL serve different purposes.

Related chapters:

[Booking pipeline \(p. 12\)](#) / [Engagements \(p. 19\)](#) / [Chron and email \(p. 61\)](#)

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CHAPTER 06 / ILLUSTRATED WALKTHROUGH

Edit standard and custom resource links

Links

Website URL

https://example.org

Bio URL

https://example.org/bio

Donation URL

https://example.com

Connection URL

https://example.org/connect

Blog URL

https://example.org/articles

Books URL

https://example.org/books

Custom Links

Add named links for resources, videos, social profiles, or other pages. Saving creates a unique link and QR code for each associated presentation.

Published links keep their names, destinations, and statistics when you edit or remove a profile link. Manage a published destination from its Statistics page under Presentations.

Link Name

Resource Library

URL

https://example.org/resources

Remove

Add Custom Link

AS

Speaker Photo

Choose File

No file chosen

Figure 40. Edit standard and custom resource links. Actual preview screen; fictional records.

01

Add full web addresses for audience resources.

02

Choose Add Custom Link, provide its name and URL, and save the speaker.

03

New links receive separate codes on existing and future presentations.

04

Removing or renaming a profile link preserves already-published codes. Manage an individual code in Statistics.

Related chapters: [Booking pipeline \(p. 12\)](#) / [Engagements \(p. 19\)](#) / [Chron and email \(p. 61\)](#)

CHAPTER 07

Work Queue

Tasks turn relationship context into an explicit next action with an owner, timing, priority, and state. They may stay attached to an Inquiry until selected work moves during booking.

Related chapters: [Daily dashboard \(p. 10\)](#) / [Engagements \(p. 19\)](#) / [Map and calendar \(p. 68\)](#)

My Reminders

The sidebar badge and Dashboard personal cards always count your own reminders. In the queue, choose Mine, Everyone, or Unassigned; every queue summary count uses that same ownership scope, search, and related-record filter.

Queue Views

Choose an ownership scope, then filter Active, Overdue, Due Today, Next 7 Days, Waiting, or Completed. The Dashboard's My Overdue Work card opens Overdue limited to your assignments. Search matches task content, related records, and assignees.

Due Dates and Overdue Pulses

The due date appears as a short date, such as **Sep 4**, with a second line showing the number of days overdue, **Due today**, or **Upcoming**. Days are counted by calendar date in the application's time zone. Priority and workflow status appear separately.

Active tasks **more than 3 days overdue** - starting at 4 days overdue - have a pulsing red row background. When several items qualify, the pulse moves down the list in a repeating cascade, with each pulse overlapping the next. Tasks 1-3 days overdue keep a steady red highlight; tasks due today have a blue highlight. Completed and canceled tasks do not pulse.

The same attention cues appear in the Dashboard's **My Work** and **Inquiry Next Actions** lists. An active inquiry qualifies when its **next-action due date** is more than 3 days overdue.

Hovering over a row, or moving keyboard focus to a control within it, shows an **orange highlight** over the pulse so you can clearly follow the item you are using. The pulse becomes visible again when the pointer and focus leave the row. If your device or browser prefers **reduced motion**, overdue items use a steady stronger red highlight instead of blinking.

Create a Useful Task

- 01 Write a concrete title.** Use notes for supporting detail, not as a substitute for the next action.
- 02 Link the right record.** Search for an engagement, organization, or contact; choose General MOED work when no record applies.
- 03 Name an owner and due date.** Tasks may be unassigned or have no date, but explicit ownership keeps them out of limbo.
- 04 Set priority and status.** Choose Low, Normal, High, or Urgent and one of the workflow states below.

Status	Use It When	What Happens Next
Open	The task is known but not started.	Start it, complete it, or edit the owner/timing.
In Progress	Someone is actively working it.	Complete it or move it to Waiting if blocked.
Waiting	Progress depends on a person, organization, or decision.	The "Waiting on" description becomes required.
Completed	The action is finished.	It moves to Completed and can be reopened.
Canceled	The action is no longer needed.	It leaves active views and can be reopened.

Duplicate a Task to Another Event

- 01 **Open the task for editing.** Select **Duplicate to Another Event** at the bottom of the Edit Task page.
- 02 **Choose the destination event.** Enter at least three characters in **Search events** , then select a different event from the results. The source event cannot be selected as its own destination.
- 03 **Review the copied work.** The title, notes, assignee, due date, and priority are copied so you can adjust them before saving. The duplicate starts **Open** , and any "Waiting on" text is cleared.
- 04 **Create the duplicate.** Select **Duplicate task** to add a new task to the destination event. The original task remains unchanged.

Fast Actions and Record-Level Work

From the queue, editors and administrators can assign an unowned task to themselves, start, complete, reopen, edit, or - administrators only - delete it. Engagement, organization, and contact detail pages show their open tasks in **Follow-Up Work**, where you can add a task or open a filtered queue.

Standard Event Tasks

Standard tasks are reusable definitions copied when a new active engagement is first saved. Each has content, priority, display order, and a due date offset from event start or end.

If the new engagement has a selected **Caller**, every initial standard task is assigned to that Caller. If no Caller is selected, the tasks are assigned to the user creating the engagement. This ownership decision happens only once: changing the engagement's Caller later does not reassign its initial tasks, which remain independently editable.

Changing an engagement's dates reschedules its active generated tasks from the standard schedule. A task whose due date was manually overridden keeps that date. Editing a standard definition does not rewrite an existing task's title, notes, priority, or current due date.

Editors and administrators manage reusable definitions from **Standard Event Tasks**. Use **Add Missing Checklist Tasks** on an active engagement to generate any standard items that are absent; tasks added this way are assigned to the user performing the action. The built-in financial closeout reminder is fixed at one week after event end and cannot be edited, archived, or deleted.

Sending a booking, reconfirmation, or thank-you message from an engagement does not complete its related checklist task automatically. Check the Correspondence result, then complete the task when the follow-up itself is finished.

CHAPTER 07 / ILLUSTRATED WALKTHROUGH

Work the queue

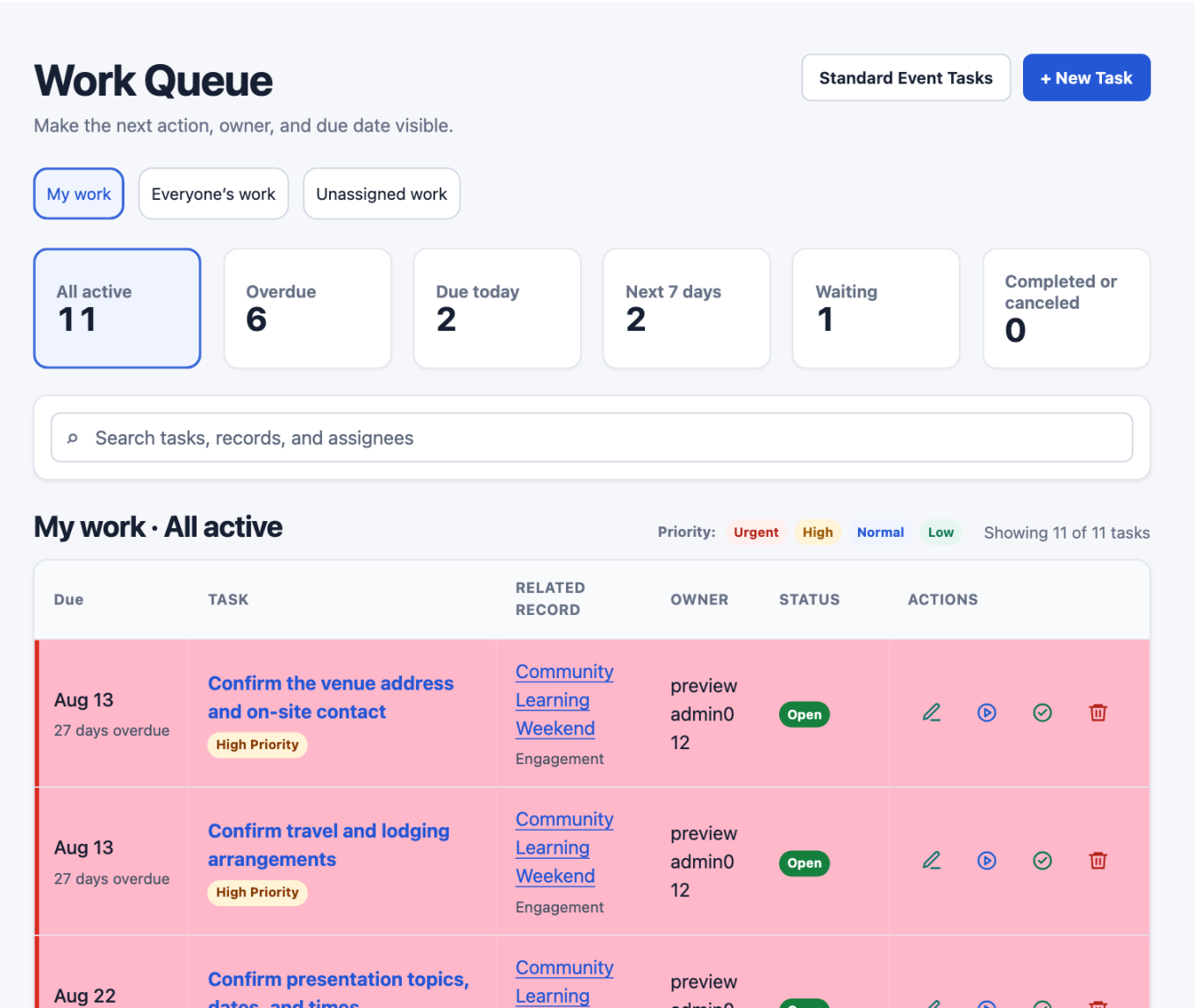


Figure 41. Work the queue. Actual preview screen; fictional records.

- 01

Choose Mine, Everyone, or Unassigned.
- 02

Use status, due-date, priority, search, and related-record controls to focus the list.
- 03

Use row actions to assign, start, complete, reopen, or edit work.
- 04

Overdue and due-today highlights draw attention; the sidebar badge counts your active assigned tasks.
- Related chapters:

Daily dashboard (p. 10) / Engagements (p. 19) / Map and calendar (p. 68)

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Edit a task

Work Queue / Edit Task

Edit Task

Update the next action, owner, timing, or status.

* Required fields

Task

Title

Confirm host arrival and room setup

Notes

Call Jordan, verify the arrival time, and record the agreed room layout in Chron.

Related record

Selected record

Community Learning Weekend

Change record

Clear relationship

Type at least three characters, then choose a result. Searching does not change the selected record. Use General MOED work when no record applies.

Figure 42. Edit a task. Actual preview screen; fictional records.

01 Write a concrete title and notes.

02 Search at least three characters and select the intended related record, or use general work.

03 Set assignee, due date, priority, and status. Waiting work should name the person or decision it needs.

04 Save task to persist the change.

Related chapters: [Daily dashboard \(p. 10\)](#) / [Engagements \(p. 19\)](#) / [Map and calendar \(p. 68\)](#)

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CHAPTER 07 / ILLUSTRATED WALKTHROUGH

Duplicate a task to another event

Work Queue / Duplicate Task

Duplicate Task

Copy this task to a different event, then adjust its owner or timing.

* Required fields

Task

Title

Confirm host arrival and room setup

Notes

Call Jordan, verify the arrival time, and record the agreed room layout in Chron.

Destination event

Selected record

No record selected

Change record

Find a related record

Search events

Type at least three characters, then choose an event different from the source task.

Figure 43. Duplicate a task to another event. Actual preview screen; fictional records.

01

Choose Duplicate to Another Event from the task editor.

02

Search events and select a destination different from the source event.

03

Review copied title and notes, ownership, due date, priority, and status before saving.

04

The original task remains unchanged.

Related chapters:

Daily dashboard (p. 10) / Engagements (p. 19) / Map and calendar (p. 68)

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CHAPTER 07 / ILLUSTRATED WALKTHROUGH

Manage reusable event checklists

Work Queue / Standard Event Tasks

Standard Event Tasks

Control the reusable work assigned automatically when new events are created.

Active (9)

Archived (0)

Every active definition is added and assigned to the creator automatically with each new event. "Add missing checklist tasks" remains available for older events. Existing event tasks remain unchanged.

ORDER	STANDARD TASK	DUE RULE	PRIORITY	GENERATED	ACTIONS
10	Confirm the venue address and on-site contact	30 days before event start	High	1 event task	  
20	Confirm travel and lodging arrangements	30 days before event start	High	1 event task	  
30	Confirm presentation topics, dates, and times	21 days before event start	High	1 event task	  
40	Confirm book table, brochures, and event materials	14 days before event start	Normal	1 event task	  
50	Reconfirm final details with the host	7 days before event start	High	1 event task	  
60	Prepare presentations, handouts, books, and supplies	3 days before event start	High	1 event task	  

Figure 44. Manage reusable event checklists. Actual preview screen; fictional records.

- 01

Open Standard Event Tasks from the Work Queue.
- 02

Create or edit the content, priority, display order, and offset from event start or end.
- 03

Changes affect generation; existing task content is independently editable.
- 04

The required financial reminder is fixed at one week after event end.
- Related chapters:

Daily dashboard (p. 10) / Engagements (p. 19) / Map and calendar (p. 68)

CHAPTER 07 / ILLUSTRATED WALKTHROUGH

Define a standard event task

Work Queue / Standard Event Tasks / New Task

New Standard Event Task

Add reusable work that will be assigned automatically when a new event is created.

* Required fields

Task

Title

Notes

These notes are copied into each event task generated from this definition.

Scheduling

Priority

Normal

Due date relative to

Event start

Figure 45. Define a standard event task. Actual preview screen; fictional records.

01 Enter the reusable title and notes.

02 Choose the date anchor and number of days before or after it.

03 Set priority and display order, then save.

04 Add Missing Checklist Tasks on an active engagement creates absent items assigned to the person performing that action.

Related chapters: [Daily dashboard \(p. 10\)](#) / [Engagements \(p. 19\)](#) / [Map and calendar \(p. 68\)](#)

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CHAPTER 08

Chron and Email

Chron is the durable communication history. Add concise Chron Log Entries directly, send tracked engagement correspondence, or route copied inbound email into the relevant records.

Related chapters: [Booking pipeline \(p. 12\)](#) / [Engagements \(p. 19\)](#) / [Mattermost \(p. 82\)](#)

Use the Chron Log

- Use **Add Chron Log Entry** on an inquiry, engagement, organization, or contact to save activity without editing the full record. Contact Chron records communication with that person only; Organization Chron records organization-level history; Engagement Chron records event-specific planning.
- Entries show created time and author, newest first. Edited entries also show the last update time and editor.
- Editors and administrators add, edit, and archive entries from the record's edit page. Administrators can permanently delete them.
- Restore archived entries in batches from the Restore page. An archived parent record must be active before its Chron can be restored.
- Email-generated entries link to their retained inbound or outbound source message while it exists.

Start from an Engagement

Open an active engagement and select **Send Message**. Choose assigned contacts, review the message, and send. The engagement marker and related Chron history are handled for you. Check the Correspondence section for the result.

Route Outside Email

When using another mail application, copy the inbound address and keep the exact engagement marker in the subject or plain-text body: `[MOED#123.<signed-token>]`. Use the copy icon beside the marker on the engagement page.

How Inbound Email Finds Its Records

One unique, valid signed engagement marker routes directly to that active engagement only when the sender also uniquely matches an active record. Unknown senders and unsigned legacy markers require review. When participant matching is uncertain, only the signed engagement receives the Chron entry.

Address Matching

MOED matches exact email addresses for active verified users, contacts, and organizations. A unique contact participant routes to the Contact and its Organization. A direct organization address routes there. **Cc** and **Bcc** delivery both work.

What Inbound Mail Retains

The Chron entry includes normalized headers, subject, timestamps, inert plain-text body, attachment names, and a source link. Attachment contents are not stored. Duplicate delivery of the same message ID is ignored.

Review the Inbound Queue

Editors and administrators see messages grouped as Needs review, Pending, Processing, Failed, Processed, or Rejected. Without a valid signed marker, every message requires review. Unknown or ambiguous senders, shared addresses, messages with no unique target, and invalid, conflicting, unknown, or archived engagement markers also require review.

01 Inspect the source. Open Inbox in the sidebar. On a wide screen, the message list is on the left, the reader is in the middle, and the filing panel is on the right. Search by subject, sender, or content; use Order and the queue pages to find a message. Needs review starts oldest first. Selecting a message keeps Search messages visible at the same vertical position. Expand Message details for recipients, sent time, and attachment names, or Routing details for matching reasons. On a phone, use Back to messages to return to the list.

02 Choose the Chron Logs. In the filing panel, review the reason under Review needed. Select the intended Contact, Organization, or Inquiry suggestions when available. Find an active engagement by name, ID, or email marker and choose it from the Engagement list. Check the count of selected Chron logs before saving.

03 Choose an outcome. **Save to Chron logs** saves the conversation to the selected records. **Find matches again** checks routing again after record corrections.

Expand **Other actions** to **Reject message** while retaining the source without writing Chron. For a processed message, review **Message status** and **Saved to Chron** to confirm where it was filed.

Treat Address Matching as a Routing Aid

An exact visible From address is not independent proof of identity. Respect the mailbox provider's spam and SPF/DKIM/DMARC signals. Leave suspicious messages for review. Administrators can expand Manage retained email below the reader and choose Purge Mail Entry after confirmation. Purge removes the retained MOED mail card and source links, but preserves Chron entries and never deletes the original IMAP message.

Email delivery states and retries

Queued messages can be Pending or Processing before becoming Sent, Failed, or partially delivered across recipients. Open the delivery record to inspect individual recipient outcomes and timestamps.

Retry Failed Deliveries retries the failures rather than re-sending successful recipients. Correct a bad destination before retrying. Inquiry delivery retries are available while the inquiry is active.

After booking conversion, replies to earlier inquiry email route into the resulting engagement. The original inquiry retains the pre-booking record and any tasks not moved.

The Inbox filing panel saves an email to existing destinations. Create a new booking request from New Inquiry in Booking Pipeline when needed, then return to the Inbox and review the appropriate destinations.

Archive and restore Chron in batches

Open the parent record's edit/Chron controls to archive an obsolete entry. Archive preserves history; administrator-only deletion removes it permanently after fresh confirmation.

Open Restore Archived Chron Log Entries, select the entries to return, and choose Restore Selected. The parent must be active. Restored entries retain their historical authorship and dates.

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Add a Chron entry

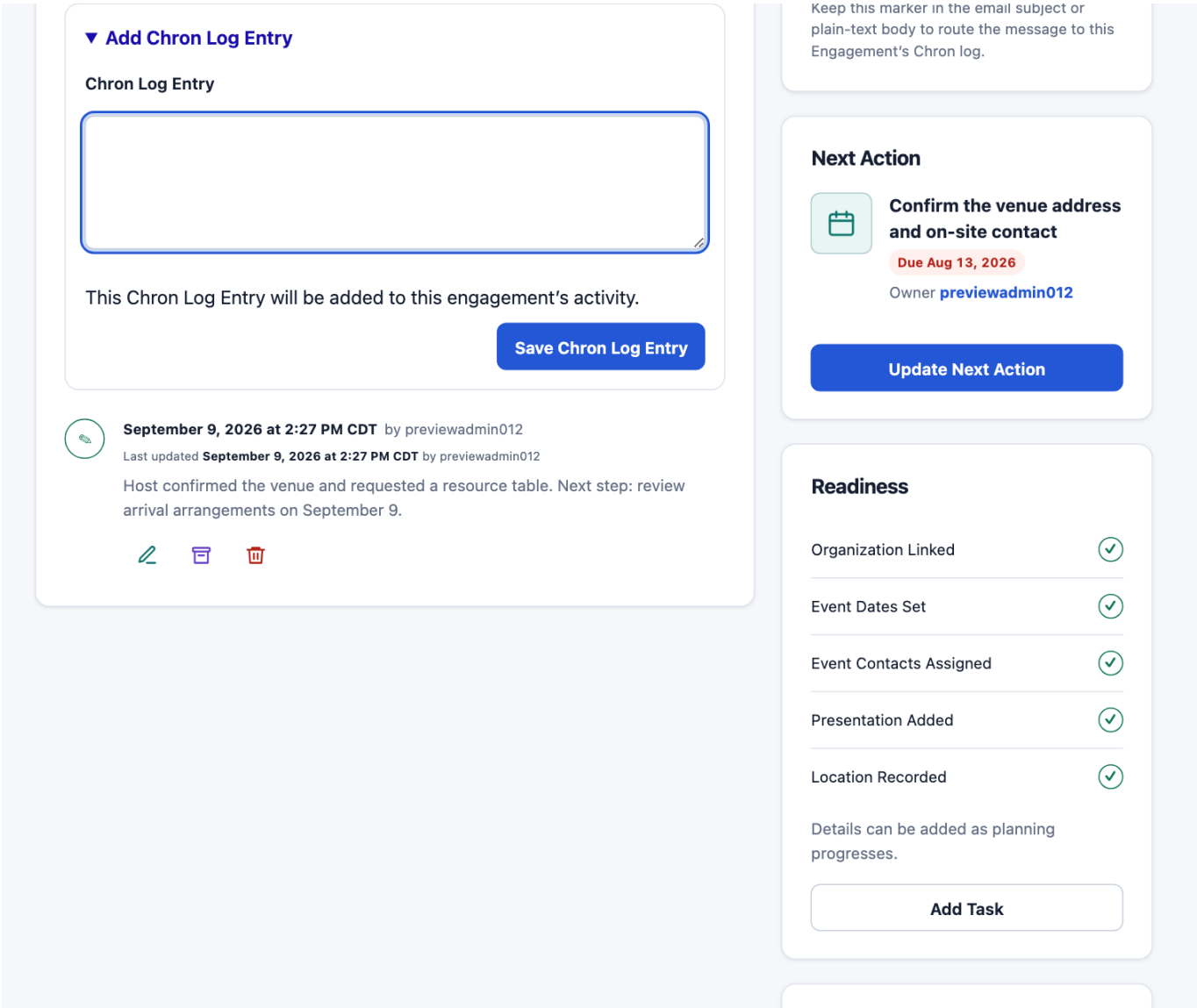


Figure 46. Add a Chron entry. Actual preview screen; fictional records.

- 01 Use Add Chron Log Entry to record a decision without opening the complete edit form.
- 02 Include the context, outcome, and agreed next step.

- 03 Save the entry; its author and time appear in reverse chronological history.
- Related chapters: [Booking pipeline \(p. 12\)](#) / [Engagements \(p. 19\)](#) / [Mattermost \(p. 82\)](#)

CHAPTER 08 / ILLUSTRATED WALKTHROUGH

Choose event recipients and a template

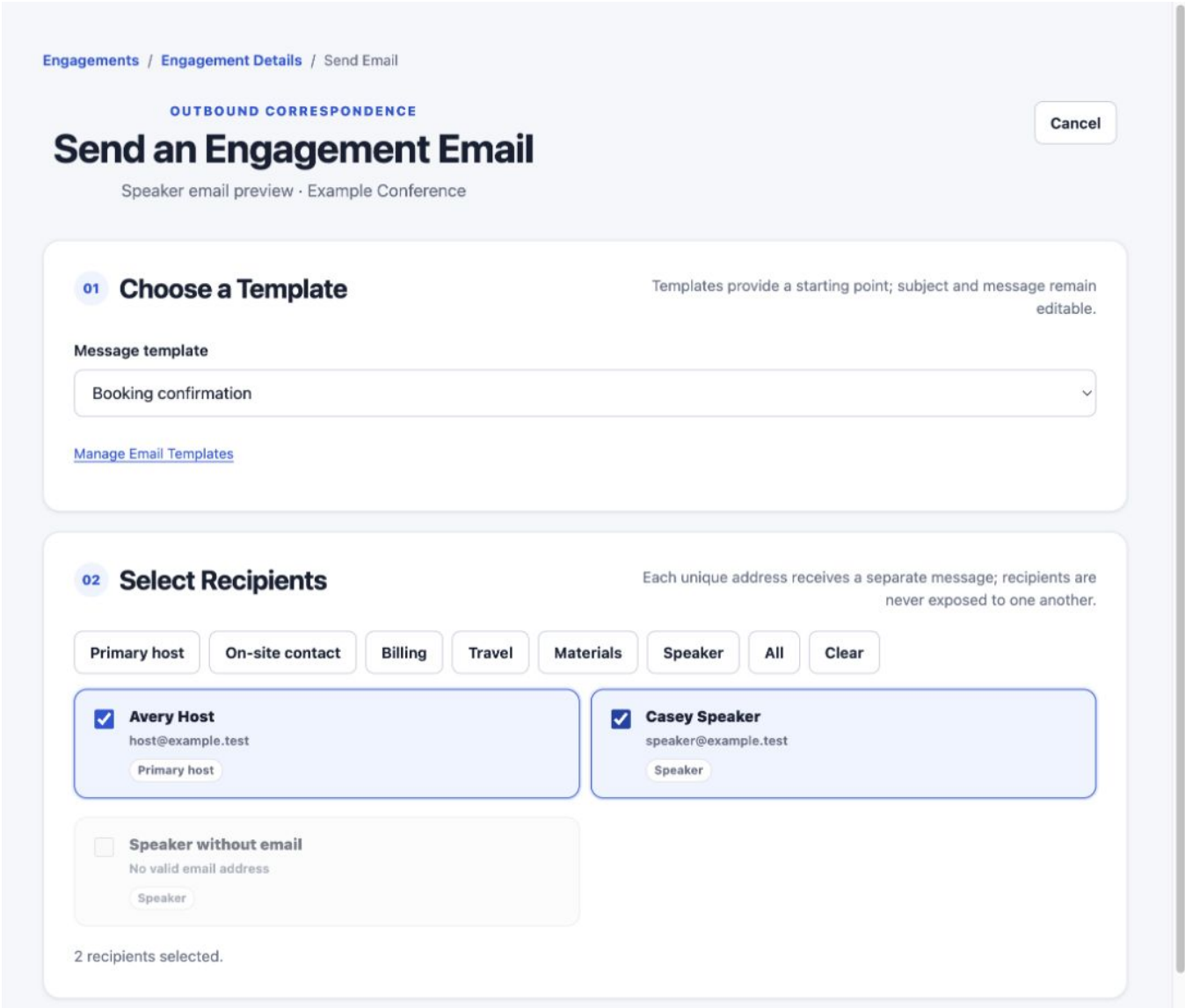


Figure 47. Choose event recipients and a template. Actual preview screen; fictional records.

- 01** Choose an active template or Custom message. Review event fields after they are filled.

02 Select event contacts, then use Speaker or an individual speaker card to include an assigned speaker. Speakers are unchecked by default.
- 03** Only speakers assigned to active presentations with valid email addresses can receive mail. Speaker-only messages are allowed.

04 Each unique address receives one separate email. Changing templates preserves selected speakers. Review the message and brief before Queue Email.

Screenshot context: This example selects Casey Speaker explicitly. No email was queued during capture.

Related chapters: [Booking pipeline \(p. 12\)](#) / [Engagements \(p. 19\)](#) / [Mattermost \(p. 82\)](#)

CHAPTER 08 / ILLUSTRATED WALKTHROUGH

Review the message and brief

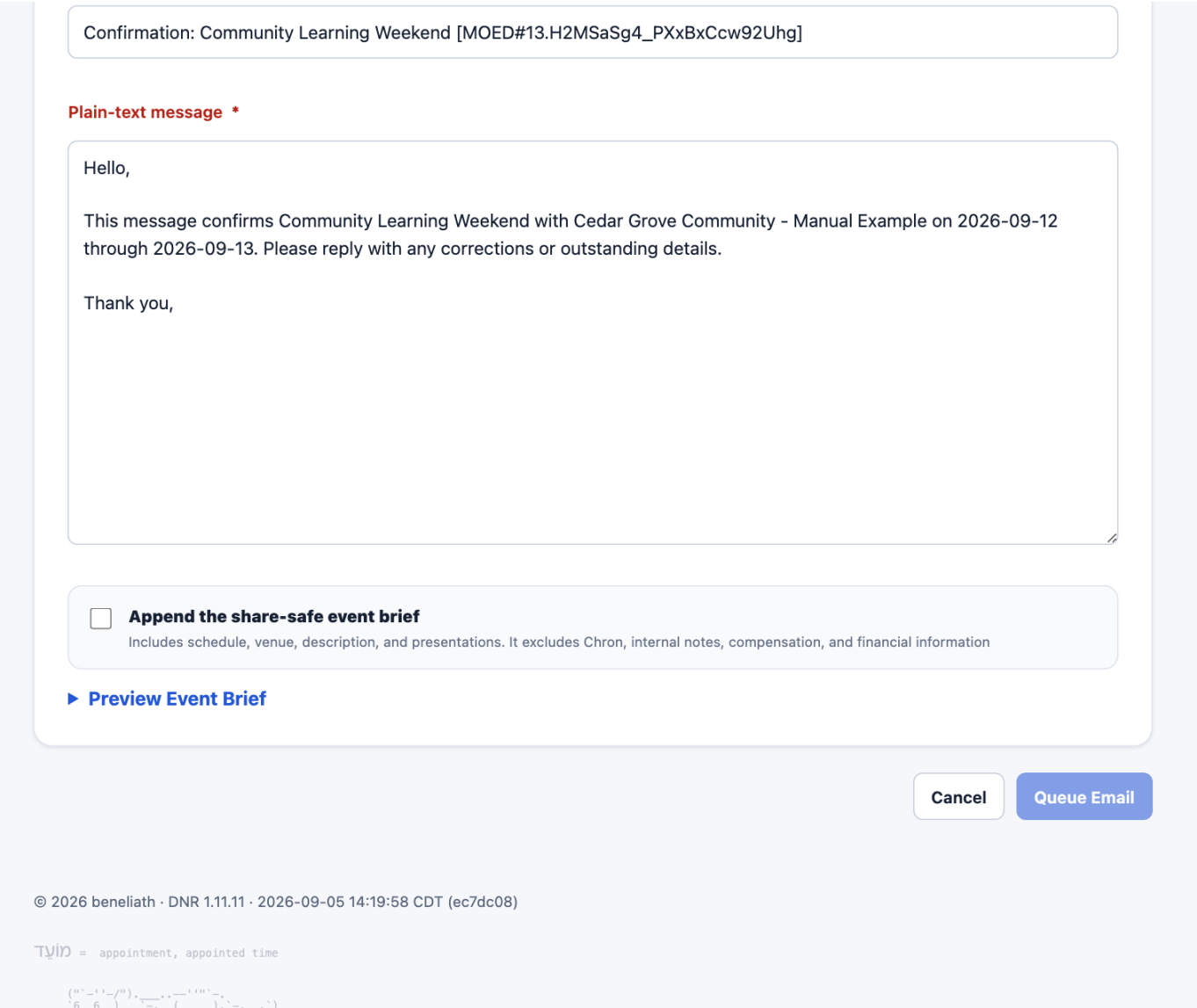


Figure 48. Review the message and brief. Actual preview screen; fictional records.

- 01** Edit the plain-text message for the event.

02 Expand Preview Event Brief to inspect the optional share-safe information before including it.
- 03** The event marker is maintained for routing. A sent checklist email does not complete its related task automatically.

Related chapters: [Booking pipeline \(p. 12\)](#) / [Engagements \(p. 19\)](#) / [Mattermost \(p. 82\)](#)

CHAPTER 08 / ILLUSTRATED WALKTHROUGH

Review copied inbound email

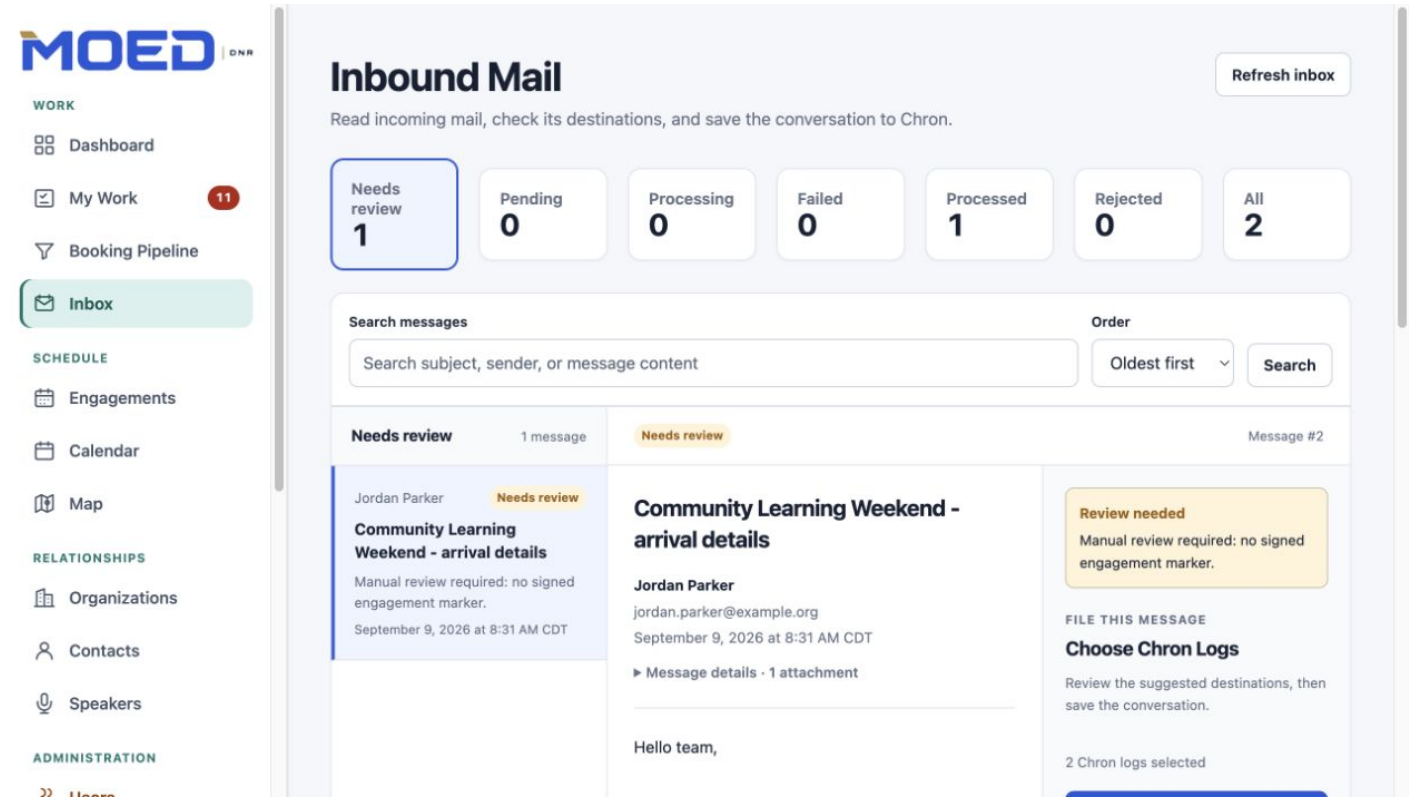


Figure 49. Review copied inbound email. Actual preview screen; fictional records.

- 01 Open Inbox in the sidebar. Use Needs review, Pending, Processing, Failed, Processed, Rejected, or All to choose a queue.

02 Search by subject, sender, or message content. Change Order or use the queue pages when needed. Needs review starts oldest first.
- 03 On a wide screen, the message list is on the left, the reader is in the middle, and the filing panel is on the right.

04 Select a message to read it. Search messages stays visible at a consistent vertical position. On a phone, use Back to messages to return to the queue.

Related chapters: [Booking pipeline \(p. 12\)](#) / [Engagements \(p. 19\)](#) / [Mattermost \(p. 82\)](#)

CHAPTER 08 / ILLUSTRATED WALKTHROUGH

Read a message and choose its Chron Logs

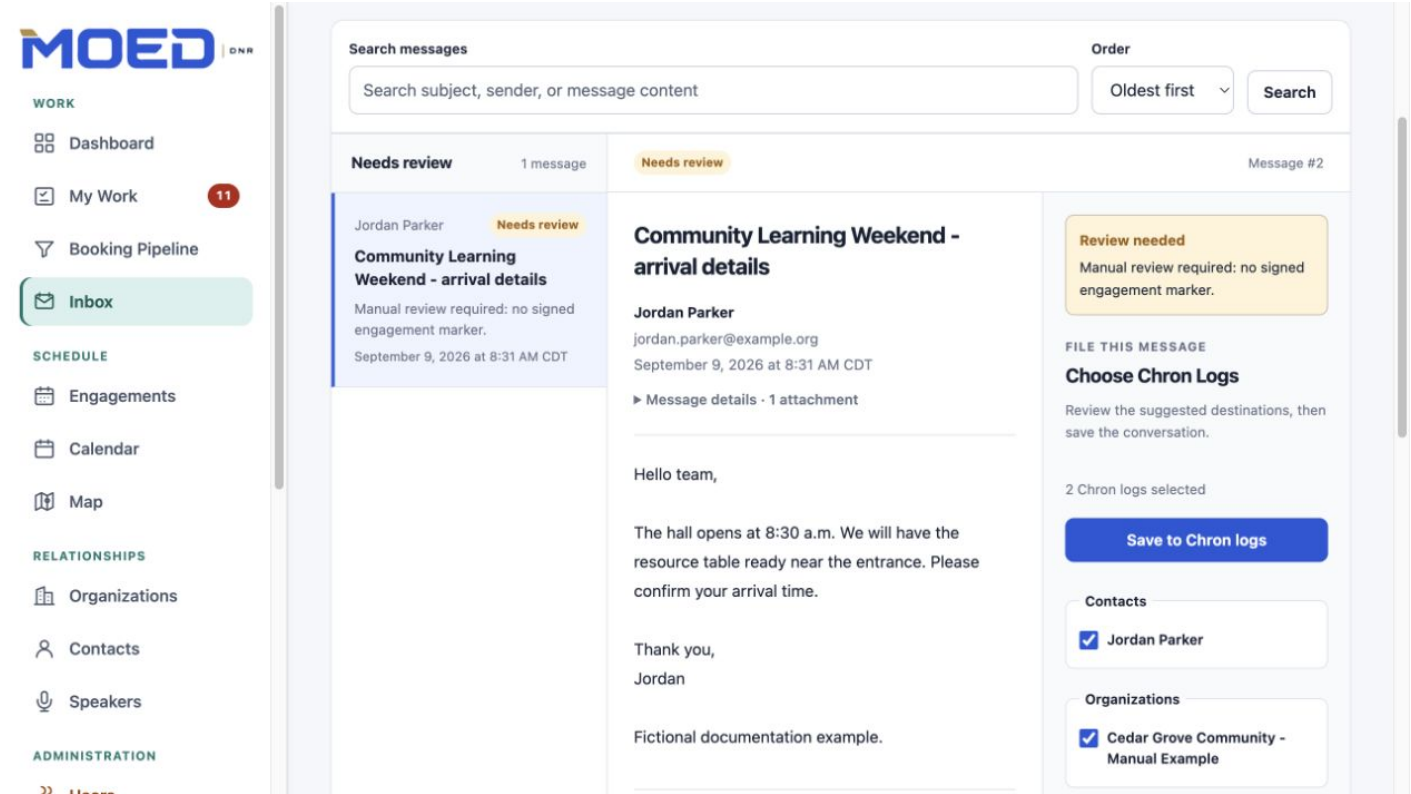


Figure 50. Read a message and choose its Chron Logs. Actual preview screen; fictional records.

- 01 Read the sender, subject, and message. Expand Message details for recipients, sent time, and attachment names; open Routing details for matching reasons.

02 Review the reason in the right-hand filing panel. Choose the intended Contact, Organization, or Inquiry suggestions when available. Scroll down to search and select an active Engagement.
- 03 Check the selected Chron log count, then choose Save to Chron logs. Processed messages show Message status and Saved to Chron so you can review their destinations.

04 Find matches again checks routing after record corrections. Other actions contains Reject message. Administrators can expand Manage retained email below the reader to purge the retained source after confirmation.

Related chapters: [Booking pipeline \(p. 12\)](#) / [Engagements \(p. 19\)](#) / [Mattermost \(p. 82\)](#)

CHAPTER 09

Map and Calendar

Use the map for geographic planning, the graphical month calendar for events and due work, and private subscription links for an external calendar application.

Related chapters: [Engagements \(p. 19\)](#) / [Work queue \(p. 54\)](#) / [Profile and security \(p. 75\)](#)

Engagement Map

The initial view shows active engagements in a bounded date window. Filter by lifecycle, confirmation, and date. Pin colors show confirmation, with lifecycle shown by the outline or symbol. Select a pin for the event, organization, dates, lifecycle, and a link to details.

Use **Fit visible pins** after filtering. The companion list includes On map, Needs address, Awaiting lookup, and Not located views for the current page, with Edit location links. Find missing addresses searches the selected date and lifecycle scope; Previous and Next reach additional pages. A new address may take time to resolve.

Monthly Calendar

The Calendar page opens to the current month. Use **Previous** and **Next** to move one month at a time, or **Today** to return to the current month.

The current day has its own rose-colored cell and **Today** badge, distinct from every event and task color.

Select an event to open its details. Tasks with due dates link to task work; editors and administrators return to the same calendar view after editing.

Private Calendar

Create a separate subscription for each device or service. Choose Events, Presentations, My Active Work, All Active Work, and/or Birthdays (from Contacts). Copy the private URL when it is shown; it is shown only once.

Revoke one link without affecting the others; revoked records can be purged. A link grants access to its selected feed content, including work when selected. It does not grant application access to internal Chron or financial records.

Choose What the Month Calendar Shows

Selector	Calendar Content	Color Coding
Events	Engagements that overlap the displayed dates.	Event color, with distinct border tones for tentative/postponed, canceled, and completed events.
My Tasks	Your active tasks that have a due date in the displayed calendar grid.	Your task accent color.
All Tasks	Every active due-dated task in the grid, including unassigned work.	Your tasks use the accent color; other or unassigned tasks use a contrasting color.
Birthdays	Annual birthday reminders from active contacts.	Birthday styling; open the contact for details.
Everything	Events, all active due-dated tasks, and contact birthdays.	All event and task colors appear together.

The selected filter remains active as you move between months. The summary above the grid reports how many matching events and tasks fall in that month.

Choose the content of each private feed

Each new subscription has a Device or Service label and one or more content options: Events, Presentations, My Active Work, All Active Work, and Birthdays (from Contacts). The defaults are Events, Presentations, and Birthdays.

My Active Work includes assigned work; All Active Work includes everyone and unassigned work. Active work appears on its due date. A subscription can therefore reveal task titles and schedule context: share it only with the intended device or service.

The private URL is shown only once. Save it immediately or create a new subscription if it is lost. Revoke one link without changing others, and purge revoked token records when no longer needed.

The in-app month selector and a private feed are separate controls. Changing the month selector does not change the saved subscription content.

Correct a map location

Use Edit location to correct the written event address. If the address is right but the lookup pin is wrong, open the pin editor, choose the intended coordinates, and save the manual confirmation.

A manual override is tied to the address it confirms. Check it again after changing the address. Clear the override to return to normal address lookup. The map/list can still show an event awaiting lookup or without enough address information.

Birthdays and the mobile daily agenda

The month selector includes Birthdays. Everything includes events, active due-dated work, and birthdays. Enter contact birthdays as MM/DD; a birth year is not collected. February 29 birthdays recur on the last day of February in non-leap years.

On a phone-sized display the calendar becomes a daily agenda. Use the previous/next day controls or Today, keep the desired content selector, and open the linked record. Desktop uses the month grid. The agenda and month view preserve the chosen content scope.

CHAPTER 09 / ILLUSTRATED WALKTHROUGH

Locate engagements

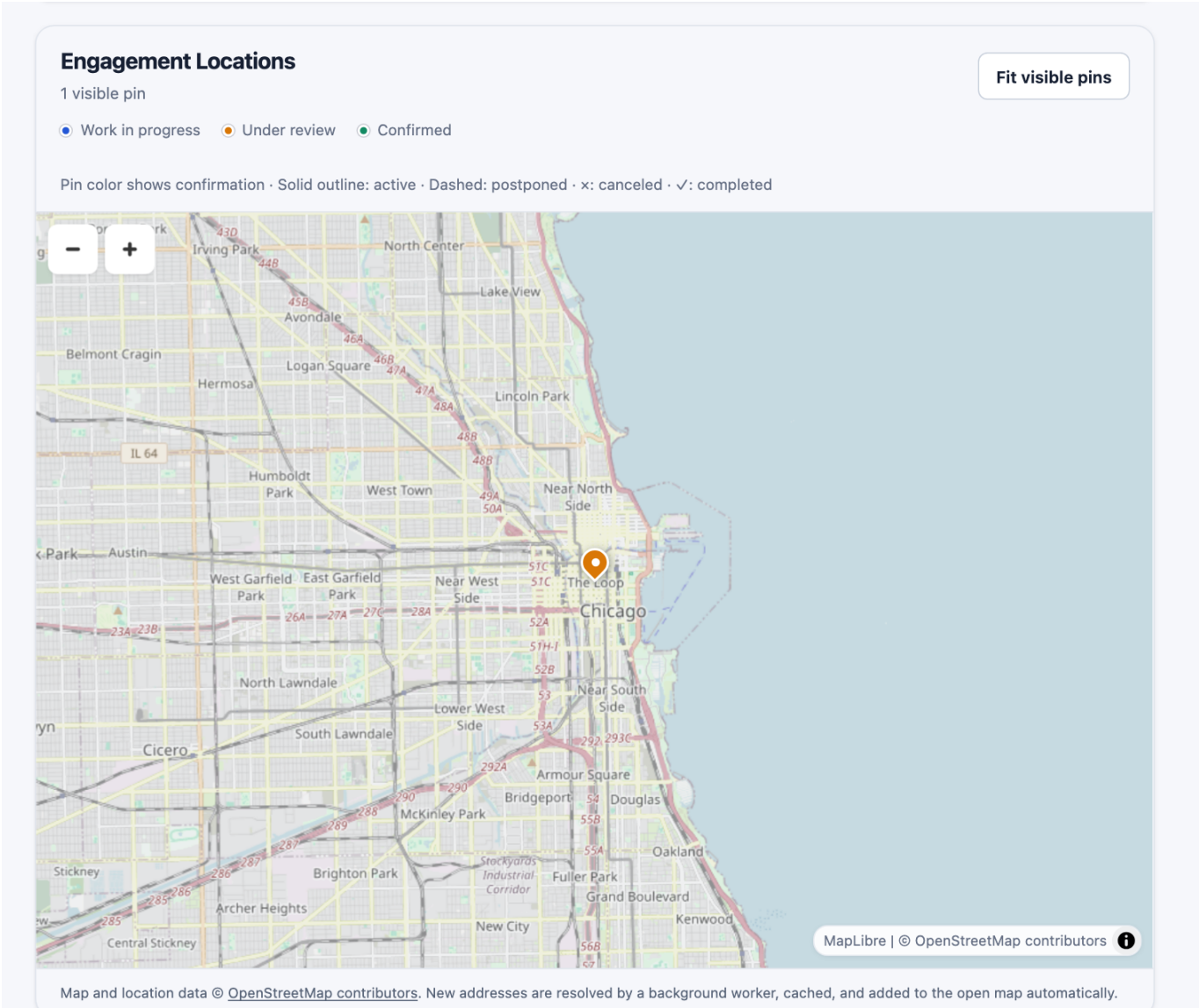


Figure 51. Locate engagements. Actual preview screen; fictional records.

- 01 Filter the date window, lifecycle, and confirmation state.
- 02 Select a pin for event context; use Fit visible pins after filtering.

- 03 Use the list views to find missing addresses, pending lookups, or unlocated records, then Edit location.
- Related chapters: [Engagements \(p. 19\)](#) / [Work queue \(p. 54\)](#) / [Profile and security \(p. 75\)](#)

CHAPTER 09 / ILLUSTRATED WALKTHROUGH

Confirm an event pin

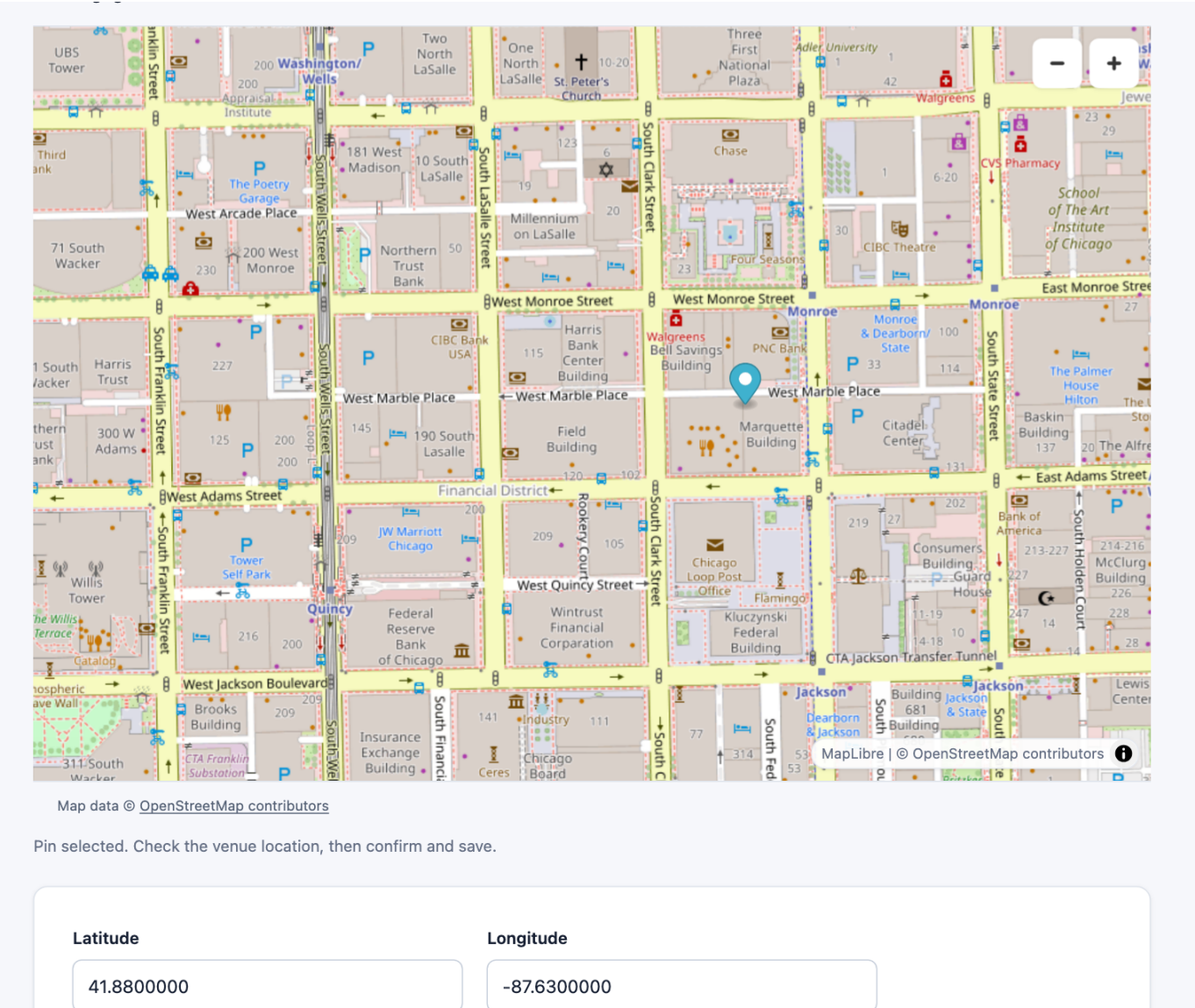


Figure 52. Confirm an event pin. Actual preview screen; fictional records.

- 01 Check the written event address before changing a pin.

02 Zoom and click the map, drag the pin, or enter latitude and longitude.
- 03 Check the venue confirmation, then Save confirmed pin.
Use automatic lookup clears an override.

04 A later address change requires checking the pin again.

Related chapters: [Engagements \(p. 19\)](#) / [Work queue \(p. 54\)](#) / [Profile and security \(p. 75\)](#)

CHAPTER 09 / ILLUSTRATED WALKTHROUGH

Browse the month calendar

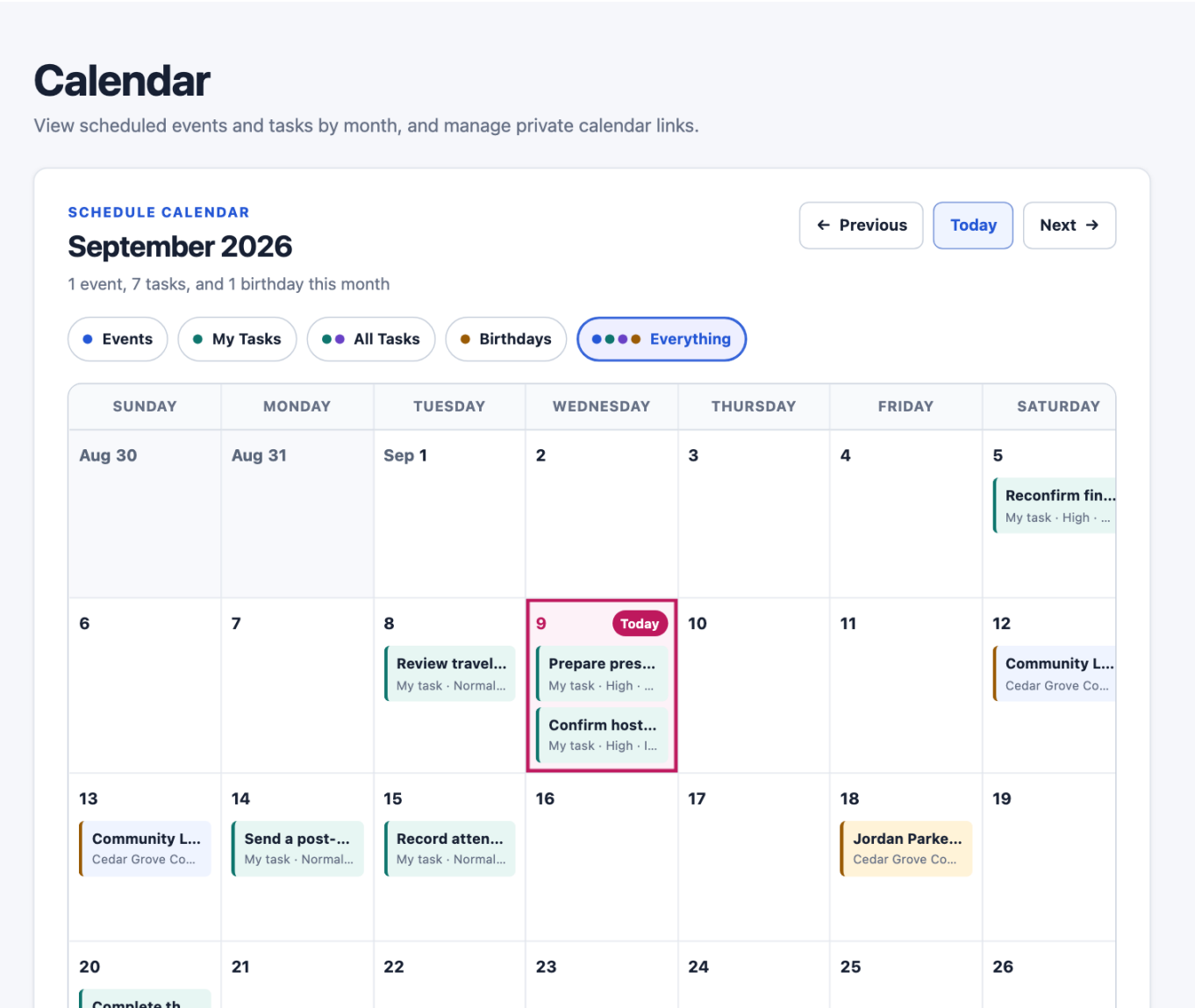


Figure 53. Browse the month calendar. Actual preview screen; fictional records.

- 01 Use Previous, Next, or Today to navigate months.
- 02 Choose Events, My Tasks, All Tasks, Birthdays, or Everything.
- 03 Tasks need an active status and a due date to appear.
- 04 Click an event, task, or birthday to open its record. The rose Today cell is separate from event/task colors.
- Related chapters: Engagements (p. 19) / Work queue (p. 54) / Profile and security (p. 75)

CHAPTER 09 / ILLUSTRATED WALKTHROUGH

Private calendar subscriptions

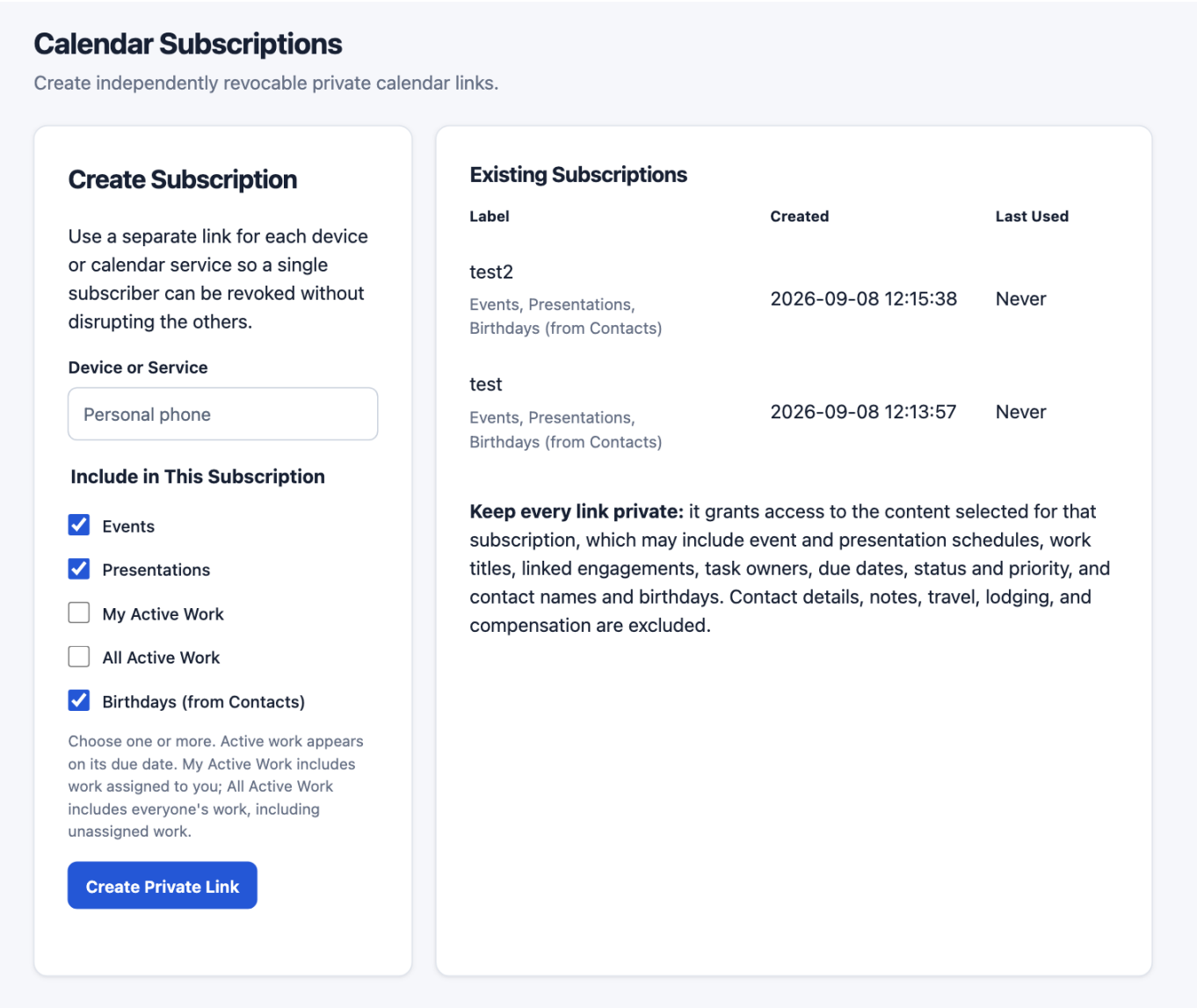


Figure 54. Private calendar subscriptions. Actual preview screen; fictional records.

- 01** Name each device or service, and choose Events, Presentations, My Active Work, All Active Work, and/or Birthdays (from Contacts).

02 Create the link, copy its private URL immediately, and add it in the calendar application as a subscribed calendar.
- 03** Treat the URL as an access credential for its selected content, including task information when work is selected.

04 Revoke one device link when no longer needed; purge revoked records separately.

Related chapters: [Engagements \(p. 19\)](#) / [Work queue \(p. 54\)](#) / [Profile and security \(p. 75\)](#)

CHAPTER 09 / ILLUSTRATED WALKTHROUGH

Use the daily agenda on a phone

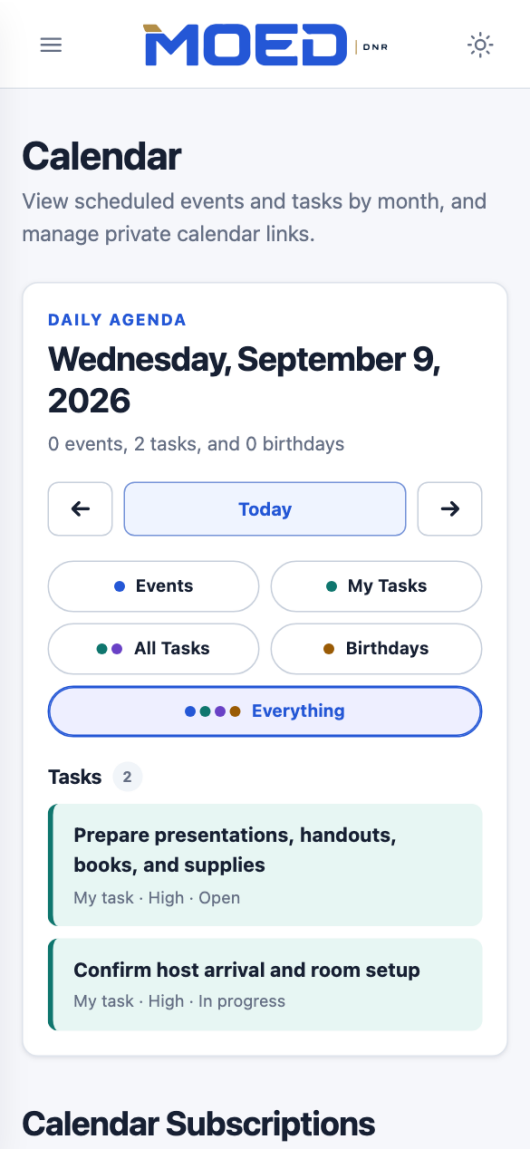


Figure 55. Use the daily agenda on a phone. Actual preview screen; fictional records.

- 01 On a narrow screen, Calendar presents a daily agenda.
- 02 Use the day arrows or Today to change dates and the content selector to focus the list.

- 03 Open an event, task, or contact birthday to continue with that record.

Related chapters: [Engagements \(p. 19\)](#) / [Work queue \(p. 54\)](#) / [Profile and security \(p. 75\)](#)

CHAPTER 10

Profile and Security

Your profile makes ownership recognizable; the security controls protect your identity and provide safe recovery paths.

Related chapters: [Roles and access \(p. 8\)](#) / [Map and calendar \(p. 68\)](#) / [Administration \(p. 87\)](#)

Profile and Notifications

Maintain your name, phone, optional profile picture, and notification schedule in My Profile. Use Change Recovery Email for a protected email change: the current address remains active until the new address is verified. Verification signs out existing sessions and pauses the daily digest. Resending verification preserves your draft and chosen photo.

A verified email enables password recovery and a Dashboard-style Daily Digest of upcoming engagements, My Work, event readiness, and financial closeouts. Overdue and due-today tasks are highlighted, and editor/admin digests also include inbound mail awaiting review. The digest defaults to enabled on weekdays at 7:00 a.m.; disable it or choose a different time and set of days at any time.

Password Rules

Passwords must contain at least 12 characters and no more than 72 UTF-8 bytes. Changing your password signs out every other session. If an administrator gives you a temporary password, you must replace it before using other areas.

Set Up Two-Factor Authentication

- 01 Confirm your password.** If replacing an existing authenticator, also enter a current code.
- 02 Add the account.** Scan the QR code in your authenticator app or enter the manual setup key.
- 03 Verify enrollment.** Enter the six-digit code generated by the app.
- 04 Save recovery codes.** Each code works once and the page will not show them again. Store them in an approved password manager or another secure location.

At sign-in, enter either the current six-digit authenticator code or one unused recovery code. Generating new recovery codes invalidates all old ones. Administrators must keep 2FA

enabled; other roles may disable it after password and current-code confirmation.

Recover Access

- **Forgotten password:** choose "Forgot your password?" on sign-in and enter the verified email on the active account. The single-use link lets you set a new password and invalidates existing sessions.
- **Lost authenticator:** use one saved recovery code at the 2FA prompt, then replace the authenticator and generate a fresh set of codes.
- **No recovery factor:** contact an administrator. They can issue a temporary password or reset another user's 2FA after elevated confirmation; they cannot reveal an old password or recovery code.
- **New account:** open the invitation link within seven days, create a private password, and complete required authenticator enrollment.

Change the recovery email securely

Use the separate Change Recovery Email panel. Enter the new address, current password, and a fresh authenticator code when 2FA is enabled, then select Verify New Email Address.

The current address stays in use while the new address is pending. Follow the verification message sent to the new address to finish. Successful verification signs out all sessions and pauses the daily digest. Sign in again and review notification preferences.

Cancel Pending Email Change abandons a pending request. Resend email verification requests a new message while preserving the profile draft and chosen photo. Editing the ordinary profile form does not directly replace the recovery destination.

CHAPTER 10 / ILLUSTRATED WALKTHROUGH

Profile and daily work digest

My Profile

Manage your personal details and profile picture.

Profile Picture

Choose a new picture

Choose File

No file chosen

JPEG, PNG, or WebP. Maximum file size: 5 MB.

Personal Details

First name

Last name

Email address

Phone number

+1

(555) 555-0123

Username

previewadmin012

Role

Admin

Notifications

☒ Daily work digest

Email me on the selected days with a Dashboard-style snapshot of upcoming engagements, My Work, event readiness, and financial closeouts, plus inbound mail awaiting review. Overdue and due-today tasks are highlighted

Digest delivery is off. Your previous schedule is kept for when you turn it on.

Verify your email address to enable daily digests.

Figure 56. Profile and daily work digest. Actual preview screen; fictional records.

- 01 Maintain name, email, phone, and optional picture.

02 Verify an email change to re-enable recovery and the digest.
- 03 Choose digest time and days, or turn the digest off.

04 Resending verification preserves the current profile draft and chosen photo.
- Related chapters:

Roles and access (p. 8)

Map and calendar (p. 68)

Administration (p. 87)

CHAPTER 10 / ILLUSTRATED WALKTHROUGH

Manage account security

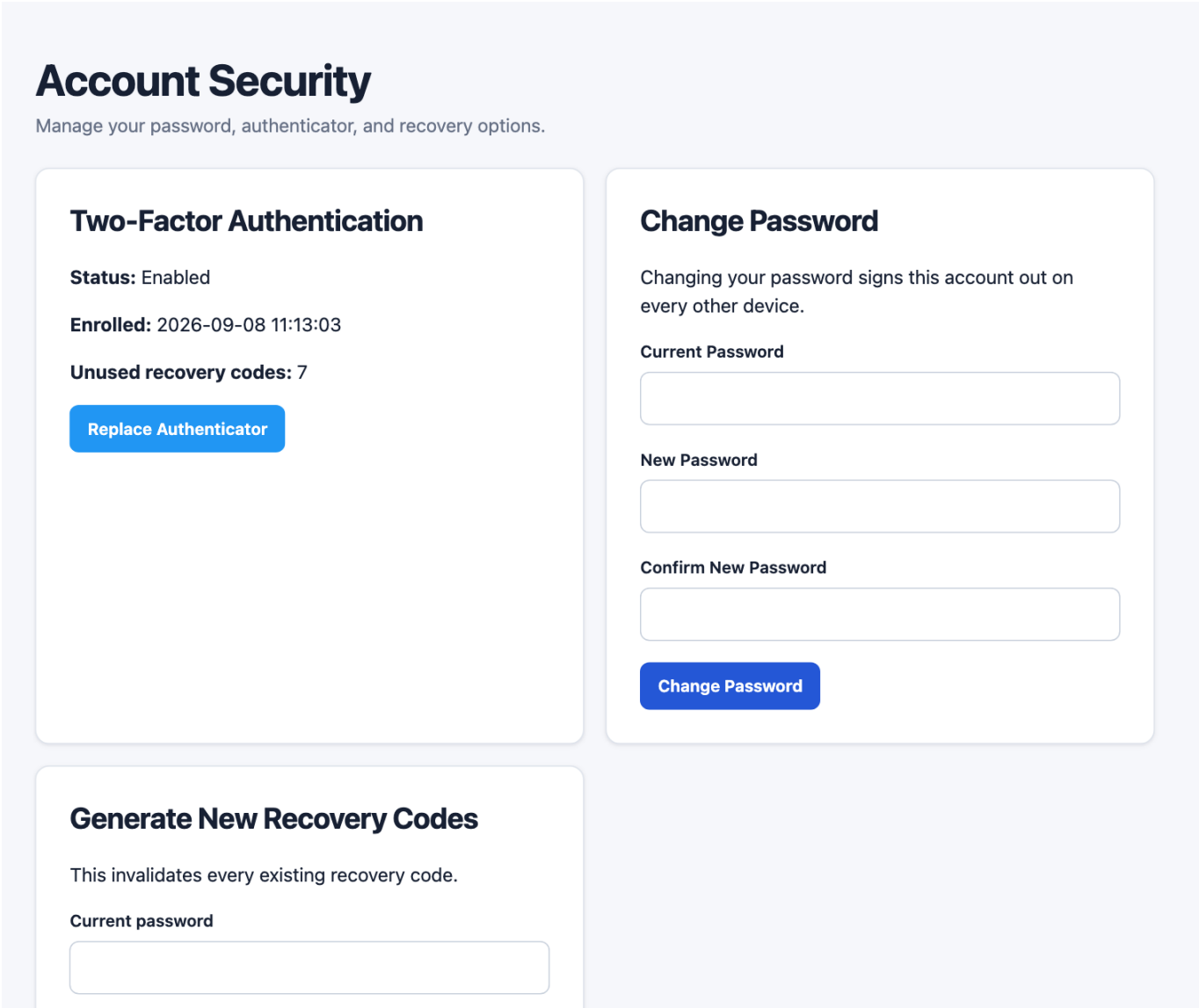


Figure 57. Manage account security. Actual preview screen; fictional records.

- 01** Enroll or replace the authenticator as needed.
Administrators must use 2FA.

02 Change your password with the current password and required factor.
- 03** Generate fresh recovery codes when needed; save them securely because each works once.

04 Administrator resets and password changes can invalidate other sessions.

Related chapters: [Roles and access \(p. 8\)](#) / [Map and calendar \(p. 68\)](#) / [Administration \(p. 87\)](#)

Confirm your password and current authenticator code before replacing the enrolled authenticator.

Current password

Current authenticator code

Continue

τ_{yin} = appointment, appointed time

Figure 58. Start authenticator replacement. Actual preview screen; fictional records.

- 01** Confirm your password and, when replacing an enrolled factor, your current authentication code.
 - 02** Scan the enrollment QR code in an authenticator or enter its setup key.
 - 03** Enter the generated six-digit code to verify enrollment, then store the recovery codes.

Screenshot context: Only the initial confirmation screen is illustrated. Enrollment secrets and recovery codes are intentionally absent.

Related chapters: [Roles and access \(p. 8\)](#) / [Map and calendar \(p. 68\)](#) / [Administration \(p. 87\)](#)

CHAPTER 10 / ILLUSTRATED WALKTHROUGH

Verify a new recovery email

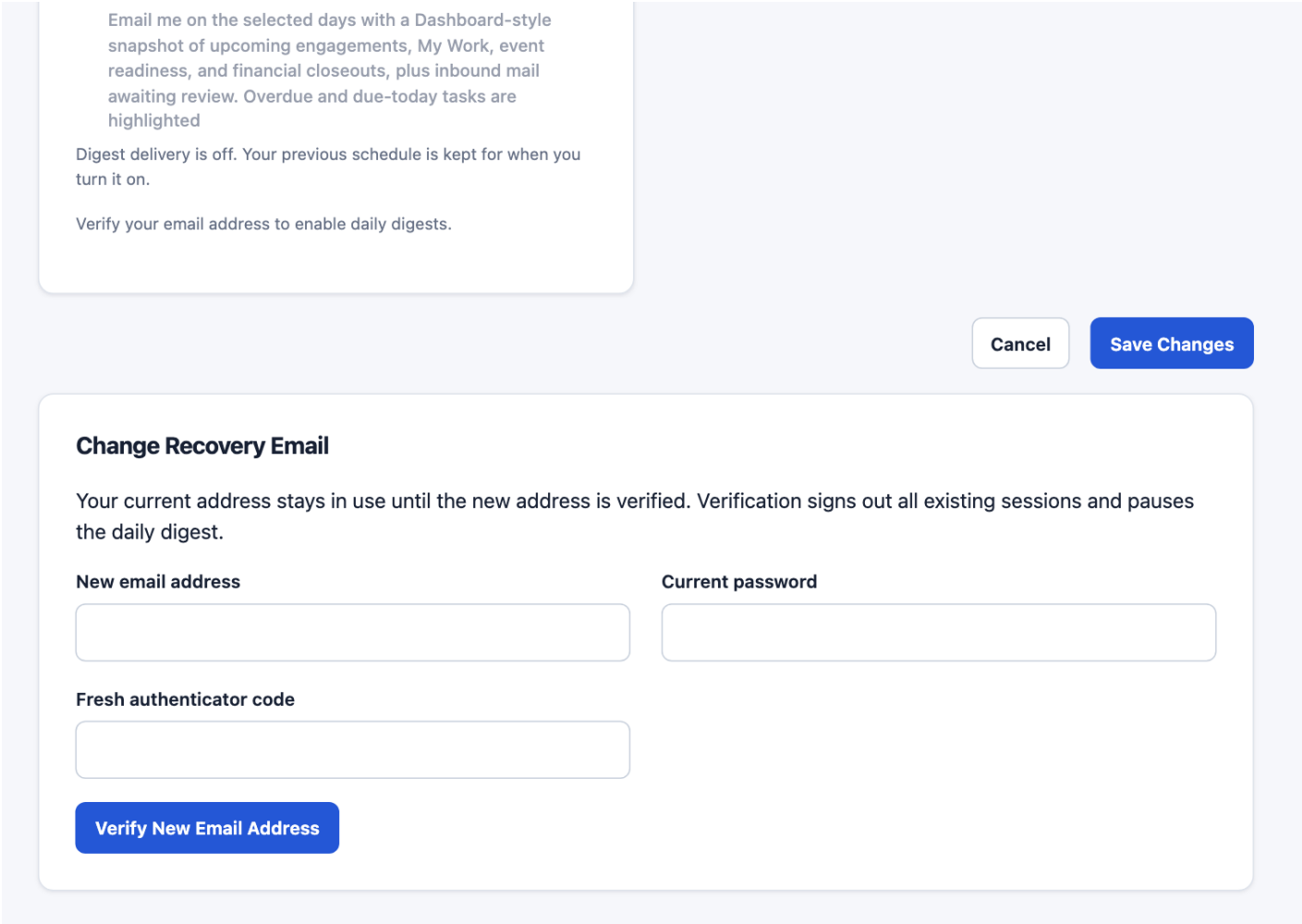


Figure 59. Verify a new recovery email. Actual preview screen; fictional records.

- 01** Open Change Recovery Email and enter the new email, current password, and a fresh authenticator code when required.

02 Select Verify New Email Address, then follow the verification message at the new address.
- 03** Your old address stays active until verification. The change signs out all sessions and pauses the digest.

04 Cancel a pending change or resend verification from My Profile if needed.

Related chapters: [Roles and access \(p. 8\)](#) / [Map and calendar \(p. 68\)](#) / [Administration \(p. 87\)](#)

CHAPTER 10 / ILLUSTRATED WALKTHROUGH

Sign in to MOED

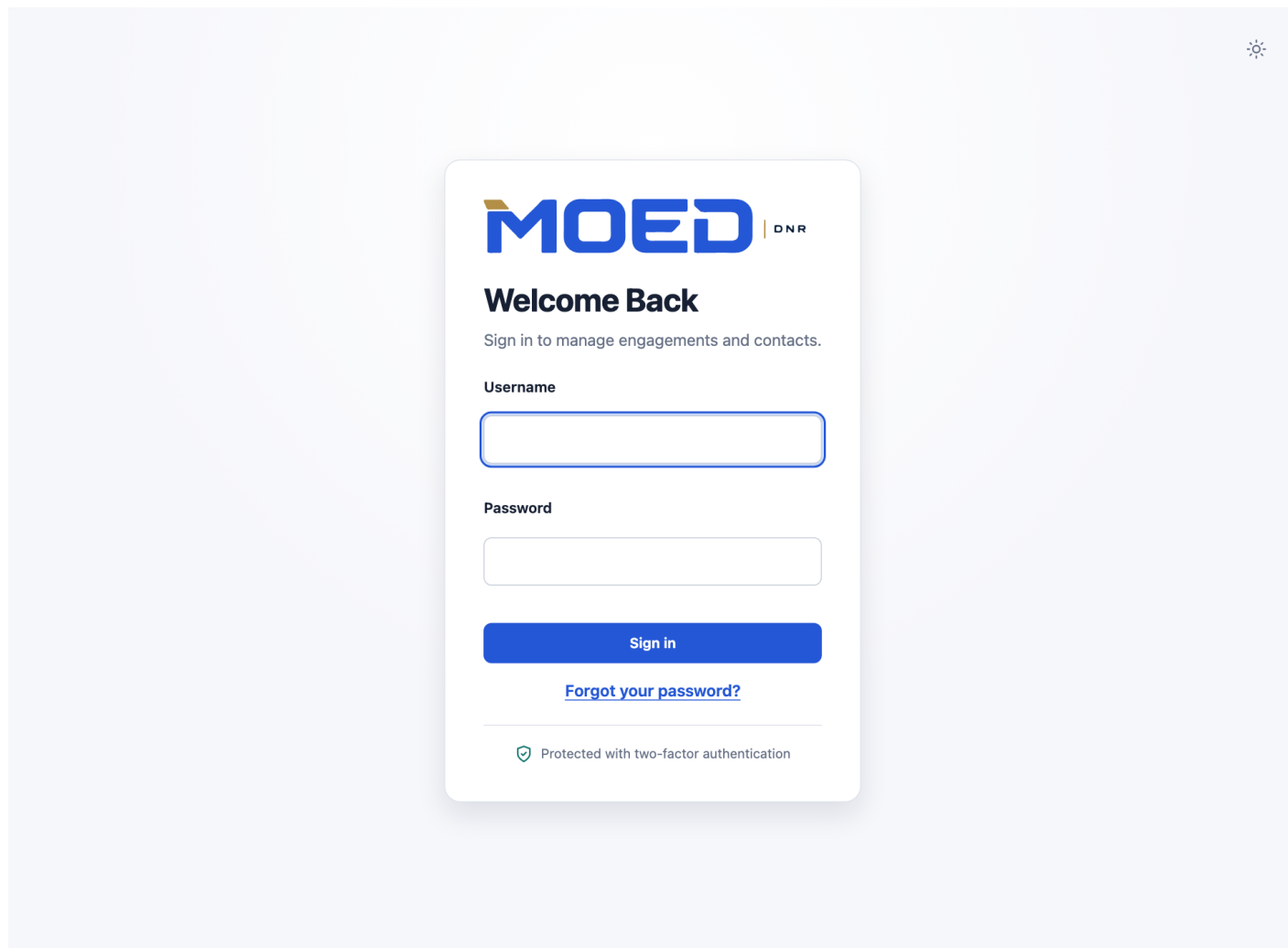


Figure 60. Sign in to MOED. Actual preview screen; fictional records.

01 Enter the username and your private password.

02 Complete the authenticator or recovery-code challenge if your account uses 2FA.

03 If prompted to change a temporary password or enroll an authenticator, complete that step before continuing.

04 Use the account menu to sign out when finished.

Related chapters: [Roles and access \(p. 8\)](#) / [Map and calendar \(p. 68\)](#) / [Administration \(p. 87\)](#)

CHAPTER 10 / ILLUSTRATED WALKTHROUGH

Request password recovery

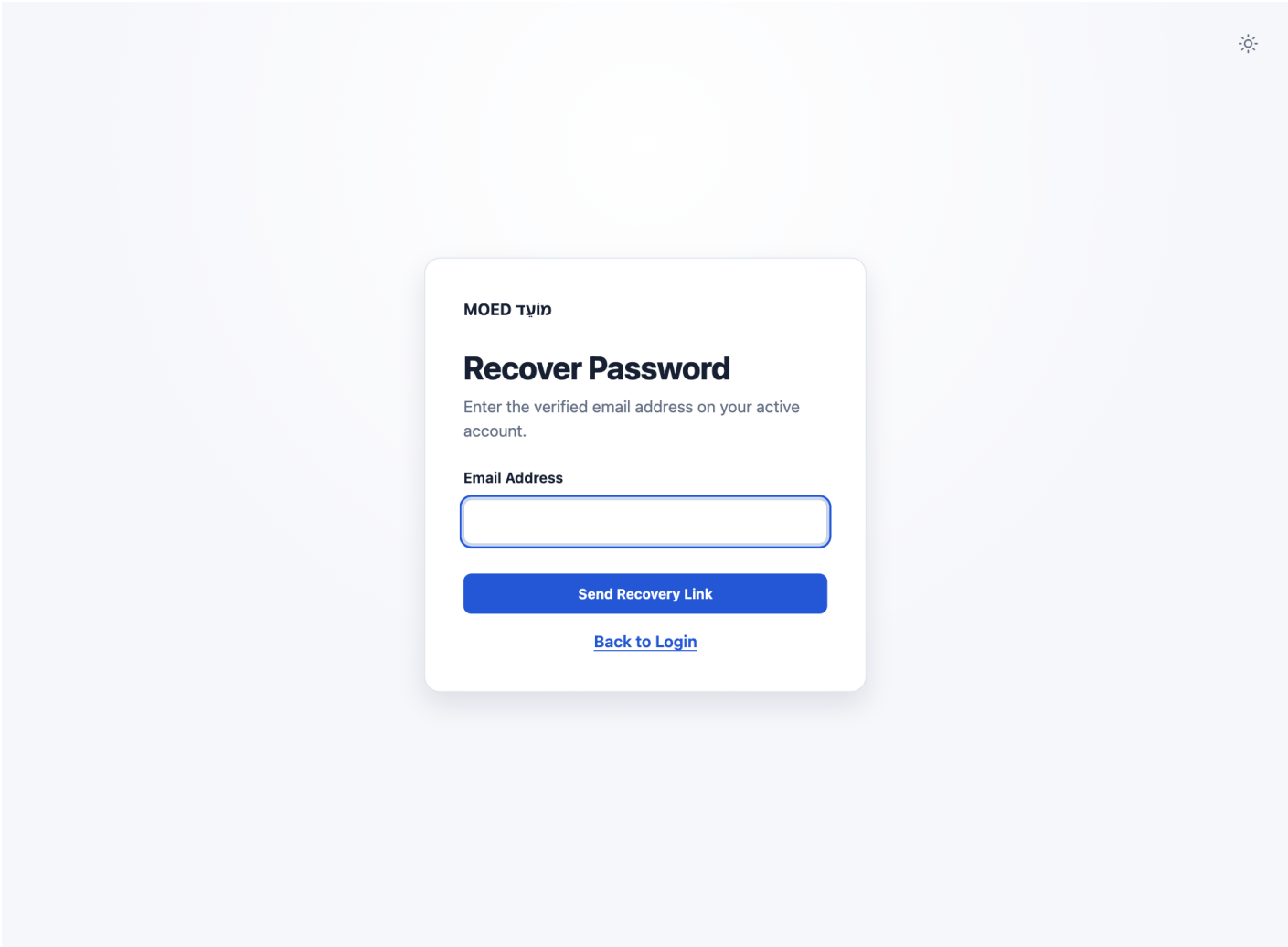


Figure 61. Request password recovery. Actual preview screen; fictional records.

- 01** Choose Forgot your password? on sign-in.

02 Enter the verified email address on the active account and request a reset link.
- 03** Follow the single-use link to choose a new password. Existing sessions are invalidated.

04 If no verified email or recovery factor is available, contact an administrator.

Related chapters: [Roles and access \(p. 8\)](#) / [Map and calendar \(p. 68\)](#) / [Administration \(p. 87\)](#)

CHAPTER 11

Mattermost

The MOED plugin brings daily summaries, engagement context, follow-up actions, and carefully reviewed engagement email into Mattermost while MOED remains the system of record.

Related chapters: [Engagements \(p. 19\)](#) / [Work queue \(p. 54\)](#) / [Chron and email \(p. 61\)](#)

Link Your Account First

Open Mattermost in MOED and generate a one-time code. The code is shown once, expires after 10 minutes, and becomes invalid after use. In Mattermost, run `/moed connect CODE`. Command responses are private to you. Never paste the code into an ordinary channel message.

Commands

Command	What It Does	Who Can Use It
<code>/moed help</code>	Shows the command reference and account-linking reminder.	Everyone
<code>/moed status</code>	Checks the plugin connection and reports which MOED account is linked.	Everyone
<code>/moed connect CODE</code>	Consumes a one-time code from the MOED Mattermost page and links your identities.	Everyone
<code>/moed today</code>	Shows a private dashboard with Overdue, Due today, Next 7 days, and Waiting counts, active assigned tasks, and a link back to MOED.	Linked users
<code>/moed tasks</code>	Shows the same task dashboard with up to 15 active tasks assigned to you. Editors and administrators also receive permitted action buttons.	Linked users
<code>/moed event search TEXT</code>	Searches active and historical non-archived engagements by event or organization name.	Linked users
<code>/moed event show ID</code>	Shows a private engagement card with dates, status, location, work counts, and an email routing marker you can copy.	Linked users
<code>/moed link-event ID</code>	Binds the current channel to an engagement, adds its routing marker to the channel name, and posts its card.	Editors and administrators with native channel-management and posting permission
<code>/moed unlink-event</code>	Removes the channel's engagement binding and its channel-name marker.	Editors and administrators with native channel-management and posting permission

Engagement Cards

Run `/moed event show ID` for a private engagement card. Linking a channel posts the same useful engagement context for the channel. Both cards show the email routing marker with a **Copy** control so it can be kept in a related email subject or message. Use **Open in MOED** for the complete record.

Task Buttons

Editors and administrators can assign an unowned task to themselves, start it, complete it, or reopen it. Each button checks the task version before changing MOED; if someone already changed the task, refresh with `/moed tasks`. Waiting details, reassignment to someone else, editing, cancellation, and deletion remain in MOED.

Recognize the Engagement Linked to a Channel

On desktop and mobile, a linked channel's sidebar label begins with a compact marker such as `[MOED#17]`, leaving more room for the channel title. The full signed routing marker remains available in engagement cards and email tools wherever it is needed for routing. The colored chain control provides the same link signal; select it, or choose **MOED engagement** from the channel menu, to see the engagement, open it in MOED, copy its routing marker, or begin an email. Unlinking removes the marker from the channel name.

Send Engagement Email From Mattermost

- 01 Start from the right engagement.** In a linked channel, select the MOED chain icon and choose **Send MOED email**. To include a particular conversation, open that post's **Message actions** and choose **Send via MOED email**.
- 02 Choose a starting point.** Select an active template from the shared Email Templates library, or Custom message, then choose from the contacts already assigned to the engagement. To include an assigned speaker, use the engagement email composer in MOED. Each unique address receives a separate message.
- 03 Decide what to include.** Edit the subject and message. You may append the share-safe event brief and, when you started from a post, the selected post or a short thread excerpt.
- 04 Review before sending.** Check the recipients, routing marker, subject, message, and any added context on the review screen. Nothing is sent until you choose **Send and add to Chron**.
- 05 Confirm the result.** The form shows whether each delivery is pending, sent, or failed and links to the full delivery record in MOED. The message is also recorded in the Engagement, Organization, and selected Contact Chron logs.
- 06 Watch for replies.** When a marked reply is routed back to the engagement, the MOED bot privately notifies the Mattermost user who most recently emailed that person about the engagement. Open the notification to continue in the engagement Chron.

Turn a Mattermost Post Into MOED Work

- 01 Link the channel.** An editor or administrator runs `/moed link-event ID`. The link establishes which engagement receives work from that channel.
- 02 Open Message actions.** Hover over a channel post and choose **Message actions** using the grid/apps icon - not the three-dot menu. Then choose **Add MOED task**, **Add to MOED Chron**, or **Send via MOED email**.
- 03 Review before saving or sending.** Confirm the task, Chron entry, or email details in the form. Nothing changes in MOED and no email is sent until you submit the final review.
- 04 Continue in MOED.** The resulting task, Chron entry, or email delivery record appears on the linked engagement, with the selected Mattermost context when you chose to include it.

Create Follow-Up Work

Add MOED task creates follow-up work on the engagement linked to the channel. MOED suggests a title and notes from the post. Review them, choose a due date and priority if needed, then save. The task appears with the engagement's other work in MOED.

Preserve Context as History

Add to MOED Chron adds the post to the linked engagement's history. Review or edit the suggested entry, then choose **Save to Chron**. Use this for decisions, outcomes, commitments, and useful conversation context that should remain with the engagement.

Continue the Conversation

Send via MOED email creates a reviewed message to assigned engagement contacts. You can include the selected post or a short thread excerpt, and MOED keeps the routing marker, delivery result, and Chron history together.

Read the Receipt Reactions

- **Chron Chron receipt:** the MOED bot adds the `:memo:` reaction after **Add to MOED Chron** successfully writes the selected post to the linked engagement.
- **Email Email receipt:** the MOED bot adds the `:email:` reaction when a MOED email includes the selected post or thread and at least one recipient delivery succeeds.

- **Both may appear:** the reactions are independent, so one post can show both. They are visual receipts rather than counters; MOED remains the authoritative record.
- **New actions only:** reactions appear for Chron saves and email deliveries completed after this feature is deployed. Earlier Chron entries and emails are not marked retroactively.

Before You Save

Task, Chron, and email actions all show what will be saved or sent. They are available in channels linked to a MOED engagement when your account has permission. Opening, reviewing, or canceling a form changes nothing.

Permissions, Privacy, and Unlinking

- **MOED decides access:** linking does not grant a new role. Reviewers stay read only; editors and administrators receive only the task actions their MOED role permits.
- **Private by default:** personal summaries, searches, email forms, delivery results, reply alerts, errors, and action confirmations are visible only to the requesting user. [link-event](#) deliberately posts a channel-visible engagement card, including its routing marker and Copy control.
- **MOED is authoritative:** task, Chron, and email actions use MOED's permissions, delivery service, and records. The plugin does not connect to the MOED database or keep a second copy of engagement email.
- **Post actions are scoped:** the plugin verifies that you can read the selected post and obtains the engagement from the channel's link. MOED permits task, Chron, and email actions only for linked editors and administrators.
- **Remove access:** open [Mattermost](#) in MOED and choose **Unlink**. The Mattermost account immediately loses command access until it is linked again.

CHAPTER 11 / ILLUSTRATED WALKTHROUGH

Connect MOED and Mattermost

Mattermost

Link your Mattermost identity to use MOED summaries, engagement cards, and follow-up actions in chat.

Integration Not Configured

An administrator must install the Mattermost plugin and configure the matching MOED service secret before accounts can be linked.

Linked Mattermost Accounts

No Mattermost account is linked.

Available Commands

`/moed today` shows your daily summary; `/moed tasks` shows assigned work; and `/moed event search TEXT` or `/moed event show ID` finds safe engagement details.

Editors and administrators can bind a channel with `/moed link-event ID`, remove that binding with `/moed unlink-event`, and use task action buttons. Reviewers keep read-only access.

[Read the Mattermost User Guide](#)

Figure 62. Connect MOED and Mattermost. Actual preview screen; fictional records.

01 When configured, generate a one-time linking code here and run `/moed connect CODE` in Mattermost.

02 Use `/moed today`, `/moed tasks`, and event search/show for private context.

03 Editors and administrators may link channels and review task, Chron, and email actions.

04 Unlink an account here to revoke its Mattermost command access.

Screenshot context: The disposable preview is not connected to Mattermost. The chapter explains the configured plugin workflow; this screen shows its setup prerequisite.

Related chapters: [Engagements \(p. 19\)](#) / [Work queue \(p. 54\)](#) / [Chron and email \(p. 61\)](#)

CHAPTER 12

Administration

Administrator tools control identity, accountability, continuity, and deployment health. Use them deliberately and keep a second administrator available.

Related chapters: [Roles and access \(p. 8\)](#) / [Profile and security \(p. 75\)](#) / [Troubleshooting \(p. 97\)](#)

Unlock Sensitive Actions

On Users - or when a destructive action redirects for confirmation - enter the administrator password plus a fresh authenticator or recovery code. Elevation lasts five minutes, and some lifecycle actions consume it immediately. Locked user controls remain hidden until confirmation succeeds.

Users

Invite a username, verified-on-acceptance email, and Reviewer, Editor, or Admin role. Review account status, profile, email verification, 2FA, password-change requirement, and activity timestamps.

Elevated actions can resend invitations, edit username, role, and daily work digest settings, set a temporary password, reset another user's 2FA, deactivate/activate, or delete an invited or inactive account. Deactivation revokes sessions and calendar links and unassigns tasks; activation does not restore those links or assignments.

Audit Log

Inspect login, security, and database-change activity with actor, affected record, IP address, and local/UTC time. Filter by category, text, date range, and exact IP. Entries are newest first and append-only during normal use; individual records cannot be edited or deleted.

In the Retention panel, administrators choose how many days to keep and preview the exact UTC cutoff and affected count. Pruning requires freshly confirmed administrator access, the literal **PRUNE** phrase, and a final themed in-app confirmation. Deployment operators can use `./scripts/prune_audit_log.sh DAYS` for the equivalent preview-first terminal workflow. Either path records the completed operation in the log.

Encrypted Backup

Confirm the administrator password and a fresh factor, then choose and confirm a unique backup password of at least 16 characters. MOED downloads the encrypted snapshot as a `.dnrbackup` file.

Keep the file, its password, and the separate DNR 2FA encryption key securely backed up. None can be recovered from the others. Database restore is intentionally a deployment-host procedure, not a web action.

Operations

The internal operations view summarizes task backlog, geocoding retries, inbound mail review/failures, recent authentication failures, migration readiness, and the last encrypted backup. Use it to identify operational work that needs attention before it becomes a user-facing problem.

Permanent Means Permanent

Deleting an organization preserves contacts who have another organization affiliation, and removes its remaining contacts and engagements together with their Chron, tasks, presentations, contact assignments, and financial reports. Deleting an engagement removes the same event-specific data. Deleting a contact removes that contact's Chron, tasks, and event role assignments. Active users must be deactivated before deletion; pending invitations can be deleted directly. Use archive for ordinary record retirement.

User lifecycle effects

Deactivating a user revokes their sessions and private calendar links and leaves their assigned work unassigned. Reactivating them does not recreate subscriptions or restore task ownership; review both explicitly.

An administrator can issue a temporary password or reset another user's authenticator after fresh confirmation. The old password, authenticator secret, and recovery codes cannot be displayed.

Active accounts must be deactivated before deletion. Pending invitations may be deleted directly. Account removal is distinct from archiving operational records.

Deployment notices and recovery boundaries

When a deployment is announced, finish or preserve your current work and follow the on-screen preparation/countdown message. Wait for service to return before retrying a save.

The PDF documents browser operations. Deployment, secret management, database restoration, migration repair, and infrastructure configuration belong to the deployment operator. Use the Exact database restore runbook in README.md and docs/release-workflow.md for those procedures.

The internal operations/readiness views help diagnose service state. They are not a substitute for a verified recovery backup or a successful restore check.

Reset and retire an account

Unlock sensitive actions on Users before selecting a reset or lifecycle control. Reset User Password asks for a new temporary password and confirmation. The user must replace it at sign-in.

Reset 2FA uses the explicit RESET 2FA confirmation phrase. Resetting another user's factor is an account recovery action; the old secret and recovery codes remain unavailable.

Review the deactivation confirmation before proceeding. To delete an invited or inactive account, use its Delete action and the requested confirmation phrase. Do not use deletion as the ordinary way to suspend access.

CHAPTER 12 / ILLUSTRATED WALKTHROUGH

Review users and access state

Users

Manage invitations, account access, verified recovery email, roles, passwords, and two-factor authentication.

Sensitive Administrator Actions

Confirm your password and a new authenticator or recovery code before inviting users, changing account access or roles, resetting authentication, or deleting a user. Locked controls remain hidden until elevation succeeds.

Administrator Password

Fresh Authentication Code

Unlock for Five Minutes

Audit Log

+ Invite User

P

preview007 (Editor) — Active —

2FA Not Enabled

Email: preview007@example.invalid · Verified

Phone: Not provided

Created: 2026-09-07 11:37

Last Modified: 2026-09-07 11:37

Last Logged In: 2026-09-08 06:12

Activated: 2026-09-07 11:37

P

previewadmin012 (Admin) — Active —

2FA Enabled

Email: Not provided Phone: Not provided

Created: 2026-09-08 06:13

Last Modified: 2026-09-09 14:24

Last Logged In: 2026-09-09 14:24 Activated: N/A

Figure 63. Review users and access state. Actual preview screen; fictional records.

01 Review role, activation, verified email, 2FA, and activity.

02 Invite a user with their own email and role. The invitation is valid for seven days.

03 Unlock sensitive actions with your administrator password and a fresh factor before changing another account.

Related chapters: [Roles and access \(p. 8\)](#) / [Profile and security \(p. 75\)](#) / [Troubleshooting \(p. 97\)](#)

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CHAPTER 12 / ILLUSTRATED WALKTHROUGH

Invite a new user

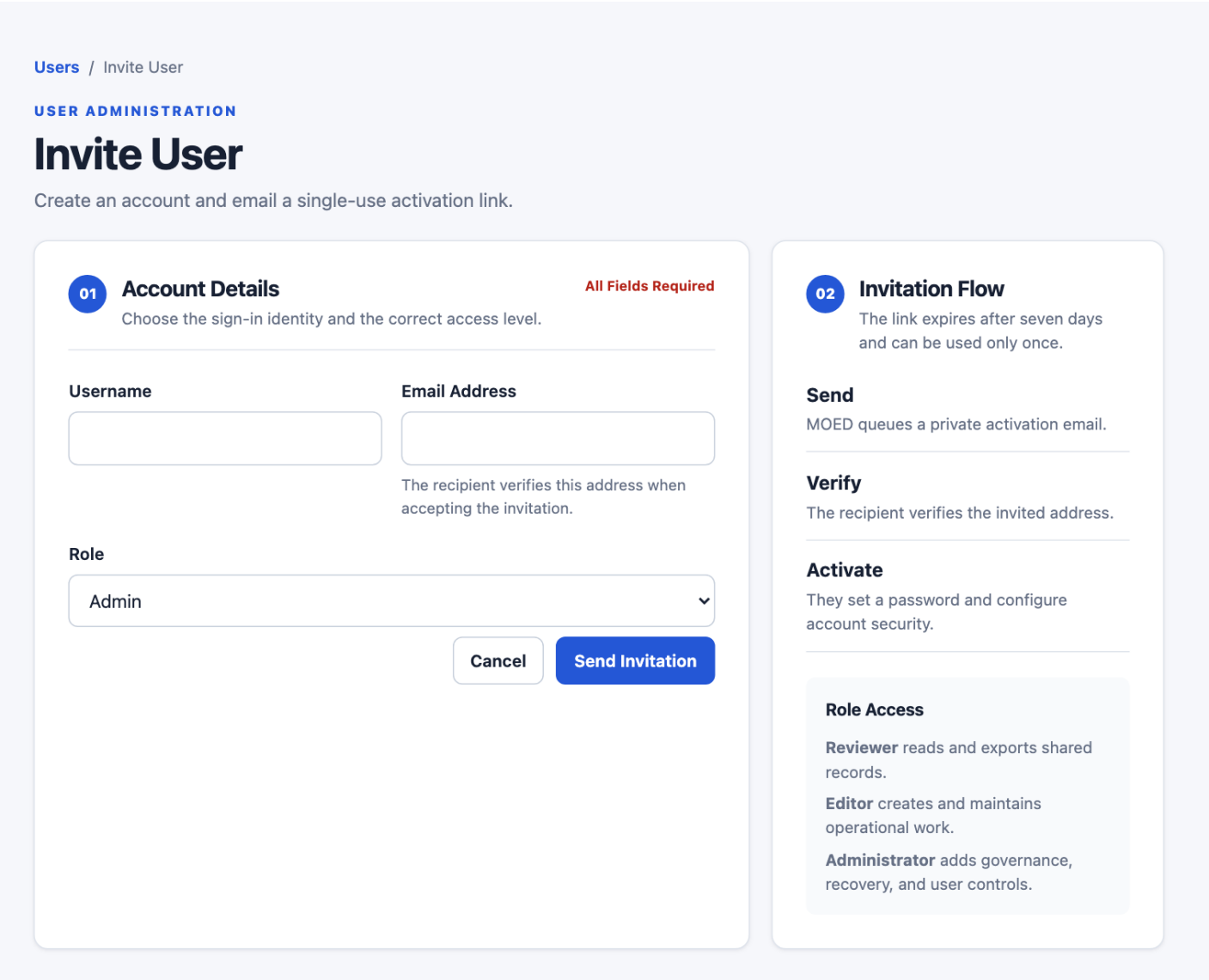


Figure 64. Invite a new user. Actual preview screen; fictional records.

- 01 Confirm administrator access, then choose a unique username, email address, and role.
- 02 Review the recipient and access level before sending the invitation.

- 03 The recipient creates their own password through the activation link. Configure digest preferences from the user editor afterward.

Screenshot context: This form is shown without submitting an invitation.

Related chapters: [Roles and access \(p. 8\)](#) / [Profile and security \(p. 75\)](#) / [Troubleshooting \(p. 97\)](#)

CHAPTER 12 / ILLUSTRATED WALKTHROUGH

Confirm administrator access

Confirm Administrator Access

Enter your current password and a fresh authenticator or recovery code. Sensitive administrator actions remain unlocked for five minutes.

Administrator password

Fresh authenticator code or recovery code

Unlock Sensitive Actions

Cancel

© 2026 beneliath · DNR 1.11.11 · 2026-09-05 14:19:58 CDT (ec7dc08)

Τῡν = appointment, appointed time

Figure 65. Confirm administrator access. Actual preview screen; fictional records.

- 01 Enter your administrator password and a fresh authenticator or recovery code.
- 02 Continue to the sensitive action and review the target.

- 03** Elevation lasts five minutes; some lifecycle operations consume it immediately.

Related chapters: Roles and access (p. 8) / Profile and security (p. 75) / Troubleshooting (p. 97)

CHAPTER 12 / ILLUSTRATED WALKTHROUGH

Inspect the audit trail

Audit Log

Login, logout, and database activity. Newest entries appear first.

Audit records remain append-only during normal application use. Administrators can preview and prune expired entries below; deployment operators can use the equivalent terminal command.

Back to Users

RETENTION

Fresh confirmation required to prune

Prune Old Entries

Keep the most recent number of days you choose and permanently delete only older entries.

Days to keep

365

Preview Pruning

Search audit log

mm/dd/yyyy

mm/dd/yyyy

Exact IP

Apply

All

Login

Database

Security

Records per page:

2050100

< Previous

1

2

3

4

...

27

Next >

Showing 1–20 of 525 entries

TIMESTAMP	CATEGORY	USER	EVENT	RECORD	IP ADDRESS
2026-09-09 14:36:42 CDT 2026-09-09 19:36:42 UTC	Database	System	Created	User: casey.manual (#24)	—

Figure 66. Inspect the audit trail. Actual preview screen; fictional records.

- 01

Filter by category, search text, date range, and exact IP address.
- 02

Review actor, action, affected record, and local/UTC timestamps.
- 03

Individual audit records cannot be edited or removed through ordinary record controls.

Related chapters: [Roles and access \(p. 8\)](#) / [Profile and security \(p. 75\)](#) / [Troubleshooting \(p. 97\)](#)

CHAPTER 12 / ILLUSTRATED WALKTHROUGH

Preview audit retention

RETENTION

Fresh confirmation required to prune

Prune Old Entries

Keep the most recent number of days you choose and permanently delete only older entries.

Days to keep

365

Preview Pruning

Search audit log

mm/dd/yyyy

mm/dd/yyyy

Exact IP

Apply

All

Login

Database

Security

Records per page: 2050100

< Previous

1

2

3

4

...

27

Next >

Showing 1–20 of 525 entries

TIMESTAMP	CATEGORY	USER	EVENT	RECORD	IP ADDRESS
2026-09-09 14:36:42 CDT 2026-09-09 19:36:42 UTC	Database	System	Created	User: casey.manual (#24)	—
2026-09-09 14:36:42 CDT 2026-09-09 19:36:42 UTC	Database	System	Created	Engagement_financial_report: Engagement 14 financial report (#14)	—
2026-09-09 14:36:42 CDT 2026-09-09 19:36:42 UTC	Database	System	Created	Engagement: Summer Learning Day - Example (#14)	—
2026-09-09 14:36:42 CDT 2026-09-09 19:36:42 UTC	Database	System	Updated	Contact: Jordan Parker (#54)	—

Figure 67. Preview audit retention. Actual preview screen; fictional records.

- 01

Open the retention panel and choose the number of days to keep.
- 02

Preview the UTC cutoff and affected count before pruning.
- 03

Fresh administrator confirmation, the exact PRUNE phrase, and the final in-app confirmation are required.
- Related chapters:

Roles and access (p. 8) / Profile and security (p. 75) / Troubleshooting (p. 97)

CHAPTER 12 / ILLUSTRATED WALKTHROUGH

Export an encrypted database backup

Database Backup

Download a password-encrypted snapshot. Restore access is isolated from the web application.

Backups are encrypted with the password you choose, but should still be kept private. They contain user accounts, password hashes, encrypted authenticator secrets, recovery-code hashes, contacts, and all engagement data. A lost backup password cannot be recovered. The separate DNR two-factor encryption key is not included and must also be backed up securely.

Export Backup

Estimated backup size: 17 MB of 512 MB. This estimate uses database statistics; the actual export may be larger.

Creates a consistent snapshot of every DNR table, encrypts and authenticates the complete archive with your password, and downloads it as a `.dnrbackup` file.

Your Administrator Password

Fresh Authenticator Code or Recovery Code

New Backup Encryption Password

Confirm Backup Encryption Password

Restore Procedure

Restore is intentionally unavailable to the web process.
From the deployment host:

1. Confirm that the matching DNR release, Docker Compose, the original 2FA encryption key, all database secret files, the encrypted backup, and its exact password are available.
2. Place the backup in `backups/`, create `secrets/backup_password` with the exact password and no trailing newline, then set both files to mode `600`.
3. Create a separate root-level SQL safety dump of the current database.
4. Stop `web` and `geocoder` so no request can write during restoration.
5. Run `docker compose --profile maintenance run --rm --no-deps maintenance /backups/FILE.dnrbackup RESTORE`.
6. Run the migration command and database privilege configuration command, start `web` and `geocoder`, and wait for the web health check to pass.
7. Sign in with an account contained in the restored

Figure 68. Export an encrypted database backup. Actual preview screen; fictional records.

01 Review estimated size against the 512 MB browser export limit.

02 Provide your administrator password and fresh factor.
Choose and confirm a unique backup password of at least 16 characters.

03 Encrypt and Download Backup creates a `.dnrbackup` snapshot. Keep its password and the separate 2FA encryption key securely available.

04 Restoration is a deployment-host procedure described in the repository runbook.

Related chapters: [Roles and access \(p. 8\)](#) / [Profile and security \(p. 75\)](#) / [Troubleshooting \(p. 97\)](#)

CHAPTER 12 / ILLUSTRATED WALKTHROUGH

Review operational health

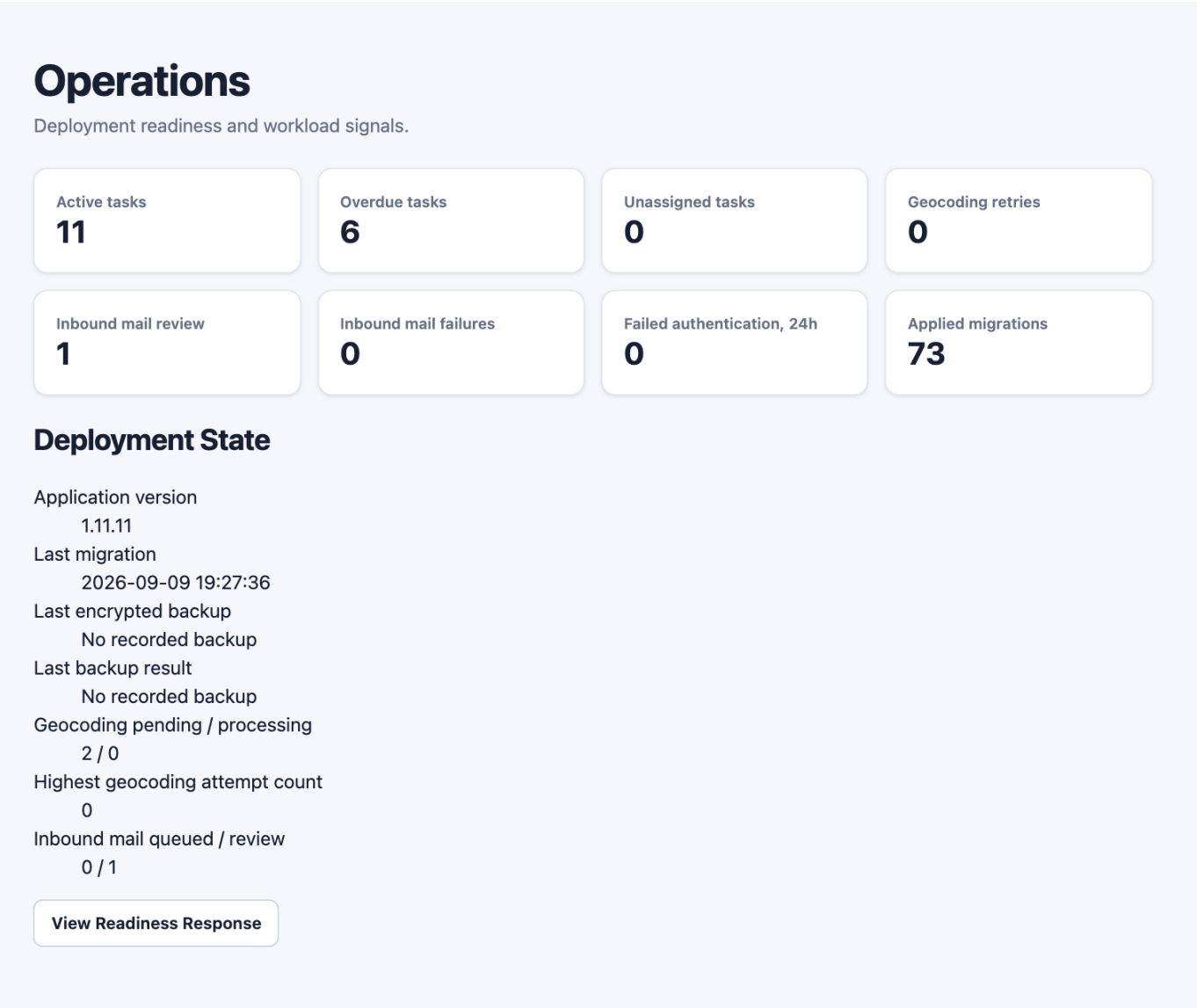


Figure 69. Review operational health. Actual preview screen; fictional records.

- 01

Inspect task backlog, geocoding queues, mail review/failures, and authentication failures.
- 02

Check migration readiness and the latest recorded encrypted backup.
- 03

Follow the linked application view for work you can resolve; give deployment faults to the operator.
- Related chapters:

[Roles and access \(p. 8\)](#) / [Profile and security \(p. 75\)](#) / [Troubleshooting \(p. 97\)](#)

CHAPTER 12 / ILLUSTRATED WALKTHROUGH

Edit role and digest preferences

Users / Edit User

Edit User

Change account access and daily work digest settings.

Username

casey.manual

Role

Editor

Daily Work Digest

☒ Enable daily work digest

Delivery requires an active account with a verified email address

Delivery time

07:00

⌚

Delivery days

Weekdays

Weekends

Every day

M

T

W

Th

F

Sa

Su

Cancel

Save changes

Figure 70. Edit role and digest preferences. Actual preview screen; fictional records.

01

After confirming administrator access, open the user editor.

02

Change username or role and review the daily digest time and days.

03

Save the changes. Recheck the intended role before granting administrator access.

Related chapters:

[Roles and access \(p. 8\)](#) / [Profile and security \(p. 75\)](#) / [Troubleshooting \(p. 97\)](#)

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CHAPTER 13

Troubleshooting and Good Practice

Most surprises come from role limits, archived parent records, delayed background work, or intentionally strict safety checks.

Related chapters: [Getting oriented \(p. 4\)](#) / [Engagements \(p. 19\)](#) / [Chron and email \(p. 61\)](#) / [Administration \(p. 87\)](#)

I Cannot See an Edit or Delete Button

Reviewers are read only. Editors can edit and archive but cannot permanently delete. Administrators must freshly unlock sensitive access before many delete and user-management controls appear. Archived records are viewed separately and generally must be restored before editing.

An Organization Will Not Archive

Archive or move every active contact and engagement belonging to it first. MOED prevents archiving the parent while active child records would become stranded.

Search Did Not Find an Engagement

Use terms of at least three characters. Remove quotation marks to match any word; quote a group when all its words must match. Check the Archived view and the lifecycle filter. Search indexes active Chron entries, not archived ones.

A New Address Has No Map Pin

Confirm that the engagement contains a usable location and fits the current lifecycle/date filters. New addresses are processed asynchronously; keep the Map page open and the pin will appear automatically when the lookup finishes. If it remains pending, ask an administrator to check the geocoding queue and provider health.

An Email Still Needs Review

Confirm the sender or participant address exactly matches one active record. Correct missing contact/organization email data, preserve a single valid `[MOED#ID.<signed-token>]` marker in the subject or plain-text body where applicable, then choose **Find matches again**, or select the intended destinations and choose **Save to Chron logs**.

An Outbound Email Failed or Cannot Be Queued

Confirm that the engagement has a selected contact or a speaker assigned to an active presentation with a valid email address. Open the outbound message to see which recipient failed. Correct the address if needed, then select **Retry Failed Deliveries**. If sending is unavailable for every message, ask an administrator to check the mail setup.

A Template Is Missing or My Changes Cannot Be Saved

Clear the library search and check **Archived**. Restore a template to make it available for new emails; permanently deleted templates cannot be restored. If another session changed the template, keep a copy of your edits, reload, and compare with the current version. Use only the listed event fields with matching double braces; Presentation schedule is allowed in the message only.

A Speaker Is Missing from Email Recipients

Check that the speaker is assigned to an active presentation on this engagement. Add or correct their email on the speaker record, then reload the composer and select the speaker. Archived presentations do not supply recipients. Putting a speaker's name in template text does not select that person as a recipient.

A Calendar Stopped Refreshing

Check whether that device's subscription is still Active. A revoked URL cannot be recovered; create a new device-specific link and replace the old subscription in the calendar application.

An Event or Task Is Missing from the Month Calendar

Confirm the displayed month and selector. **Events** does not include tasks, **My Tasks** includes only work assigned to you, and task views include only active tasks with due dates. Use **Everything** when you want events and all due-dated work together.

Changing the Caller Did Not Move Existing Tasks

The Caller is used only to choose the owner of the initial standard checklist when the engagement is created. Later Caller changes do not alter existing task ownership. If responsibility changes, reassign those tasks in the Work Queue.

A QR Code Will Not Copy

Use the PNG or SVG download links to save a generated code. You can also select the QR preview to copy it; if clipboard access is unavailable, the application opens a preview. Save the presentation first if its QR codes are not yet available.

The Speaker Notes QR Code Is Missing

Open the engagement for editing, check the speaker selected for that presentation, and choose a Speaker Notes PDF. Save the engagement before looking for the code. A code is not shown when there is no saved PDF for that speaker. If an upload is too large, reduce the PDF to 100 MB or less; when adding several files, keep the combined upload under 120 MB.

A Save Reports an Expired Request

Reload the page and repeat the change. MOED rejects stale or missing request tokens, and concurrent edits may require you to reopen the newest record before saving.

I Was Signed Out Unexpectedly

Password changes, password recovery, administrator resets, account changes, deactivation, and database restoration invalidate existing sessions by design. Sign in again with the current credentials or contact an administrator if the account is unavailable.

Record the Why

Chron should preserve the decision and context a future teammate needs, not just "called."

Name the Next Action

Turn commitments into tasks with an owner and date while the conversation is fresh.

Archive Before Deleting

Relationship history and event outcomes are usually more valuable than a perfectly tidy active list.

Protect Private Links

Calendar URLs, invitation links, verification links, recovery links, and recovery codes are credentials.

CHAPTER 13 / ILLUSTRATED WALKTHROUGH

Find help quickly

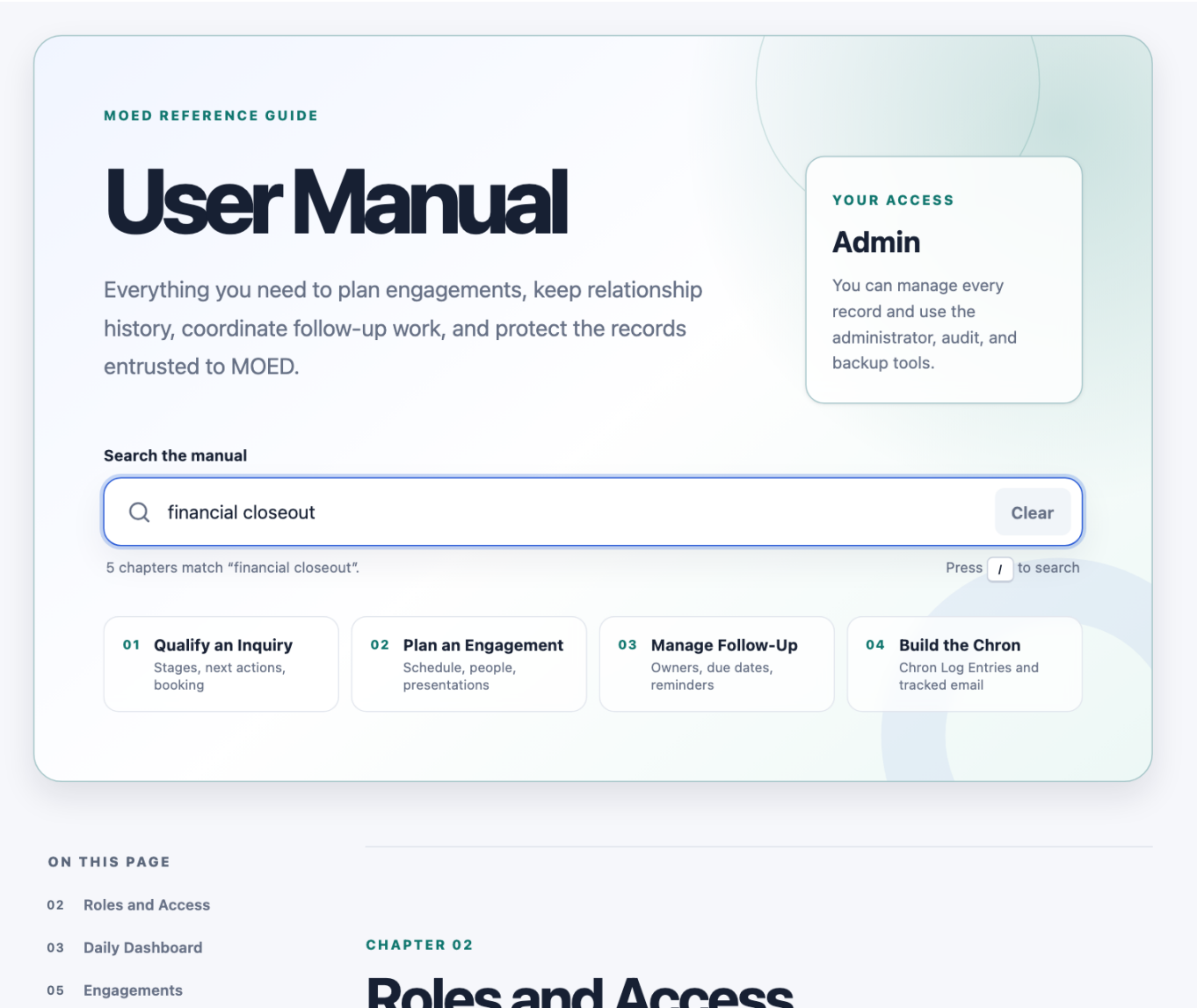


Figure 71. Find help quickly. Actual preview screen; fictional records.

01 Search the online manual for a term such as financial closeout.

02 Use Clear to return to all chapters.

03 In this PDF use the contents, bookmarks, related-topic links, topic finder, or your reader's text search.

Related chapters: [Getting oriented \(p. 4\)](#) / [Engagements \(p. 19\)](#) / [Chron and email \(p. 61\)](#) / [Administration \(p. 87\)](#)

Topic finder

Select a topic to open the relevant procedure. Use the PDF bookmarks for the full section and screenshot outline.

[Account activation and invitations](#) - p. 90

[Account recovery](#) - p. 75

[Administrator confirmation](#) - p. 91

[Archive, restore, and permanent deletion](#) - p. 22

[Audit log and retention](#) - p. 92

[Backup and restoration](#) - p. 94

[Birthdays](#) - p. 69

[Booking conversion](#) - p. 17

[Booking stages and board filters](#) - p. 13

[Calendar content selectors](#) - p. 72

[Calendar subscriptions and revocation](#) - p. 73

[Caller and generated task ownership](#) - p. 54

[Change recovery email](#) - p. 75

[Chron log entries](#) - p. 63

[Closeout prerequisites](#) - p. 33

[Confirmation vs. lifecycle](#) - p. 30

[Contact affiliations and event roles](#) - p. 50

[Contact photographs](#) - p. 50

[Countries, addresses, states, and provinces](#) - p. 47

[Daily digest](#) - p. 76

[Dashboard and readiness](#) - p. 11

[Dates, rescheduling, and task offsets](#) - p. 54

[Decline and reopen an inquiry](#) - p. 14

[Deployment notices](#) - p. 88

[Duplicate a task](#) - p. 58

[Email delivery and retry](#) - p. 62

[Email recipients and templates](#) - p. 64

[Email routing and signed markers](#) - p. 67

[Email templates: add and edit](#) - p. 35

[Email templates: archive, restore, and delete](#) - p. 34

[Event contacts](#) - p. 26

[Event fields in email templates](#) - p. 36

[Event logistics and planning estimates](#) - p. 29

[Financial drafts and final reports](#) - p. 33

[Financial history by organization](#) - p. 46

[Inbound mail triage](#) - p. 66

[Inquiry correspondence](#) - p. 18

[Keyboard navigation](#) - p. 4

[Manual map pin](#) - p. 71

[Map and missing addresses](#) - p. 70

[Mattermost commands and account linking](#) - p. 82

[Mattermost message actions](#) - p. 84

[Mobile sidebar and theme](#) - p. 7

[Next action on an inquiry](#) - p. 16

[Operations and readiness](#) - p. 95

[Passwords and 2FA](#) - p. 77

[PDF speaker notes](#) - p. 28

[Presentation export: text, Markdown, PDF](#) - p. 27

[Presentation schedule and attendance](#) - p. 28

[QR code resources and downloads](#) - p. 27

[QR code statistics and link controls](#) - p. 31

[Recovery codes](#) - p. 75

[Roles: reviewer, editor, administrator](#) - p. 8

[Search and records per page](#) - p. 4

[Speaker biography and links](#) - p. 52

[Speaker custom resource links](#) - p. 53

[Speaker email recipients](#) - p. 64

[Standard event checklists](#) - p. 59

[Task status, ownership, priority, and due date](#) - p. 57

[Troubleshooting](#) - p. 97

[User deactivation and deletion](#) - p. 87

[Work queue and reminders](#) - p. 56

Edition and source notes

Application source: MOED / DNR 2.0.8. Guide updated: September 10, 2026. Inbox, engagement email, and email template screenshots were refreshed for this edition; other screenshots were captured on September 9, 2026. Source basis: src/help.php, application page/forms/controllers, and the Mattermost integration guide.

The PDF expands the online guide with current private-calendar content options, birthdays, recovery-email verification, map-pin correction, and additional form walkthroughs. Infrastructure-only procedures are referenced at the point where a deployment operator is required.

Document build and screenshots: scripts/manual/ and docs/user-manual/. Rebuild with the included instructions

when the application changes. Navigation is internal to this PDF; links do not depend on the sample preview being available.

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APPENDIX A / CURRENT REPOSITORY README

README.md

Complete README for application source 2.0.8

The following pages reproduce the full current README, including deployment procedures and command examples. Long code lines wrap to fit the page. The original, unmodified README.md is also embedded as a PDF attachment for copying commands exactly.

Source: README.md · 89,266 bytes · September 10, 2026. The build records its SHA-256 checksum. Select a section below, or expand Appendix A in the PDF bookmarks.

DNR: MOED מודען - p. 103	User lifecycle and invitations - p. 115
Description - p. 103	Usage - p. 115
Installation - p. 103	Audit-log retention - p. 116
Upgrading an existing database - p. 103	Operational mail retention - p. 116
Configuration - p. 104	Exact database restore runbook - p. 116
Front-end assets - p. 104	Mattermost plugin - p. 118
Architecture and quality checks - p. 104	Daily operations dashboard - p. 118
Health and operations - p. 105	Event contacts and roles - p. 118
Outbound email and task digests - p. 105	Engagement lifecycle - p. 118
Inbound email to Chron - p. 106	Inquiry and booking pipeline - p. 118
Proton sender authentication: incident, fix, and verification - p. 107	Financial closeout and giving history - p. 119
Follow-up: automatic Chron filing permissions - p. 109	Worldwide address lookup with Geoapify - p. 119
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DNR: MOED דנר

Description

DNR (MOED דנר) is a web-based application for managing speaking engagements, presentations, and organizational contacts.

Installation

To set up the project on your local machine, follow these steps:

01 Clone the repository

Copy the clone URL from the repository's **Code** menu and replace <repository-url> below:

```
git clone '<repository-url>' DNR
```

02 Navigate to the project directory

```
cd DNR
```

03 Create deployment secrets

```
cp .env.example .env
install -d -m 700 secrets backups
umask 077
openssl rand -hex 32 > secrets/mysql_root_password
openssl rand -hex 32 > secrets/mysql_app_password
openssl rand -hex 32 > secrets/mysql_backup_password
openssl rand -hex 32 >
secrets/mysql_maintenance_password
openssl rand -hex 32 > secrets/mysql_geocoder_password
openssl rand -hex 32 >
secrets/mysql_mail_ingest_password
openssl rand -hex 32 >
secrets/mysql_mail_dispatch_password
openssl rand -base64 32 >
secrets/dnr_2fa_encryption_key
openssl rand -base64 32 >
secrets/dnr_inbound_routing_key
openssl rand -hex 32 >
secrets/deployment_backup_password
: > secrets/backup_password
chmod 600 secrets/*
```

Store protected offline copies of dnr_2fa_encryption_key and deployment_backup_password separately from database backups. Do not rotate these keys casually: enrolled factors and encrypted deployment archives depend on them. The separate empty backup_password file is populated only for a one-shot browser-format restore and is removed afterward.

04 Build and run the application using Docker Compose

```
# Load the images.env from the qualified release
artifact first.
set -a
. /path/to/release/images.env
set +a
./scripts/compose_with_provenance.sh production
```

Production mode requires HTTPS from a trusted reverse proxy and uses only source code baked into the immutable image. For local HTTP development, run ./scripts/compose_with_provenance.sh development and browse to http://localhost:8080. The published port binds to 127.0.0.1 even when the production DNR_BIND_ADDRESS is configured for a LAN interface. Exposing plaintext development mode remotely requires both a non-loopback DNR_DEV_BIND_ADDRESS and DNR_DEV_ALLOW_REMOTE_HTTP=1; use that opt-in only on an explicitly trusted network, never as production ingress. A one-shot migrator applies the baseline and every forward migration before the web or worker services can start, so no separate fresh-install migration command is required.

05 Create the first administrator on a fresh database

```
docker compose exec web dnr-create-admin admin
```

The command securely prompts for a password. The administrator must enroll an authenticator during the first login. DNR no longer installs accounts with known default passwords.

A server administrator can securely reset an existing account password from the container console:

```
docker compose exec web dnr-set-password USERNAME
```

Upgrading an existing database

Follow the release and recovery workflow (docs/release-workflow.md). On s1, scripts/deploy_s1.sh pauses writers, creates and restore-verifies a fresh encrypted database backup, and only then advances the application checkout, database image and schema. Manual upgrades must enforce the same backup-before-upgrade directive (.cursor/rules/backup-before-database-upgrade.mdc). Verify the recovery copy before changing any active version; a successful dump command alone is not a restore test.

For the one-time upgrade from a legacy installation that still exposes database passwords through container environment variables, run:

```
# Production deployment (default)
sh scripts/secure_existing_deployment.sh

# Local development deployment
DNR_COMPOSE_MODE=development sh
scripts/secure_existing_deployment.sh
```

The helper makes a timestamped database backup, generates private database credentials, writes the ignored .env with mode 600, rotates the root account and six service-specific MySQL accounts, applies tracked migrations and grants, and rebuilds the services with immutable footer provenance. An existing .env is accepted only when it has the

two recognized legacy MySQL password variables. The file is never sourced: only an explicit allowlist of non-secret application settings is copied to the replacement. Deprecated password and calendar-token values are discarded. The helper refuses an already-secured, unrecognized, or partially migrated setup. This legacy credential-conversion helper is not the release deployment command. Before using it for a structural or database-version upgrade, complete the fresh, restore-verified backup procedure in the runbook; do not rely on its unverified SQL dump alone.

The migration runner obtains database-administrator access only in the isolated migrator/database services. It holds a MySQL advisory lock, records applying, applied, or failed state plus each filename and SHA-256 checksum, and refuses to guess after interrupted DDL. Fresh installs and upgrades both start with migrations/20260813_baseline.sql and then follow the same ordered forward migrations. Add a new migration instead of editing an applied file. Existing administrators will be required to enroll 2FA after their next password login; other roles can enable it from **Account Security**.

If the upgraded database predates password hashing and still contains plaintext user passwords, run the one-time CLI migration. It leaves recognized password hashes unchanged, invalidates sessions for migrated accounts, and requires those users to choose a new password after login:

```
docker compose exec web php
/opt/dnr/bin/migrate_passwords.php
```

There is intentionally no browser-accessible password migration page.

The audit-log and Chron-entry migrations use the database administrator account because they install triggers; MySQL requires elevated privileges for that operation when binary logging is enabled. The application continues to connect with the restricted dnruser account. The migrations record successful logins, security events, and row-level inserts, updates, and deletes for users, organizations, contacts, engagements, Chron entries, and presentations. Administrators can review this history from **Users > Audit Log**. Audit entries identify the actor, affected record, IP address, and UTC timestamp without storing passwords, authentication secrets, recovery codes, Chron contents, or before/after field values.

Configuration

For installations outside Docker, Composer validates the PHP runtime used by the application. Install the cURL, Fileinfo, GD, mbstring, mysqli, OpenSSL, and Sodium extensions before running composer install.

Front-end assets

The application serves committed minified CSS and JavaScript while retaining readable source files in src/assets. After changing a source asset, rebuild the production files with:

```
npm ci
npm run build:assets
```

Page heads and asset cache keys are generated by renderPageHead() and assetUrl(). The asset build also emits a deterministic PHP fingerprint manifest, allowing OPcache-backed requests to avoid hashing large bundles; development falls back to direct hashing for files not yet present in the manifest. Page-specific source styles live in src/assets/css/pages; executable page behavior lives in src/assets/js/page-actions.js. Inline executable scripts and style attributes are intentionally avoided so the application can enforce a strict content security policy. The map alone permits runtime inline positioning because MapLibre GL requires it.

Architecture and quality checks

HTTP entry points load src/bootstrap.php, which initializes Composer/local autoloading, structured request logging, database configuration, and shared application helpers. Reusable validation and reference data live under src/app/Domain; create and edit routes use the same normalizers. PHPStan checks migrated request/controller routes at level 3, the remaining HTTP surface at level 0, the legacy helper layer at level 5, and the typed domain/runtime layer at level 6 so newly extracted code can move progressively into the stricter configurations.

Telephone numbers are validated with libphonenumber metadata and stored in canonical E.164 form. National and international formatting is applied only when values are rendered or returned to an edit form.

Run the complete local quality gate before opening a pull request:

```
composer install
npm ci
composer validate --strict --no-check-publish
composer audit --no-interaction
npm audit --audit-level=high
composer check
npm test
```

composer check runs PHP syntax checks, PHPStan level 6 on the typed domain/runtime layer, PHPStan level 5 on the legacy helper layer, level 3 on migrated request/controller routes, level 0 on the remaining HTTP surface, and all unit/feature tests. npm test rebuilds every committed production asset and runs JavaScript syntax and behavior tests. GitHub Actions runs the PHP checks on PHP 8.4 and

8.5 while frontend checks, locked-dependency audits, plugin checks, and integration checks run once in parallel. Dependency audits have bounded retries so an advisory-service outage fails promptly without weakening the security threshold, and superseded runs for the same branch or pull request are cancelled. CI also validates every Compose overlay combination and starts a disposable Docker/MySQL environment through the documented automatic-migration path before exercising HTTP and database behavior. Dependabot proposes weekly Composer, npm, Docker, and pinned GitHub Actions updates.

Database integration suites are discovered automatically from `tests/*_integration_test.php` and `tests/integration/*_test.php`. Run them only against the disposable Compose environment with `sh scripts/run_integration_tests.sh disposable`; the runner sends destructive backup/restore coverage to the isolated maintenance container.

VERSION is the single source of release-version metadata. DNR uses the project ontology `[super].[major].[minor] = x.y.z`: a minor release is `[x].[y].[z+1]` (for example, 1.10.3 becomes 1.10.4), a major release is `[x].[y+1].0`, and a super release is `[x+1].0.0`. Update it once when preparing a release; runtime responses, fallback asset cache keys, the footer, backups, and container images read that value automatically.

Health and operations

- `/health.php` is a dependency-free liveness response.
- `/ready.php` verifies database connectivity and every migration filename/checksum before reporting ready.
- Application errors are logged as structured JSON with a request ID. Public error responses omit database and exception details and include the request ID for correlation.

Every DNR Compose service opts out of host-wide Watchtower polling. Production images are built and qualified by CI on the final merged commit, then promoted by digest through the guarded deployment workflow (`docs/release-workflow.md`). The opt-out prevents Watchtower from treating Compose's local project-service:latest image names as Docker Hub repositories or trying to update dependencies outside the reviewed deployment path.

Outbound email and task digests

Invitation, verification, and recovery messages use an encrypted transactional outbox. Creating a token and its queued message is one database change; a dedicated mail-dispatch worker owns the SMTP credential and the only SMTP egress path. The web service never connects to SMTP. A previous single-use link remains valid while its replacement is pending and is invalidated only after the relay accepts the new message. Failed deliveries retry with bounded exponential backoff, and the encrypted payload is erased after success or terminal failure.

Verified users receive a daily work digest by default on weekdays at 7:00 a.m. under **My Profile > Notifications**. Each user can disable it, choose a different local delivery time, or select any combination of days; weekday, weekend, and every-day presets are available. After the selected time on a selected day, the same worker queues at most one message per user and business date with a light-only HTML snapshot of the Dashboard: upcoming engagements, My Work, event readiness, financial closeouts, and, for administrators and editors, inbound mail awaiting review. Overdue and due-today rows retain the Dashboard's semantic highlighting, and every record includes the same role-appropriate navigation link as the Dashboard. A matching plain-text alternative is included for clients that do not render HTML. Digests use a separate encrypted notification_outbox so account-token delivery remains isolated; messages are discarded if the recipient opts out, changes their email, becomes inactive, no longer has a verified address, or removes the queued day from their schedule. Sent and terminal payloads are erased. `php scripts/generate_daily_digest_preview.php` refreshes the deterministic sample at `docs/daily-digest-preview.html` through the production renderer.

Administrators with fresh elevation can also manage any user's digest enablement, delivery time, and delivery days from **Users > Edit User**.

Email Templates in the sidebar lets editors and administrators create and edit reusable engagement email subjects and messages, suggest event contact roles, and set the display order. The five original templates are seeded into the library and can be edited, archived, or restored. Archived templates are excluded from both the web and Mattermost composers. Administrators can permanently delete an archived template after confirming their access. Reviewers can view the library. **Custom message** remains available for writing from scratch. Template names and reviewed content are retained with sent messages even if a template is later edited or deleted; overlapping edits are rejected instead of overwriting newer changes.

Templates support `{{event_name}}`, `{{organization_name}}`, `{{event_dates}}`, `{{event_start_date}}`, `{{event_end_date}}`, `{{event_location}}`, `{{speaker_names}}`, and `{{presentation_schedule}}` (message only). Use the editor's **Personalize with event fields** buttons to insert them. Values are filled when composing; the sender can still edit the subject, message, and recipients before sending. The engagement routing marker is added automatically. Apply `20260910_add_email_message_templates.sql` and the corresponding application-table grant before deploying this feature.

Editors and administrators can also send tracked plain-text correspondence from an active Engagement. The composer provides booking-confirmation, travel/lodging, final-reconfirmation, presentation-schedule, post-event thank-you, and custom starting templates. Recipients are selected from the Engagement's assigned Primary host, On-site contact, Billing, Travel, and Materials roles, plus speakers assigned to its active presentations. Speakers are optional and unchecked by default; select their recipient cards or use the Speaker shortcut to include them. Each speaker appears once, even when assigned to multiple presentations, and can also receive a message without any event contacts. Every unique address receives an independent delivery so recipients are never disclosed to one another. The authoritative Engagement marker is appended to the subject automatically, allowing a reply that preserves the subject to return through the normal inbound-mail routing workflow when `DNR_INBOUND_ADDRESS` is configured. In that mode, tracked Engagement messages set the shared inbound mailbox as their Reply-To address; other application email keeps the ordinary sender.

The optional share-safe event brief contains only the public event schedule, location, description, and presentations. It deliberately excludes Chron entries, internal notes, compensation, giving, and financial-closeout data. Queuing a message creates linked source entries in the Engagement and Organization Chron logs and in each selected Contact's Chron log. The retained correspondence record stores the reviewed plain-text subject and body as business history, consistent with Chron and retained inbound mail. The separate SMTP delivery payload is sealed with the application key while pending and erased after successful or terminal delivery. Each recipient retries independently with bounded exponential backoff; editors and administrators can explicitly reconstruct and re-queue a terminal failed delivery from the retained source record.

Create `secrets/smtp_password`, configure the sender and relay in `.env`, and use `production-smtp` or `development-smtp`. Combine inbound and outbound mail with `production-mail-smtp` or `development-mail-smtp`.

SMTP relays with a private or self-signed certificate can be pinned without disabling TLS verification. Put only the relay's public certificate in `secrets/smtp_ca.pem`, set `DNR_SMTP_PEER_NAME` to a name or IP address covered by that certificate, and use the matching `*-smtp-ca` Compose mode. The trust anchor is mounted only into `mail-dispatch`; the web and other workers cannot read it. If the relay rotates its certificate, delivery fails closed until the pinned certificate is replaced.

For Proton Mail Bridge on Docker Desktop, use the Bridge-generated username and password - not the Proton account password - and keep Bridge bound to the host loopback interface. Bridge's exported public TLS certificate covers `127.0.0.1`, while the container reaches the same-Mac listener through `host.docker.internal`, so a typical combined configuration is:

```
DNR_SMTP_HOST=host.docker.internal
DNR_SMTP_PORT=1025
DNR_SMTP_ENCRYPTION=starttls
DNR_SMTP_PEER_NAME=127.0.0.1
DNR_SMTP_CA_SECRET_FILE=./secrets/smtp_ca.pem
```

Export Bridge's TLS files under **Settings > Advanced > Export TLS certificates**, retain the private key outside DNR, and copy only the public certificate to `secrets/smtp_ca.pem`. Start both mail workers with

```
./scripts/compose_with_provenance.sh production-mail-smtp-ca.
```

Do not publish the Bridge IMAP or SMTP ports to the LAN or Internet.

Inbound email to Chron

DNR can poll a dedicated IMAP mailbox and turn messages copied to it into Contact, Organization, and Engagement Chron entries. It uses exact email-address rules plus an explicit Engagement marker; it does not send message content to an AI service.

The routing policy is deliberately conservative:

- The sender must uniquely match an active DNR user with a verified email address, an active Contact, or an active Organization.
- DNR considers the message's From, To, and Cc addresses, excluding the configured gateway address and recognized internal users. A unique Contact match adds the message to that Contact and the Contact's Organization. A direct Organization email match adds it to the Organization.
- Put the exact signed marker shown on the Engagement page in the subject or plain-text body to route the message. In the tracked MOED deployment profile its shape is `[MOED#123.<signed-token>]` for Engagement

123. A valid signed marker routes automatically only when the visible sender also uniquely matches an active DNR user, Contact, or Organization. Unknown senders, unsigned or altered markers, unknown or archived Engagements, and messages containing multiple different markers require review. Participant mismatches route only to the signed Engagement; clean exact-address matches retain the usual Contact and Organization routes too.

- Duplicate delivery of the same RFC Message-ID is idempotent. Without a valid signed marker, all messages remain in **Inbound Mail** for review; ambiguous senders, shared email addresses, unknown senders, and messages with no matched target go to **Inbound Mail** for an administrator or editor to review. Reviewers may choose any active Engagement from the searchable selector, in addition to approving matched Contact and Organization routes. If an address is missing from DNR, update the record and use **Retry routing**.
- The Chron entry contains the normalized headers, subject, timestamps, plain-text body, attachment names, and a link to the retained inbound record. Attachment contents are not stored. When an HTML body is available, it is converted to inert text, retaining HTTP(S) link destinations next to their labels. Otherwise the plain-text body is used. Retained web URLs are clickable in both Inbound Mail and the activity entry; email HTML itself is never rendered.
- Automatic routing requires authenticated sender results by default. The bundled Proton Bridge adapter supplies signed assertions from Proton's API metadata for internal mail and external mail that passed DMARC, as described below. For other IMAP providers, list the exact trusted mailbox authserv-id values in `DNR_INBOUND_TRUSTED_AUTH_SERVERS`; until then, messages remain in **Inbound Mail** for review. The topmost Authentication-Results header must come from a listed server and report an aligned `dmARC=pass` for the visible From domain. Missing, forged, failing, or mismatched results fail closed. Set `DNR_INBOUND_REQUIRE_AUTHENTICATED_FROM=0` only as an explicit compatibility exception for a trusted mailbox that cannot expose provider-generated results. DNR intentionally does not trust a later sender-supplied Authentication-Results header.

Proton sender authentication: incident, fix, and verification

Issue identified on September 6, 2026 (application 1.11.23). An incoming message had a valid signed Engagement marker and uniquely matched an active user's verified email address, but Routing details reported: "The visible sender does not have a trusted aligned DMARC pass from the mailbox provider." The message had arrived successfully; the missing evidence prevented automatic Chron filing. That warning did not establish that the sender had failed DMARC or was spoofed.

Two independent conditions explained the behavior. Production's `DNR_INBOUND_TRUSTED_AUTH_SERVERS` was empty, so no ordinary Authentication-Results header could satisfy the original authentication gate. In addition, the inspected message was internal, end-to-end encrypted Proton mail. Its retained headers included `X-Pm-Origin: internal` and `X-Pm-Content-Encryption: end-to-end`, but no Authentication-Results header. Adding a trusted server name alone could not fix this internal-mail case. The 22-character signature in `[MOED#123.<signed-token>]` authenticates the routing capability; it does not authenticate the person named in From.

Implementation. The Proton sidecar retains the checksum-pinned official 3.25.0 package and uses a narrowly adapted backend built from the separately checksum-pinned 3.25.0 source archive. The adapter lives in `docker/proton-bridge-auth/`; its installation script inserts one call at the end of upstream `getMessageHeader()`. The backend reports 3.25.0+dnr.1. Its decision uses the authenticated Proton API's `MessageMetadata.Flags` and `Sender`, rather than interpreting incoming `X-Pm-*` or sender-supplied Authentication-Results headers:

- A received message marked both **Internal** and **E2E** gets an internal assertion.
- Other received messages marked **DMARCPass** get a `dmARC` assertion.
- Imported mail, messages with **DMARCFail**, and messages flagged as spam or phishing get an unverified assertion, even if other flags claim success. Drafts and sent-only items also remain unverified. SPF failure alone does not override a DMARC pass, because DKIM can satisfy DMARC.

Release qualification also found fixable high-severity vulnerabilities in the upstream source's older Go dependency graph. The checked-in dependencies patch pins `golang.org/x/crypto 0.55.0`, `golang.org/x/net 0.57.0`, `golang.org/x/text 0.41.0`, and `gRPC 1.83.1`, plus their required transitive updates and verified Go checksums. It is applied

strictly to the pinned source; the image does not resolve floating latest versions. Message, IMAP-service, and synchronization-service tests run against that patched graph before the backend is built. Release scans remain mandatory.

With the Compose-provided key setting present, the adapter replaces every existing X-Dnr-Sender-Authentication field with its own result. Even when authentication or the key is unavailable, it installs a failure value so the upstream decrypted MIME-header merge cannot reintroduce a forged assertion. The result is bound to the normalized sender address and RFC Message-ID and authenticated with a full HMAC-SHA256. A versioned, domain-separated key is derived from the existing inbound routing key; event-tag signatures cannot be reused as authentication assertions. No private key or message content is exposed in the header.

`src/app/Security/InboundBridgeAuthentication.php` verifies that assertion, its signature, sender, and Message-ID. Duplicate, malformed, modified, or mismatched assertions fail closed. A failed Bridge assertion never falls back to ordinary DMARC headers. The existing topmost-header, exact-trusted-server, aligned-DMARC policy remains available for generic IMAP providers and retained messages without a Bridge assertion. Raw X-Pm-Origin and encryption headers alone never authorize processing. Successful Routing details now identify **Verified by Proton (internal message)** or **DMARC verified by Proton**; missing evidence and configuration failures have distinct explanations.

This fixes both Proton cases without turning off `DNR_INBOUND_REQUIRE_AUTHENTICATED_FROM`: the Proton API assertion supplies authenticated evidence directly, so an empty generic trusted-server list does not block verified Bridge mail. A recognized active sender and a valid, unambiguous signed marker for an active Engagement are still required. Existing participant routing, Email Gateway attribution, duplicate-delivery protection, and manual approval remain in force.

Rollout requirements. This change includes the Bridge image, Compose secret mount, and PHP application. Deploy all of them through the normal release workflow; updating PHP alone cannot create authentication evidence that an older Bridge never supplied. There is no database migration. Before the first rollout, run `scripts/prepare_linux_secrets.sh` on the Linux host (with `DNR_INBOUND_ROUTING_KEY_FILE` set explicitly if a nondefault host path is used). It grants UID 10001 read access to only the inbound routing key, while retaining the existing `root/www-data` ACLs. `docker-compose.proton-bridge.yaml` mounts that key read-only into Bridge. The entrypoint refuses to start if the configured key cannot be read or is not a base64-encoded 32-byte key. Bridge does not receive database or outbound SMTP credentials. Keep the existing key; do not generate a

replacement as part of this fix. Rotation invalidates previously issued event tags and retained Bridge assertions.

Keep Bridge's automatic updates disabled as described in its setup instructions, so an upstream binary cannot silently replace the adapter. Upgrading Bridge requires updating both pinned source and package, reviewing the metadata flags and header-merge behavior, and rerunning adapter tests. Generic IMAP installations continue to configure their real provider's exact `authserv-id` values; do not guess a server name or disable authentication to suppress a warning.

Validation and operational check. The Bridge image build runs the upstream message package tests plus `dnr_auth_test.go`. Adapter cases cover internal and external DMARC passes, unverified external mail, imports, drafts, missing encryption, spam, phishing, contradictory pass/fail flags, forged duplicate headers, and unavailable keys. PHP's `inbound_bridge_authentication_test.php` checks the same fixed cross-language HMAC fixture and rejects changed senders, Message-IDs, signatures, versions, duplicates, body-only assertions, and fallback attempts. The disposable `inbound_email_integration_test.php` suite exercises parsing, storage, automatic processing, actual Engagement Chron writes, and deduplication with an empty trusted-server list. It also verifies that unknown senders, altered tags, and invalid assertions remain in review. These tests use synthetic mail and do not send messages or modify the production mailbox.

Local implementation validation passed composer check, the disposable inbound integration suite, production Compose configuration validation, and the Linux/amd64 Bridge image build (including the upstream and adapter tests). A network-disabled container reported **Proton Mail Bridge 3.25.0+dnr.1**; a separate startup check correctly refused an unreadable authentication key. The disposable database was removed after testing. These results do not constitute a production rollout.

After deployment, check new internal Proton mail and new external DMARC-passing mail from known senders carrying a valid event tag. Confirm the authentication method in Routing details, a final **Processed** status, and exactly one matching Engagement Chron entry. This live delivery check is separate from local automated tests and is not implied by a successful image build. Existing retained messages and Bridge-cached messages may lack the new assertion: the update does not invent evidence, reset mailbox state, refetch old mail, or automatically retry the review queue. Review those messages manually. Already processed messages remain terminal and are not filed again.

Primary implementation references: Proton's pinned message metadata flags (https://github.com/ProtonMail/go-proton-api/blob/6bf7f5a61eb8/message_types.go), Bridge header construction (<https://github.com/ProtonMail/proton-bridge/blob/v3.25.0/pkg/message/build.go>), and sender verification explanation (<https://proton.me/support/digital-signature>).

Follow-up: automatic Chron filing permissions

After deploying 1.11.24, a new Proton test message passed sender authentication and its signed Engagement route, but automatic filing failed with UPDATE command denied on contact_chron_entries. **Retry Routing** then succeeded because that action uses the web application's database account. The background worker uses dnrmailingest, which intentionally has only SELECT and INSERT access to Contact, Organization, Engagement, and Inquiry Chron.

The filing queries used ON DUPLICATE KEY UPDATE id = id to avoid duplicate Chron entries. MySQL requires UPDATE permission for that clause when preparing the statement, including a first delivery and even when the Contact target list is empty. The error therefore occurred before the intended Engagement entry could be inserted. The transaction rolled back, retaining the source message for retry. This was a filing-permission defect after successful authentication.

Filing now uses INSERT ... SELECT ... WHERE NOT EXISTS for all four Chron tables. The existing FOR UPDATE lock on the inbound source row serializes filing attempts for that message throughout the transaction. Existing entries are skipped without changing their text, attribution, or timestamps; the unique source/target indexes remain in place, and other database errors still abort the transaction. This requires no additional worker grants or schema migration. Processed messages remain terminal.

tests/inbound_worker_filing_integration_test.php reproduces the old permission denial and verifies the fix through a real dnrmailingest connection in a disposable database. It confirms that Chron updates remain forbidden, tests an Engagement-only message with no Contact match, covers all four Chron destinations, preserves pre-existing entries, and checks duplicate delivery and completed-message protection. Fixture setup and cleanup use the web account; filing always uses the restricted worker. Both ordinary integration CI and exact-release-image qualification run this regression. Previously, filing integration used the web account, while worker-privilege tests only checked account-column access, which allowed the mismatch to escape testing.

An Engagement marker is a signed routing capability, so do not expose it outside the intended correspondence. Unsigned legacy markers intentionally require review after this upgrade. Exact address matching remains a routing aid, not proof of sender identity, and inbound entries are always attributed to the Email Gateway rather than the visible From identity. The

mailbox provider should still enforce its normal spam and SPF/DKIM/DMARC checks.

Both Cc and Bcc delivery to the configured DNR_INBOUND_ADDRESS work. A Bcc delivery normally omits the gateway address from the stored message headers, which is expected; DNR routes using the remaining participants. For example, mail from a verified DNR user to a unique Contact routes to the Contact and its Organization, while a reply from that Contact is recognized from From.

Apply the tracked database migration before enabling the worker. Then create the ignored IMAP password secret, set the inbound variables in .env, and start a mail-enabled Compose mode:

```
install -m 600 /dev/null secrets/imap_password
# Put the exact IMAP/Bridge password in
secrets/imap_password without committing it.
./scripts/compose_with_provenance.sh production-mail
```

For local development, use development-mail. The worker imports unseen messages in bounded batches, marks a message seen only after DNR has stored it, and retries transient routing failures. Oversized or unparseable poison messages are recorded in inbound_email_quarantine before being marked seen so they cannot starve later UIDs; transient mailbox or database failures remain unseen. It does not delete or move the source message, including after successful routing; the IMAP mailbox remains a recoverable source of record. The web and worker accounts retain only the database privileges needed for this workflow.

An administrator with a recent elevated session can purge an individual retained mail entry from its **Inbound Mail** detail view. Purging removes the DNR mail card and its retained source content, but preserves every associated Contact, Organization, and Engagement Chron Log entry; only the source-email link on those Chron entries is cleared. The original message in the IMAP mailbox is not deleted or moved.

Proton Mail accounts require Proton Mail Bridge (<https://proton.me/support/imap-smtp-and-pop3-setup>) and a paid Proton plan. Configure DNR with the IMAP hostname, port, username, and generated password shown by Bridge - not the Proton account password. Bridge uses a local, self-signed IMAP endpoint and is designed for clients on the same computer. Set DNR_IMAP_VERIFY_PEER=0 only for that local or tightly isolated Bridge connection; keep certificate verification enabled for any remote IMAP server. Because a

Docker container may not be able to reach a Bridge listener bound only to the host's loopback interface, confirm network reachability from the mail-ingest service before using the production mail mode. If Bridge cannot be made reachable without exposing it, run the worker beside Bridge or use a dedicated standards-compliant IMAP mailbox rather than publishing Bridge to an untrusted network. Proton documents the local-only design in its Bridge overview (<https://proton.me/support/why-you-need-bridge>) and Bridge CLI guide (<https://proton.me/support/bridge-cli-guide>).

Ubuntu 24.04 deployment with Proton Mail

The Linux deployment is supported on the existing Ubuntu 24.04 host; it does not require an Ubuntu release upgrade. Proton Bridge runs as a non-root, headless sidecar built from Proton's checksum-verified official Debian package. Only mail-ingest shares its network namespace, so Bridge's IMAP listener remains on 127.0.0.1 and is never published to the host, LAN, or Internet. Its account, keychain, and generated credentials persist in the `proton_bridge_data` volume.

Outbound mail should use a Proton SMTP submission token (<https://proton.me/support/smtp-submission>) directly at `smtp.protonmail.ch:587` with STARTTLS. This keeps mail-dispatch independent of Bridge and leaves Bridge responsible only for inbound IMAP.

The Ubuntu ingress also differs from the Docker Desktop deployment: it publishes no host port. Traefik and MOED share a dedicated /29 network, while Cloudflared receives one stable address on the existing proxy network. DNR trusts those exact hops. Do not replace the exact Cloudflared address with the entire shared proxy subnet; another container on that subnet could otherwise present forged Cloudflare client headers.

Prepare the network and merge the tracked proxy overlay with the existing Traefik stack. Replace `/path/to/traefik` with that stack's directory:

```
docker network create --driver bridge --subnet
172.29.255.0/29 moed_edge
install -m 600 deploy/traefik-moed-edge.yaml \
/path/to/traefik/docker-compose.moed-edge.yml
cd /path/to/traefik
docker compose -f docker-compose.yml -f
docker-compose.moed-edge.yml config --quiet
docker compose -f docker-compose.yml -f
docker-compose.moed-edge.yml up -d traefik cloudflared
```

Use both Traefik Compose files for every future update to that stack. The overlay pins Traefik to 172.29.255.2 on `moed_edge` and Cloudflared to 172.18.0.254 on `proxy`; the matching defaults are already in `.env.example`. If either subnet conflicts on another host, change all matching values together before

creating the network.

Those commands preserve the current MOED host layout. For a separate deployment, merge `deploy/traefik-edge.yaml` instead and set the same `DNR_EDGE_NETWORK`, Traefik address, and Cloudflared address in both Compose projects; its defaults match the generalized Ubuntu overlay.

Create the normal database/application secrets, an empty mode-600 IMAP password file, and a mode-600 SMTP token file. Copy `.env.example` to `.env`, set the public URL and mail identities, and leave `DNR_TRAEFIK_ENABLE=false` during staging. Native Linux Compose mounts local secrets with their host ownership, while MOED deliberately drops the capability that would bypass file modes. Run `./scripts/prepare_linux_secrets.sh` after creating or replacing any secret. It keeps the host directory private and grants read-only ACLs only to the container root and `www-data` identities; do not make the secret files world-readable. For Proton SMTP Submission, set:

```
DNR_MAIL_FROM=dnr@example.org
DNR_SMTP_HOST=smtp.protonmail.ch
DNR_SMTP_PORT=587
DNR_SMTP_ENCRYPTION=starttls
DNR_SMTP_USERNAME=dnr@example.org
DNR_SMTP_PASSWORD_SECRET_FILE=./secrets/smtp_password
DNR_IMAP_USERNAME=
DNR_IMAP_PASSWORD_FILE=./secrets/imap_password
DNR_TRAEFIK_ENABLE=false
```

Put the SMTP token - not the Proton account password - in `secrets/smtp_password`. Then initialize the Bridge volume and sign in interactively:

```
./scripts/proton_bridge_cli.sh configure
```

At the Bridge prompt run `login`, complete Proton authentication, run `updates autoupdates disable`, and run `info`. Put the generated IMAP username in `DNR_IMAP_USERNAME` and its generated password in `secrets/imap_password`; do not use the Proton account password. Exit the prompt and start the staged stack:

```
./scripts/compose_with_provenance.sh
production-ubuntu-proton up -d --build
./scripts/proton_bridge_cli.sh status
./scripts/proton_bridge_cli.sh logs
```

Before cutover, restore and verify the encrypted DNR backup on Ubuntu and confirm Traefik can reach the staged ingress with `docker exec traefik wget -qO- http://172.29.255.3/health.php`. The final cutover has two coordinated changes: remove the old `moed-router` and `moed-service` entries from Traefik's file-provider configuration, then set `DNR_TRAEFIK_ENABLE=true` and recreate the Ubuntu ingress service. Keeping the label disabled until that moment prevents the staged router from competing with the live Mac route. Rollback is the reverse: disable the label and restore

the old file-provider route.

Database initialization uses only the ordered files under migrations/. The Compose migrator service applies the minimal baseline and every forward migration, records immutable checksums and state, and then applies the single privilege manifest. There is no independent schema snapshot to drift from the upgrade path.

Deployment configuration profiles

Non-secret deployment identity and business defaults live in a validated YAML profile. Compose mounts `DNR_CONFIG_FILE_HOST` read-only and exposes it inside PHP as `DNR_CONFIG_FILE`. The tracked MOED profile is `deployments/moed/application.yaml`; copy `deployments/example/application.yaml` when preparing an independent deployment. Do not put passwords, encryption keys, SMTP tokens, or other secrets in these files.

The profile controls:

- display, native, mail, calendar, and authenticator names plus light/dark logo assets;
- default speaker, country, telephone country code, and business timezone;
- emitted and accepted inbound-email marker prefixes, allowing old replies to remain routable after a deployment rename;
- tile provider URL/attribution and bounded dashboard, task, map, calendar, and PDF windows;
- an explicit initial standard-task seed list.

Logo paths are relative to the public application directory and must name files included under `src/assets/` before the image is built.

Configuration precedence is built-in safe defaults, then deployment YAML, then a small set of documented environment overrides. Unknown YAML keys and invalid types, URLs, asset paths, marker prefixes, ranges, and timezones fail application startup. Validate a profile without connecting to the database:

```
DNR_CONFIG_FILE=deployments/example/application.yaml
php scripts/check_config.php
```

Standard tasks are database-managed after import. Migrations never re-read YAML, and normal container startup never reconciles administrator edits. After migrations, explicitly insert missing profile tasks and apply only required-task policy with:

```
docker compose exec web php
/opt/dnr/bin/seed_standard_tasks.php
```

Existing titles, notes, priorities, schedules, and sort order are not overwritten. Tasks absent from the profile are not deleted or archived. Historical product-default tasks came from immutable migrations, so a new deployment should archive any unwanted baseline definitions in **Standard event tasks** after running its profile seed.

Environment variables

Configure these values as needed:

- `PORT`: published HTTP port; defaults to 8080.
- `DNR_BIND_ADDRESS`: address on which Docker publishes the HTTP port; defaults to 127.0.0.1.
- `DNR_DEV_BIND_ADDRESS` and `DNR_DEV_ALLOW_REMOTE_HTTP`: separate development-mode publish controls. Development ignores `DNR_BIND_ADDRESS`, binds to 127.0.0.1 by default, and refuses a non-loopback address unless the plaintext remote-access opt-in is exactly 1.
- `DNR_MYSQL_ROOT_PASSWORD_FILE`, `DNR_MYSQL_APP_PASSWORD_FILE`, `DNR_MYSQL_BACKUP_PASSWORD_FILE`, `DNR_MYSQL_MAINTENANCE_PASSWORD_FILE`, `DNR_MYSQL_GEOCODER_PASSWORD_FILE`, `DNR_MYSQL_MAIL_INGEST_PASSWORD_FILE`, and `DNR_MYSQL_MAIL_DISPATCH_PASSWORD_FILE`: host paths to independent secret files. The authenticated export path uses the read-only full-schema backup identity; ordinary web requests, geocoding, inbound parsing, outbound delivery, migration, and destructive maintenance retain separate identities with only their required privileges.
- `DNR_BACKUP_PASSWORD_FILE`: host path to the temporary file containing the exact password of the backup being restored. It is mounted only in the maintenance profile and should be emptied or removed immediately after the restore is verified.
- `DNR_PUBLIC_BASE_URL`: externally visible HTTPS origin used to construct calendar, invitation, verification, recovery, and task-digest links.
- `DNR_MATTERMOST_TOKEN_SECRET_FILE` and `DNR_MATTERMOST_INSTANCE_ID`: host path to the shared Mattermost plugin token and stable identifier for the authorized Mattermost server. They are mounted only when a *-mattermost Compose mode is selected. See `docs/mattermost-plugin.md`.
- `DNR_MAIL_TRANSPORT`: smtp enables account and task-digest email delivery; the secure default is disabled. log acknowledges messages without logging their

bearer links and is intended only for automated tests.

- **DNR_MAIL_FROM**: validated sender address for outbound email. The sender display name comes from `brand.mail_name`; **DNR_MAIL_FROM_NAME** remains an optional highest-precedence override.
- **DNR_SMTP_HOST**, **DNR_SMTP_PORT**, and **DNR_SMTP_ENCRYPTION**: SMTP relay connection. Encryption accepts `starttls` (the default), `implicit tls`, or `none` for a trusted internal relay.
- **DNR_SMTP_USERNAME** and **DNR_SMTP_PASSWORD_SECRET_FILE**: optional SMTP authentication and the host path mounted by the SMTP overlay. The dedicated mail-dispatch service receives it as **DNR_SMTP_PASSWORD_FILE**; the web service remains backend-only and cannot reach the relay.
- **DNR_SMTP_CA_SECRET_FILE** and **DNR_SMTP_PEER_NAME**: optional pinned SMTP trust anchor and expected certificate name. The certificate is mounted as **DNR_SMTP_CA_FILE** only by an `*-smtp-ca` mode; `peer` and `chain` verification remain enabled.
- **DNR_EMAIL_OUTBOX_BATCH_SIZE** and **DNR_EMAIL_OUTBOX_IDLE_SECONDS**: bounded outbound messages per worker cycle and idle polling interval. Defaults are 20 messages and 15 seconds.
- **DNR_NOTIFICATION_OUTBOX_BATCH_SIZE**: bounded task-digest messages claimed per worker cycle; defaults to 20.
- **DNR_ENGAGEMENT_EMAIL_OUTBOX_BATCH_SIZE**: bounded Engagement-recipient deliveries claimed per worker cycle; defaults to 20.
- **DNR_NOTIFICATION_SCHEDULE_INTERVAL_SECONDS**: interval between checks for newly due task digests; defaults to 300 seconds.
- **DNR_INBOUND_ADDRESS**: required dedicated mailbox address copied on messages when a mail-ingest Compose mode is enabled.
- **DNR_INBOUND_ROUTING_KEY_FILE**: host path to the independent base64-encoded 32-byte key used to sign Engagement reply-routing markers and derive a separate HMAC key for the bundled Bridge's sender-authentication assertions. Generate it with `scripts/ensure_inbound_routing_key.sh`, back it up securely, and do not rotate it while issued reply markers or retained assertions are still in use.
- **DNR_INBOUND_MAX_BYTES**, **DNR_INBOUND_BATCH_SIZE**, and **DNR_INBOUND_IDLE_SECONDS**: maximum raw message size, bounded messages per polling cycle,

and idle polling interval. Defaults are 10 MiB, 20 messages, and 30 seconds.

- **DNR_INBOUND_REQUIRE_AUTHENTICATED_FROM** and **DNR_INBOUND_TRUSTED_AUTH_SERVERS**: fail-closed automatic-routing gate. It defaults on. The bundled Bridge supplies signed assertions from Proton's internal-mail or DMARC metadata. Other IMAP providers require a topmost Authentication-Results header from a listed trusted `authserv-id` with an aligned DMARC pass. An empty trusted-server list blocks that generic path but does not block valid Bridge assertions. Set the gate to 0 only as an explicit compatibility exception for a trusted mailbox that cannot expose provider authentication results. See the Proton authentication incident and rollout notes above.
- **DNR_IMAP_HOST**, **DNR_IMAP_PORT**, and **DNR_IMAP_SECURITY**: inbound mailbox endpoint. Security accepts `starttls` (the default), `implicit tls`, or `none` only for a trusted isolated connection.
- **DNR_IMAP_USERNAME**, **DNR_IMAP_PASSWORD_FILE**, and **DNR_IMAP_MAILBOX**: mailbox credentials and selected folder. The mail Compose overlay mounts the password as a Docker secret; **DNR_IMAP_PASSWORD** is accepted only for simple non-Compose or development execution.
- **DNR_IMAP_VERIFY_PEER**: verifies the IMAP server certificate by default. Disable it only for a local Proton Bridge endpoint using Bridge's self-signed certificate.
- **DNR_CONFIG_FILE_HOST**: host path to the selected non-secret deployment YAML; defaults to the tracked MOED profile for backward compatibility.
- **DNR_BRAND_DISPLAY_NAME**, **DNR_TOTP_ISSUER**, **DNR_CALENDAR_NAME**, **DNR_INBOUND_MARKER_PREFIX**, and **DNR_INBOUND_ACCEPTED_MARKER_PREFIXES**: optional highest-precedence identity overrides. The accepted marker value is a comma-separated list and must include the emitted prefix.
- **DNR_DEFAULT_SPEAKER**: optional name used to preselect a matching saved speaker record for new presentations; otherwise the first-created speaker is selected. The legacy **DEFAULT_SPEAKER** name remains accepted during migration.
- **DNR_DEFAULT_COUNTRY**, **DNR_DEFAULT_PHONE_COUNTRY_CODE**, and **DNR_TIMEZONE**: optional overrides for the corresponding profile defaults. Invalid values fail startup.

- **DNR_2FA_KEY_FILE**: host path to the Docker secret containing the base64-encoded 2FA encryption key; defaults to `./secrets/dnr_2fa_encryption_key`.
- **DNR_REQUIRE_HTTPS**: rejects non-HTTPS requests in production; defaults to 1. The development Compose override sets it to 0 for loopback-only HTTP.
- **DNR_SESSION_IDLE_SECONDS**, **DNR_SESSION_ABSOLUTE_SECONDS**, and **DNR_SESSION_ROTATION_SECONDS**: authenticated-session idle lifetime, absolute lifetime, and identifier-rotation interval. Defaults are 12 hours, 24 hours, and 15 minutes.
- **DNR_TRUSTED_PROXY_IPS**: comma-separated reverse-proxy IP addresses or CIDR networks whose X-Forwarded-For hops DNR may trust; defaults to Docker Desktop's published-port proxy at 192.168.65.1. DNR walks the forwarding chain from the nearest hop outward, skips only configured proxies, and uses the first untrusted address so a client-controlled leftmost value cannot override audit or throttling attribution. Other deployments can set an explicit proxy address or use `docker-gateway` to resolve the container's default route dynamically. If the published port is reachable beyond the reverse proxy, restrict it with a firewall and ensure the proxy replaces client-supplied forwarding headers, including X-Forwarded-Proto.
- **DNR_BACKEND_SUBNET** and **DNR_INGRESS_PROXY_IP**: private backend network and fixed address of the localhost ingress proxy. The defaults are 172.30.255.0/24 and 172.30.255.254. Override both together if that subnet conflicts with another Docker network. Only the fixed proxy address is added to the application's trusted proxy list; the web container remains on the internal backend without an outbound route.
- **DNR_TRUSTED_CLOUDFLARE_PROXY_IPS**: comma-separated IP addresses or CIDR networks used by the trusted Cloudflare tunnel hop in X-Forwarded-For; defaults to this deployment's 172.18.0.0/24 private proxy network so container address changes do not break client-IP detection. On that route DNR records Cloudflare's CF-Connecting-IP value instead of the tunnel container address.
- **DNR_DASHBOARD_UPCOMING_DAYS**, **DNR_TASK_UPCOMING_DAYS**, **DNR_CALENDAR_PAST_DAYS**, **DNR_CALENDAR_FUTURE_DAYS**, **DNR_PDF_MAX_CHRON_ENTRIES**, and **DNR_PDF_MAX_TASKS**: optional overrides for validated workflow windows and PDF export limits normally read from YAML.
- **DNR_DATABASE_BACKUP_MAX_BYTES**: maximum unencrypted backup size; defaults to 536870912 bytes (512 MiB), with coordinated temporary-storage and worker memory budgets. The maintenance page estimates capacity; larger databases use the native encrypted deployment backup. Backup and restore plaintext exists only in the relevant container's memory-backed `/tmp`.
- **DNR_GITHUB_REPOSITORY**, **DNR_BUILD_COMMIT**, and **DNR_BUILD_TIMESTAMP**: repository link and immutable build provenance displayed in the footer. The Compose wrapper derives the full hash and UTC commit timestamp automatically. CI builds from source archives without `.git` may export both values explicitly. Page rendering never calls GitHub or another third-party API.
- **DNR_GEOCODER_BASE_URL** and **DNR_GEOCODER_ALLOWED_HOSTS**: public HTTPS endpoint and explicit hostname allowlist used only by the background geocoder worker.
- **DNR_GEOCODER_USER_AGENT**: identifying user agent sent to the configured geocoder. Set this to the deployment name and a contact URL or email. When omitted, DNR identifies itself with its version and repository URL.
- **DNR_GEOCODER_BATCH_SIZE**, **DNR_GEOCODER_IDLE_SECONDS**, **DNR_GEOCODER_LEASE_SECONDS**, and **DNR_GEOCODER_MAX_ATTEMPTS**: bounded worker throughput, polling, stale-job lease, and retry policy.
- **DNR_MAP_PAST_DAYS**, **DNR_MAP_FUTURE_DAYS**, and **DNR_MAP_MAX_EVENTS**: optional overrides for the bounded map window and result cap normally read from YAML.
- **DNR_MAP_TILE_URL**, **DNR_MAP_ATTRIBUTION_TEXT**, **DNR_MAP_ATTRIBUTION_URL**, and **DNR_MAP_MAXIMUM_ZOOM**: optional overrides for the validated map provider. The tile origin is also used to construct the page's Content Security Policy.
- **DB_HOST**, **MYSQL_DATABASE**, **MYSQL_USER**, and **MYSQL_PASSWORD_FILE**: runtime database connection settings for non-Compose deployments. **MYSQL_BACKUP_USER** and **MYSQL_BACKUP_PASSWORD_FILE** configure the separate read-only export connection. Compose uses the fixed `dnr` database with restricted `dnruser` and `dnrbackup` accounts.

Build provenance

Use `scripts/compose_with_provenance.sh` instead of invoking a Compose build directly. It derives the checked-out commit and UTC commit timestamp, validates both values, exports them as Docker build arguments, and then runs Compose. It refuses a dirty worktree because a footer commit would not accurately identify uncommitted source files.

```
# Local HTTP development
./scripts/compose_with_provenance.sh development

# Production behind the configured HTTPS proxy
./scripts/compose_with_provenance.sh production

# Production with the inbound IMAP worker
./scripts/compose_with_provenance.sh production-mail

# Local HTTP development with the inbound IMAP worker
./scripts/compose_with_provenance.sh development-mail

# Production with encrypted queued SMTP delivery
./scripts/compose_with_provenance.sh production-smtp

# Production SMTP with a private pinned trust anchor
./scripts/compose_with_provenance.sh
production-smtp-ca

# Production with both inbound IMAP and outbound SMTP workers
./scripts/compose_with_provenance.sh
production-mail-smtp

# Ubuntu production with private Traefik ingress and headless Proton Bridge
./scripts/compose_with_provenance.sh
production-ubuntu-proton

# Production IMAP and SMTP with a private pinned SMTP trust anchor
./scripts/compose_with_provenance.sh
production-mail-smtp-ca

# Local equivalents
./scripts/compose_with_provenance.sh development-smtp
./scripts/compose_with_provenance.sh
development-smtp-ca
./scripts/compose_with_provenance.sh
development-mail-smtp
./scripts/compose_with_provenance.sh
development-mail-smtp-ca

# Forward a specific Compose command or service selection
./scripts/compose_with_provenance.sh development up
-d --build web ingress

# Display the metadata without running Docker
./scripts/compose_with_provenance.sh --print-metadata
```

For CI source archives without `.git`, export a complete 40-character `DNR_BUILD_COMMIT` and a UTC `DNR_BUILD_TIMESTAMP` in `YYYY-MM-DDTHH:MM:SSZ` format before invoking the wrapper. Both values must be supplied together.

Deploying the tracked MOED release to s1

Follow Release and recovery workflow (`docs/release-workflow.md`) to prepare the version, merge through the six protected checks, qualify immutable images, and verify main and the release tag on both remotes. Then deploy the exact final commit:

```
./scripts/deploy_s1.sh "$(git rev-parse HEAD)"
```

The command targets the configured `s1` deployment and uses the complete production-ubuntu-proton-mattermost topology. It creates and restore-verifies a fresh encrypted database backup before changing the checkout, database image/schema, or application version. This is a standing deployment directive (`.cursor/rules/backup-before-database-upgrade.mdc`), including for manual structural/version upgrades.

Configure the private `s1` backup password file first (`DNR_S1_BACKUP_PASSWORD_FILE`, default `secrets/deployment_backup_password` under the deployment's project directory). Deployment holds a host lock through backup, migration and public readiness; it uses CI-qualified digests with rebuilding disabled and retains an explicit recovery record. `DNR_S1_USER`, `DNR_S1_HOST`, `DNR_S1_PROJECT_DIR` and `DNR_S1_PUBLIC_BASE_URL` can override the deployment user, host, project directory, and public URL.

Two-factor authentication

Recovery-email changes are separate from ordinary profile edits. They require the current password and a fresh authenticator code when enabled. The old verified address remains active until the new address is verified, an old-address notice is queued, and successful promotion revokes existing sessions.

- TOTP authenticator codes use a 30-second time step and tolerate one neighboring time step for clock drift.
- A successful code cannot be reused.
- Five failed password or second-factor attempts temporarily lock that factor for 15 minutes.
- Recovery codes are single-use. DNR stores only a keyed HMAC lookup value, never the code itself.
- Administrators are required to use 2FA and cannot disable it themselves.
- Administrators can use **Preview access** in the sidebar to navigate with Editor or Reviewer permissions. A persistent banner identifies the active preview and

returns the session to Administrator access; actions taken during a preview still affect live data under the administrator's identity.

- An administrator can reset another user's 2FA from **Manage Users**. Resetting or replacing a factor invalidates that user's other sessions.
- An administrator can set a temporary password for another user from **Manage Users**. The control appears only during a five-minute sensitive-action window opened with the administrator's password plus a fresh authenticator or recovery code. The route rechecks that elevation, invalidates the target user's sessions, and forces the target to choose a private password after login.
- Users can change their own password from **Account Security**; doing so invalidates their other sessions.

Password recovery

- The login page links to a self-service password recovery flow.
- Recovery links are sent only to the unique verified email address of an active account. Responses do not reveal whether an address belongs to an account.
- Recovery links are random, single-use bearer tokens. Only their SHA-256 digests are stored, and they expire after one hour.
- A successful recovery requires a new password of at least 12 characters, invalidates every existing session, records a security audit event, and returns the user to sign-in so normal 2FA requirements still apply.
- Account passwords must contain at least 12 characters and no more than 72 UTF-8 bytes; the upper bound prevents bcrypt from silently ignoring part of a password.
- Accounts without a verified email address require another administrator to set a temporary password from **Manage Users**, or a server administrator to run `docker compose exec web dnr-set-password USERNAME`.

User lifecycle and invitations

- Freshly invited accounts cannot authenticate. The seven-day invitation link lets the recipient choose a private password, verifies the invited email address, and activates the account.
- A self-service email change retains the current verified address while a new 24-hour verification link is pending. Successful verification promotes the new address and

revokes existing sessions. Only verified addresses can receive password-recovery links.

- Deactivation retains the user and every audit reference while incrementing the authentication version, revoking all calendar subscriptions, invalidating outstanding email links, removing task assignments, and making any session-held administrator elevation unusable.
- Reactivation restores sign-in access only. Revoked calendar links and prior task assignments are intentionally not restored.

Usage

Administrators can open **Database** in the primary navigation to download a DNR database backup. Export requires the administrator's password, a fresh authenticator or recovery code, and a new backup password. That password is required to restore the file and is never stored by DNR. The complete `.dnrbackup` archive is encrypted with a key derived by Argon2id using libsodium's moderate resource profile and authenticated with XChaCha20-Poly1305 secretstream. New backup passwords must contain at least 16 characters. The versioned reader retains support for older DNRBACKUP-ENC-1 archives while new exports use the stronger DNRBACKUP-ENC-2 profile. Backups contain every application table, including user authentication and audit data. Treat encrypted files as secrets, use a strong unique backup password, and keep the password separately in a password manager. A forgotten backup password cannot be recovered.

A restore is accepted only when the backup schema exactly matches the deployed database schema. Install the matching DNR version and run its migrations before restoring. The data replacement is transactional and automatically rolls back on failure; a successful restore invalidates all existing sessions and requires everyone to sign in again. Restore is not exposed by HTTP: it runs in a one-shot, non-egress maintenance container using a database credential that the web and geocoder containers never receive. Database backups do not contain the separate `DNR_2FA_ENCRYPTION_KEY`, so keep a secure copy of that key with the backup set. For full disaster recovery, initialize and migrate an empty DNR deployment, restore the `.dnrbackup` from the maintenance profile, and restore the same two-factor encryption key.

Audit-log retention

The application keeps the audit log append-only during normal use. An administrator can open **Users > Audit Log**, enter the number of days to keep in the **Retention** panel, and preview the exact UTC cutoff and affected row count. Pruning then requires a recent administrator password and fresh 2FA confirmation, the literal PRUNE phrase, and a final themed in-app confirmation. The web database identity still has no direct delete permission; it can invoke only the migration-installed, definer-secured retention procedure, which validates the retention range and performs the deletion in 1,000-row transactions. Every committed batch writes its own audit entry atomically, so an interrupted multi-batch run retains a complete record of the deletions that finished. One invocation removes at most 50,000 rows so maintenance cannot create an unbounded transaction; if more expired entries remain, the UI asks for another reviewed run.

Deployment operators can use the equivalent preview-first terminal workflow. This example keeps the most recent 365 days:

```
./scripts/prune_audit_log.sh 365
```

After reviewing the preview, repeat the command with the literal confirmation word:

```
./scripts/prune_audit_log.sh 365 PRUNE
```

Both workflows accept retention periods from 1 through 36500 days, delete only entries strictly older than the displayed cutoff, and add an `audit_log_pruned` event with the cutoff and deleted count. If no entries qualify, they make no change. Take and retain any required backup before pruning; deleted audit entries cannot be recovered from the application.

Operational mail retention

Terminal account-email and digest outbox rows, plus processed, rejected, or failed inbound source messages, can be previewed and pruned with a separate bounded maintenance command. Pending, processing, retrying, human-review, and inbound records with an undelivered Mattermost reply alert are never eligible. Removing an inbound source message preserves its Chron Log text while its source link is cleared by the existing foreign key.

```
./scripts/prune_mail_history.sh 365
./scripts/prune_mail_history.sh 365 PRUNE
```

The command accepts 30 through 36500 retention days and removes at most 10,000 rows per category per invocation. Review the preview and retain any required backup before confirming; source mail and terminal queue history cannot be

recovered from the application.

Exact database restore runbook

The following procedure is complete for the Docker Compose deployment. Run every step from the repository root. The example deliberately renames the selected archive to `backups/restore.dnrbackup`, so the commands can be copied without shell globs or unresolved path variables.

01 Check out the exact DNR release that created the backup.

Restore the matching `secrets/dnr_2fa_encryption_key`; create or restore the independent root, application, backup-reader, maintenance, geocoder, inbound-mail, and outbound-mail database secret files; create an empty `mode-600 secrets/backup_password`; and confirm `.env` points to those nine secret files. Then build every image used by the procedure:

```
install -d -m 700 secrets backups
install -m 600 /dev/null secrets/backup_password
./scripts/compose_with_provenance.sh production build
maintenance web geocoder ingress
```

Do not generate a new 2FA key: without the original key, authenticator secrets in the restored database cannot be decrypted. If the database service and its volume already exist, start the database and one-shot migrator:

```
docker compose up -d db migrator --wait
```

For disaster recovery with no database volume, the same command builds the empty `dnr` database entirely from the baseline and forward migrations and installs the restricted service accounts. The restore refuses a schema mismatch before deleting or inserting data.

02 Copy the selected encrypted backup into the ignored

`backups/` directory under the fixed name and protect it. Replace `/absolute/path/to/selected.dnrbackup` with the actual source path:

```
install -m 600 /absolute/path/to/selected.dnrbackup
backups/restore.dnrbackup
```

03 Put the exact backup encryption password in

`secrets/backup_password` with no added newline. `read -s` keeps it off the screen and `printf` preserves the value exactly:

```
umask 077
IFS= read -r -s DNR_RESTORE_PASSWORD
printf '%s' "$DNR_RESTORE_PASSWORD" >
secrets/backup_password
unset DNR_RESTORE_PASSWORD
chmod 600 secrets/backup_password
```

04 Before stopping the application, make an independent plaintext SQL safety dump of the current database. This

file is the rollback point and contains secrets, so retain mode 600 and store it only on the deployment host:

```
docker compose exec -T db sh -c 'MYSQL_PWD="$(cat
"$MYSQL_ROOT_PASSWORD_FILE")" mysqldump
--single-transaction --routines --triggers
--no-tablespaces -uroot dnr' >
backups/pre-restore-safety.sql
chmod 600 backups/pre-restore-safety.sql
```

05 Stop every service that can mutate application data. Leave db running. If inbound or outbound mail overlays are enabled, stop mail-ingest and mail-dispatch through the same overlay set as well:

```
docker compose stop ingress web geocoder
docker compose ps
```

Verify that ingress, web, and geocoder show Exited and db shows Up (healthy) before continuing.

06 Run the one-shot restore with the literal confirmation word **RESTORE**:

```
docker compose --profile maintenance run --rm
--no-deps maintenance /backups/restore.dnrbackup
RESTORE
```

Success prints the restored row and table counts. The command authenticates and decrypts the complete archive, checks its size, format, schema fingerprint, declared row counts, content hash, and trailing data, replaces all tables in one transaction, verifies every restored table count, records database_restored, and invalidates sessions. A wrong password, damaged archive, schema mismatch, insertion error, or count mismatch rolls the transaction back.

07 If step 6 failed, do not run the remaining success steps. Because the restore transaction was rolled back, restart the stopped services. If you will not immediately correct the problem and retry, empty the password secret and remove the working copy before investigating the printed error:

```
docker compose up -d web geocoder ingress
install -m 600 /dev/null secrets/backup_password
rm -f backups/restore.dnrbackup
```

08 After a successful restore, apply any still-pending migrations, reassert table-specific runtime grants and the isolated maintenance grant, then restart the application services:

```
docker compose run --rm migrator
docker compose up -d web geocoder ingress
```

09 Wait for the health check, inspect service status, and run the schema health check directly:

```
docker compose ps
docker compose exec web php
/opt/dnr/bin/check_schema.php
```

Do not declare the restore complete until ingress, web, geocoder, and db are running; ingress, web, and db are healthy; and the schema command exits successfully.

10 Sign in with an administrator account that exists in the restored backup. Verify at least one known engagement, organization, contact, and user; open **Users > Audit Log** and verify the database_restored event. Also verify 2FA decryption by completing a fresh authenticator check. All pre-restore browser sessions should require sign-in again.

11 Empty the temporary restore password and remove the extra working copy only after verification. Keep the original encrypted backup and password in their approved long-term locations. Retain the SQL safety dump until the restored deployment has passed operational verification, then securely dispose of it according to the deployment's retention policy:

```
install -m 600 /dev/null secrets/backup_password
rm -f backups/restore.dnrbackup
```

If step 6 succeeded but later verification reveals incorrect data, stop ingress, web, and geocoder (plus enabled mail workers), import the safety dump, rerun migrations and privilege configuration, and restart the services:

```
docker compose stop ingress web geocoder
docker compose exec -T db sh -c 'MYSQL_PWD="$(cat
"$MYSQL_ROOT_PASSWORD_FILE")" mysql -uroot dnr' <
backups/pre-restore-safety.sql
docker compose exec -T db sh -c 'MYSQL_PWD="$(cat
"$MYSQL_ROOT_PASSWORD_FILE")" mysql -uroot dnr -e
"UPDATE users SET auth_version = auth_version + 1"'
docker compose run --rm migrator
docker compose up -d web geocoder ingress
docker compose exec web php
/opt/dnr/bin/check_schema.php
```

The explicit auth_version update after the safety-dump import prevents browser sessions that existed before either restore from becoming valid again. Repeat steps 9 through 11 after the rollback, using the safety-dump data for the representative-record checks. Once the approved retention period ends, remove the plaintext rollback dump with `rm -f backups/pre-restore-safety.sql`; filesystem-level secure erasure must follow the host's encrypted-storage and media-disposal policy.

Mattermost plugin

The installable server plugin under `mattermost-plugin/` exposes private daily summaries, engagement search/cards, editor/admin channel bindings, and role-checked follow-up buttons while keeping MOED authoritative. It uses single-use account-link codes and a deployment secret; it never connects to the database directly. Build it with `make -C mattermost-plugin dist` and follow the complete deployment, installation, verification, and rotation guide in `docs/mattermost-plugin.md`. End-user behavior is documented in the in-app **User Manual > Mattermost** chapter.

Daily operations dashboard

The application root and completed sign-ins open **Dashboard**, a role-aware daily operations view. It combines active engagements in the configured upcoming window, the signed-in user's active and overdue work, event-readiness gaps, and ended events that still need a financial closeout. Readiness flags identify unconfirmed events, missing venue addresses, missing presentations, and organizations without an assigned event contact. The greeting uses the signed-in user's first name when it is available. Administrators and editors also see inbound messages awaiting routing review and quick actions for creating engagements and tasks; reviewers receive the same operational context without write controls.

Event contacts and roles

Engagement create and edit forms can assign multiple contacts with one or more event-specific roles: **Primary host**, **On-site contact**, **Billing**, **Travel**, and **Materials**. Search the contact directory to add existing people, or use **Add new contact** repeatedly without leaving the event. New people and additional organization affiliations are saved together with the event; validation errors leave the draft available for correction.

A contact can belong to several organizations, each with its own role or title. Contact forms retain a primary organization and allow additional affiliations, such as Pastor at one church and Chairman at a research center. Organization contact lists, event pickers, searches, and event exports use the matching affiliation. Removing an affiliation clears only the event assignments that depended on it; assignments at other organizations remain intact. Deleting an organization retains shared contacts and promotes a remaining affiliation to primary.

PDF exports carry the configured digest-email wordmark on a white masthead for consistent graphical branding. They include only active work linked to the exported event and mirror the Work Queue's labeled overdue and due-today

colors and semantic edge.

Engagement lifecycle

Engagements track operational lifecycle separately from planning confirmation. Lifecycle states are **Active**, **Postponed**, **Canceled**, and **Completed**; confirmation remains **Work in progress**, **Under review**, or **Confirmed**. Canceling an engagement requires a reason, and a postponed or canceled engagement may link to a replacement event from the same organization. Replacement links cannot point to the same event or form a cycle.

Canceling an engagement also cancels its open, in-progress, and waiting follow-up tasks while preserving completed work. Postponed and canceled events cannot receive a new standard checklist or financial closeout. Finalizing the first financial report marks an active event completed, and an event with a final report cannot be moved back to another lifecycle state. Engagement lists, exports, maps, calendar feeds, and detail screens expose lifecycle and confirmation independently; daily operational queues default to active events so postponed, canceled, and completed records do not appear as current work.

Inquiry and booking pipeline

Potential events begin as first-class **Inquiries** instead of incomplete Engagements. The Booking Pipeline organizes active records across **New**, **Contacted**, **Qualified**, **Awaiting Details**, and **Proposal Sent**, with separate **Booked** and **Declined** outcomes. Each Inquiry can retain an optional Organization and primary Contact, preferred and alternate date ranges, request and location details, source, owner, priority, next action, Chron, outbound correspondence, stage history, and linked follow-up work. Declining requires a reason and remains reversible until the Inquiry is booked.

Create inquiries with **New Inquiry** in the Booking Pipeline. Inquiry email templates cover the initial response, request for details, date options, and proposal follow-up. Messages use the existing isolated delivery worker and add a signed Inquiry reply marker; matching replies return to the active Inquiry Chron. Delivery history is visible on the Inquiry and is also recorded on the selected Contact and Organization when present.

Review booking validates the required Organization, title, and preferred dates; surfaces overlapping active Engagements and qualified Inquiries; and lets the user choose which open Inquiry tasks should move. Acknowledged warnings do not become unexplained hard blocks. Conversion runs in one transaction: it creates the Engagement, assigns the primary Contact as Primary host when available, moves selected tasks, generates the

standard event checklist, marks the Inquiry Booked, and preserves the Inquiry as a read-only source record linked to the new Engagement.

Authenticated users can open **Work Queue** to review assignable follow-up work. Tasks may be general or linked to one inquiry, engagement, organization, or contact. Each task supports an owner, due date, priority, notes, and the states **Open**, **In progress**, **Waiting**, **Completed**, and **Canceled**. The queue provides personal, overdue, due-today, next-seven-days, waiting, unassigned, completed, and all-active views. Reviewers can inspect tasks; administrators and editors can create, edit, assign, and complete them; permanent deletion remains limited to administrators.

The navigation badge and the **My reminders** panel summarize only the signed-in user's actionable work: overdue, due today, next seven days, and waiting tasks. Administrators and editors also see their open financial closeouts. Each count links to the matching personal queue. Users with a verified email address can enable the optional daily email version under **My Profile > Notifications**.

Engagement, organization, and contact detail pages show their open follow-up work. Active engagements also offer an optional, idempotent standard checklist covering location, travel, presentations, materials, host reconfirmation, post-event thanks, outcome capture, and financial closeout. Every active standard item is automatically added and assigned to the creator when a new engagement is saved; re-running **Add missing checklist tasks** on an older engagement only adds missing active items. Open **Standard event tasks** from the Work Queue to add, view, or edit the reusable task content, priority, event-relative due rule, and order. Archived definitions are excluded from future checklists and may be restored; administrators can permanently delete archived definitions after fresh authentication. Editing, archiving, or deleting a definition does not rewrite tasks already generated for events. The required built-in **Complete the event financial closeout** task cannot be edited, archived, or deleted; it is due seven days after the event end date and directs staff to finalize giving/income, lodging, and travel received.

Financial closeout and giving history

Editors and administrators can close an active event from its detail page by finalizing the actual giving/income, lodging, and travel amounts received. These actual receipts are stored separately from the anticipated travel, lodging, and compensation fields used during planning. Before the first report can be finalized, every event task due on or before the last active presentation must be marked Completed. Tasks due after that presentation and tasks without due dates do not hold the closeout. A report can be corrected later without changing its original closer or close timestamp; concurrent corrections are rejected rather than silently overwriting newer figures. Every insert and correction is captured by the database audit log.

Organization detail pages calculate lifetime giving, latest-event giving, average giving per closed event, and aggregate lodging and travel receipts from finalized reports only. "Latest event" follows the event end date, even if its report was entered later, and archived events remain part of the historical totals. The organization list also shows latest-event and lifetime giving for quick review.

Authenticated users can open **Map** in the primary navigation to view engagements on an interactive, zoomable map. The initial view contains active lifecycle records; lifecycle, confirmation, and date filters can include other records when needed. Active pins use confirmation colors, while postponed, canceled, and completed pins display their lifecycle state. Selecting a pin opens the event summary and a link to the full engagement. Events without an address are counted but cannot be placed.

The web request never calls the geocoder. New or changed addresses enter a database queue; the dedicated egress-enabled worker resolves them at no more than one request per 1.1 seconds and caches results by normalized address, so events at the same address share one result. The initial map window and result count are bounded. The tile URL and attribution are supplied by the selected deployment profile; geocoding uses its separately allowlisted worker endpoint.

Worldwide address lookup with Geoapify

Use the optional `docker-compose.geoapify.yaml` overlay for Geoapify geocoding. Create a project at <https://myprojects.geoapify.com>, then save its API key in `secrets/geoapify_api_key` (excluded from Git). The overlay mounts the key only into the egress-enabled geocoder worker; PHP pages, browser scripts, and map payloads never receive it. The existing map renderer and tile provider remain independently configurable. Geoapify attribution is shown with its results.

After applying current migrations, activate it in the local preview with:

```
docker compose -f docker-compose.yaml -f
docker-compose.dev.yaml \
  -f docker-compose.geoapify.yaml up -d --no-deps web
geocoder
```

The deployment wrapper automatically includes this overlay when a nonempty key file is provisioned on the host, so subsequent s1 releases retain Geoapify. Provision the secret separately before deployment. On Linux, keep the file owner-only and grant the container worker (UID 33) read access with a file ACL, as for the other worker secrets. Set `DNR_GEOCODER_PROVIDER=nominatim` when invoking the wrapper to explicitly opt out, or `geoapify` to require the key. `DNR_GEOAPIFY_API_KEY_FILE` can override the host secret path. Never commit the key, place it in a file URL, or paste it into a command argument. Configure key restrictions for the worker's outgoing IP if desired; browser referrer-only restrictions do not work for background requests.

Lookups retain provider, confidence, match type, and matched-address metadata. Automatic Geoapify pins require a building or amenity result with at least 0.95 provider confidence, matching the requested house number and explicit country code when present. Confidence is a provider score, not a guarantee of accuracy. Broad or uncertain results remain unresolved instead of placing a city-center pin. Searches preserve street directions and the stored address; bounded fallbacks remove a venue prefix or normalize US highway wording. Provider errors are retried by the worker and never cached as address misses.

Show missing addresses filters for records needing address details; **Retry lookup** explicitly requeues a missed or failed lookup without resetting active jobs or their backoff. Empty results show guidance and a return action. Editors can also use **Set map pin** to click, drag, or enter coordinates and explicitly confirm the venue. Confirmed pins are stored per engagement, take priority over worker results, and apply only while the address hash still matches. **Use automatic lookup** removes the override. Both actions require CSRF and editing access, reject stale addresses, and record an audit event.

Authenticated users can open **Calendar** in the navigation to create, label, copy, and revoke private subscription URLs per device. The feed includes non-archived engagements in the configured bounded calendar window, regardless of lifecycle. Entries are all-day events covering the event date range and include lifecycle, confirmation, organization, title, type, and location. Canceled events use the calendar-standard **CANCELLED** status; postponed and canceled entries are marked transparent, and their descriptions include any cancellation reason and replacement event. Calendar clients choose their own refresh schedule, so database changes may

not appear immediately.

When creating a link, choose any combination of **Events**, **Presentations**, **My Active Work**, **All Active Work**, and **Birthdays (from Contacts)**. Select at least one category. Each link keeps its own content settings, displayed beneath its label. All Active Work disables My Active Work and includes everyone's active work, including unassigned tasks. My Active Work uses the subscription owner's assignments. Open, in-progress, and waiting tasks with a due date appear as transparent all-day reminders within the feed window; completed, canceled, and undated tasks do not appear. Task changes invalidate cached feeds. Existing links retain events, presentations, and birthdays without adding work automatically.

Each subscription URL contains a revocable bearer token and does not use a browser login. Treat it as a password; revoke only the affected device token if it is disclosed. DNR stores only a SHA-256 token digest and redacts all query strings from Apache access logs. Depending on the selected content, a feed can include contact names and birthdays or work titles, due dates, status, and priority. Contact details, task details, chronological notes, travel, lodging, and compensation remain excluded.

Contributing

Database integration tests must run only against a disposable database created for that test run. They refuse to start unless `DNR_INTEGRATION_TARGET=disposable` is set. The database-backup integration additionally requires `DNR_DESTRUCTIVE_BACKUP_TEST=isolated-restore` because it deletes and restores every application table. Never set either acknowledgement against a development or production database; dispose of the isolated container and volume afterward.

Keep work from different development computers on separate remote branches. Codex-assisted MacBook work uses `codex/macbook/<topic>` and home-computer work uses `codex/home/<topic>`. Use the same Git author identity on both computers, and add these trailers to every workstation-specific commit so its origin remains searchable after it is pushed:

```
Development-Host: MacBook
Codex-Client: Desktop
```

Push each branch with an upstream (`git push -u origin <branch>`) and merge it through a pull request. When continuing on a different computer, fetch the remote branch and create a new branch with that computer's prefix instead of committing directly to the first computer's branch. Application versions describe released behavior, not the computer that produced a commit; do not bump the version for each

workstation commit. Version positions follow [super].[major].[minor]: minor is [x].[y].[z+1], major is [x].[y+1].0, and super is [x+1].0.0.

Contributions to the DNR project are welcome. To contribute:

- fork the repository
- create a new branch for your feature or bug fix
- commit your changes with descriptive messages
- push your branch to your forked repository
- submit a pull request to the main repository

Please ensure that your contributions adhere to the project's coding standards and include appropriate tests.

Authors and Acknowledgment

License

This project is licensed under the MIT License. See the LICENSE file for more details.

Project Status

Under active development

Speaker records

Under **Relationships > Speakers**, all signed-in users can view speakers; administrators and editors can add and edit their name, email address, telephone number, and optional bio. The bio uses the same six-row text field and safe link rendering as Contact Notes. Telephone numbers use the Contacts country picker and formatting and are stored in E.164 format. Speakers also have optional website, bio, donation, connection, blog, and books URLs. These accept HTTP/HTTPS links up to 2,048 characters and appear on the speaker detail page. Speaker photos can be uploaded, replaced, or removed with the same JPEG/PNG/WebP validation, 5 MB limit, resized originals, and thumbnails as contact photos. The Speakers directory shares the Contacts theme, search/sort controls, view/edit icons, and upper/lower pagination rules. Speakers cannot be archived or deleted. Presentation forms select a saved speaker from a dropdown. Edits to a speaker apply to all associated presentations, including their calendar entries and exports.

The 20260907_add_speakers.sql migration creates an initial speaker record and assigns **every existing presentation**, including archived records, to that speaker. It then replaces the old free-text speaker name with a required foreign key. Apply it through the normal backed-up migration workflow together with the application changes. The application

database account receives SELECT, INSERT, and UPDATE rights for speakers only; the offline exact-restore account retains its recovery access.

Presentation QR codes

See Presentation QR Codes (docs/qr-code-system.md) for generated speaker links, public notes downloads, statistics, and country-detection configuration.